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# Simple and Effective Masked Diffusion Language Models

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## Abstract

While diffusion models excel at generating high-quality images, prior work reports a significant performance gap between diffusion and autoregressive (AR) methods in language modeling. In this work, we show that simple masked discrete diffusion is more performant than previously thought. We apply an effective training recipe that improves the performance of masked diffusion models and derive a simplified, Rao-Blackwellized objective that results in additional improvements. Our objective has a simple form—it is a mixture of classical masked language modeling losses—and can be used to train encoder-only language models that admit efficient samplers, including ones that can generate arbitrary lengths of text semi-autoregressively like a traditional language model. On language modeling benchmarks, a range of masked diffusion models trained with modern engineering practices achieves a new state-of-the-art among diffusion models, and approaches AR perplexity. We provide the code<sup>1</sup>, along with a blog post and video tutorial<sup>2</sup> on the project page:

<https://s-sahoo.com/mdlm>

## 1 Introduction

Diffusion models excel at producing realistic, high-quality images and have received significant attention as potential tools for generating discrete data, such as text [1, 31, 33], biological sequences [2, 47], and graphs [60, 63]. Unlike autoregressive (AR) approaches, diffusion-based methods are not constrained to generate data sequentially, and therefore have the potential to improve long-term planning, controllable generation, and sampling speed. However, discrete diffusion methods exhibit a performance gap relative to AR models [1, 23, 26, 33], especially in language modeling. The standard measure of language modeling performance is log-likelihood: when controlling for parameter count, prior work reports a sizable log-likelihood gap between AR and diffusion models.

In this work, we show that simple masked diffusion language modeling (MDLM) combined with effective training recipes is more performant than previously thought [1, 26, 69]. We develop a well-engineered MDLM implementation that significantly improves discrete diffusion log-likelihood; we

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<sup>1</sup>code: <https://github.com/kuleshov-group/mdlm>

<sup>2</sup>tutorial: <http://youtu.be/WjAUX23vgfg>

# 简单有效的掩蔽扩散语言模型

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## 抽象的

虽然扩散模型擅长生成高质量图像, 但先前的工作报告了语言建模中的扩散方法和自回归 (AR) 方法之间存在显著的性能差距。在这项工作中, 我们证明了简单的掩模离散扩散比之前想象的更高效。我们应用有效的训练方法来提高掩蔽扩散模型的性能, 并得出一个简化的 Rao-Blackwellized 目标, 从而带来额外的改进。我们的目标有一个简单的形式——它是经典掩码语言建模损失的混合——并且可用于训练仅编码器的语言模型, 该模型允许高效的采样器, 包括可以像传统语言模型一样半自回归生成任意长度的文本的模型。在语言建模基准上, 一系列经过现代工程实践训练的掩模扩散模型在扩散模型中达到了新的最先进水平, 并接近 AR 困惑。我们在项目页面上提供了代码<sup>1</sup>以及博客文章和视频教程<sup>2</sup>:

<https://s-sahoo.com/mdlm>

## 1 简介

扩散模型擅长生成真实、高质量的图像, 并且作为生成离散数据 (例如文本 [1, 31, 33]、生物序列 [2, 47] 和图形 [60, 63]) 的潜在工具而受到广泛关注。与自回归 (AR) 方法不同, 基于扩散的方法不受顺序生成数据的限制, 因此有可能改进长期规划、可控生成和采样速度。然而, 离散扩散方法相对于 AR 模型 [1, 23, 26, 33] 表现出性能差距, 尤其是在语言建模方面。语言建模性能的标准衡量标准是对数似然: 在控制参数计数时, 先前的工作报告了 AR 模型和扩散模型之间存在相当大的对数似然差距。

在这项工作中, 我们证明了简单的掩蔽扩散语言建模 (MDLM) 与有效的训练方法相结合的性能比之前想象的要更高 [1, 26, 69]。我们开发了一个精心设计的 MDLM 实现, 可以显著提高离散扩散对数似然; 我们

<sup>1</sup>code: <https://github.com/kuleshov-group/mdlm>

<sup>2</sup>tutorial: <http://youtu.be/WjAUX23vgfg>

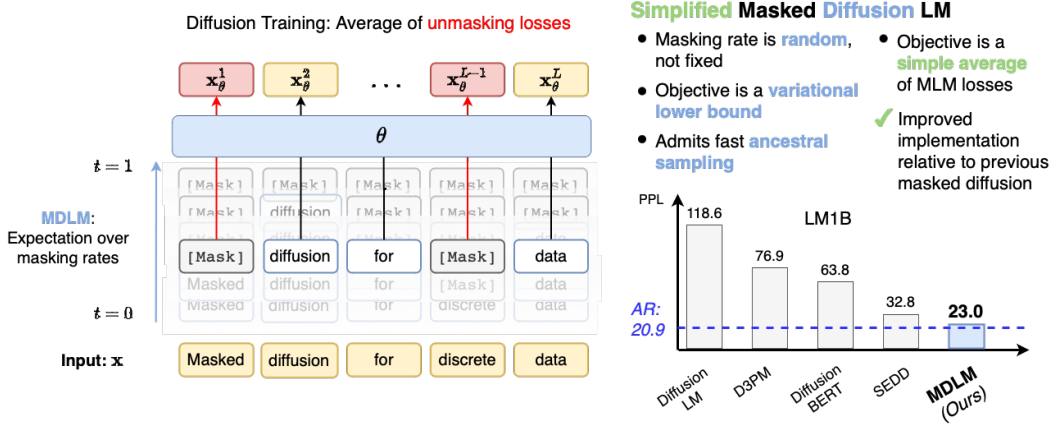


Figure 1: (Left) Our proposed masked diffusion language model (MDLM) is trained using a weighted average of masked cross entropy losses. (Top Right) In comparison to masked language models (MLM), MDLM’s objective correspond to a principled variational lower bound, and supports generation via ancestral sampling. (Bottom Right) Perplexity (PPL) on One Billion Words (LM1B) benchmark.

further improve likelihood using a simple substitution-based parameterization of the reverse diffusion process that enables deriving a Rao-Blackwellized continuous-time variational lower bound (ELBO) with improved tightness [49]. Interestingly, our objective has a simple form: it is a weighted average of masked language modeling (MLM) losses [15], and can be used to endow BERT-style, encoder-only models with principled generation capabilities. We complement this framework with efficient samplers—including ones that can generate semi-autoregressively like a typical language model.

Our masked diffusion models achieve a new state-of-the-art among diffusion models on language modeling benchmarks and approach the perplexity of AR models within 15-25%. Surprisingly, simple engineering choices significantly improve performance in both our models and simple baselines that were previously thought to perform poorly. Our framework also extends to non-language domains, including biological sequence modeling. We pre-train DNA sequence models and observe similar or higher downstream performance compared to classical BERT-style training, while also introducing generative capabilities that classical masked DNA language models lack.

**Contributions** We describe (1) a simple masked diffusion language modeling (MDLM) framework with a well-engineered implementation that outperforms all existing diffusion models across language modeling benchmarks (LM1B [8], OWT [18], DNA [12]), and that significantly improves the performance of existing baselines [1, 26]. Our MDLM framework implements (2a) a substitution-based parameterization (SUBS) of the reverse unmasking diffusion process; SUBS allows us to derive (2b) a simple, continuous-time, Rao-Blackwellized objective that improves tightness and variance of the ELBO, further increasing performance. We complement MDLM with (3) fast samplers that support semi-autoregressive (SAR) generation and outperform previous SAR models.

## 2 Background

### 2.1 Diffusion Models

Diffusion models are trained to iteratively undo a forward corruption process  $q$  that takes clean data  $\mathbf{x}$  drawn from the data distribution  $q(\mathbf{x})$  and defines latent variables  $\mathbf{z}_t$  for  $t \in [0, 1]$  that represent progressively noisy versions of  $\mathbf{x}$  [27, 54, 56, 66, 48, 19]. The standard forward process for continuous  $\mathbf{x}$  is

$$\mathbf{z}_t = \sqrt{\alpha_t} \mathbf{x} + \sqrt{1 - \alpha_t} \epsilon \quad (1)$$

where  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  and  $(\alpha_t)_{t \in [0, 1]}$  is a noise schedule, monotonically decreasing in  $t$ . The parameterized reverse diffusion model  $p_\theta$  over  $\mathbf{x}$  and  $\mathbf{z}_t$  is trained to maximize a variational lower bound on log-likelihood (ELBO). Given a number of discretization steps  $T$ , defining  $s(i) = (i-1)/T$  and  $t(i) = i/T$ ,

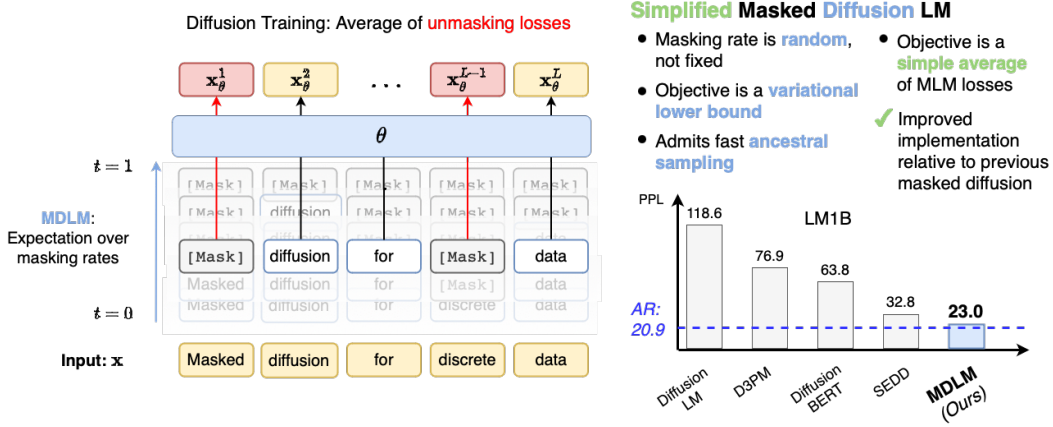


图 1: (Left) 我们提出的掩蔽扩散语言模型 (MDLM) 使用掩蔽交叉熵损失的加权平均值进行训练。 (Top Right) 与掩码语言模型 (MLM) 相比, MDLM 的目标对应于原则性变分下界, 并支持通过祖先采样生成。 (Bottom Right) 十亿字 (LM1B) 基准上的困惑度 (PPL)。

使用反向扩散过程的简单的基于替换的参数化进一步提高可能性, 该参数化能够导出具有改进的紧密性的 Rao-Blackwellized 连续时间变分下界 (ELBO) [49]。有趣的是, 我们的目标有一个简单的形式: 它是掩码语言建模 (MLM) 损失的加权平均值[15], 并且可用于赋予 BERT 式、仅编码器模型有原则的生成能力。我们用高效的采样器来补充这个框架——包括可以像典型语言模型一样半自回归生成的采样器。

我们的蒙版扩散模型在语言建模基准的扩散模型中达到了新的最先进水平, 并将 AR 模型的困惑度降低了 15-25%。令人惊讶的是, 简单的工程选择显著提高了我们的模型和以前被认为表现不佳的简单基线的性能。我们的框架还扩展到非语言领域, 包括生物序列建模。我们预训练 DNA 序列模型, 观察到与经典 BERT 式训练相似或更高的下游性能, 同时还引入了经典掩蔽 DNA 语言模型所缺乏的生成能力。

我们描述了 (1) 一个简单的掩码扩散语言建模 (MDLM) 框架, 该框架具有精心设计的实现, 其性能优于跨语言建模基准 (LM1B [8]、OWT [18]、DNA [12]) 的所有现有扩散模型, 并且显著提高了现有基线的性能 [1, 26]。我们的 MDLM 框架实现了 (2a) 反向去掩蔽扩散过程的基于替换的参数化 (SUBS); SUBS 使我们能够导出 (2b) 一个简单的、连续时间的 Rao-Blackwellized 目标, 该目标可提高 ELBO 的紧密性和方差, 从而进一步提高性能。我们用 (3) 个支持半自回归 (SAR) 生成并优于以前的 SAR 模型的快速采样器来补充 MDLM。

## 2 背景

### 2.1 扩散模型

扩散模型经过训练, 迭代地撤销前向损坏过程  $q$ , 该过程采用从数据分布  $q(\mathbf{x})$  中提取的干净数据  $\mathbf{x}$ , 并为  $t \in [0, 1]$  定义潜在变量  $\mathbf{z}_t$ , 表示  $\mathbf{x}$  [27, 54, 56, 66, 48, 19] 的渐进噪声版本。连续  $\mathbf{x}$  的标准前向过程是

$$\mathbf{z}_t = \sqrt{\alpha_t} \mathbf{x} + \sqrt{1 - \alpha_t} \epsilon \quad (1)$$

其中  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  和  $(\alpha_t)_{t \in [0, 1]}$  是噪声调度, 在  $t$  中单调递减。对  $\mathbf{x}$  和  $\mathbf{z}_t$  上的参数化反向扩散模型  $p_\theta$  进行训练, 以最大化对数似然 (ELBO) 的变分下界。给定多个离散化步骤  $T$ , 定义  $s(i) = (i-1)/T$  和  $t(i) = i/T$ ,

and using  $D_{\text{KL}}[\cdot]$  to denote the Kullback–Leibler divergence, the Negative ELBO (NELBO) equals [54]:

$$\mathbb{E}_q \left[ \underbrace{-\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)})}_{\mathcal{L}_{\text{recons}}} + \underbrace{\sum_{i=1}^T D_{\text{KL}}[q(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)}, \mathbf{x}) \| p_\theta(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)})]}_{\mathcal{L}_{\text{diffusion}}} + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)}|\mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)})]}_{\mathcal{L}_{\text{prior}}} \right] \quad (2)$$

For brevity, we drop  $i$  from  $t(i)$  and  $s(i)$  below; in general,  $s$  will denote the time step before  $t$ .

## 2.2 Discrete Diffusion Models

Applications of diffusion modeling to discrete data can be broken into two broad categories. First are works that embed discrete structures in continuous space and then perform the Gaussian diffusion defined above on these continuous representations [9, 16, 23, 24, 30, 34, 57]. More related to our method are works that define a diffusion process directly on discrete structures. D3PM [1] introduces a framework with a Markov forward process  $q(\mathbf{z}_t|\mathbf{z}_{t-1}) = \text{Cat}(\mathbf{z}_t; Q_t \mathbf{z}_{t-1})$  defined by the multiplication of matrices  $Q_t$  over  $T$  discrete time steps. This process induces marginals

$$q(\mathbf{z}_t|\mathbf{x}) = \text{Cat}(\mathbf{z}_t; \bar{Q}_t \mathbf{x}) = \text{Cat}(\mathbf{z}_t; Q_t \cdots Q_1 \mathbf{x}) \quad (3)$$

that represent the discrete-state form of (1). Extending this formalism to continuous time (as in (1)) relies on continuous time Markov chain (CTMC) theory [5]. The CTMC framework in turns leads to generalizations of the score matching perspective on diffusion modeling [55] to discrete data [33, 59]. Notably, SEDD [33] connects score-based approaches with ELBO maximization, enabling performant likelihood-based training of score-based models.

## 3 Simple Masked Diffusion Models

While previous work on discrete diffusion supports general forward processes (e.g., general  $Q_t$  in D3PM), absorbing state (i.e., masking) diffusion consistently achieves the best performance [1, 33]. In this work, instead of supporting general noise processes, we focus on masking and derive tight Rao-Blackwellized objectives that outperform general approaches and do not require CTMC theory. In this section, we first define the diffusion process for a categorical random variable. Later in Sec. 3.5, we extend this process to sequences containing multiple such categorical variables. We denote our overall approach as Masked Diffusion Language Models (MDLM).

**Notation.** We denote scalar discrete random variables with  $K$  categories as ‘one-hot’ column vectors and define  $\mathcal{V} \in \{\mathbf{x} \in \{0, 1\}^K : \sum_{i=1}^K \mathbf{x}_i = 1\}$  as the set of all such vectors. Define  $\text{Cat}(\cdot; \boldsymbol{\pi})$  as the categorical distribution over  $K$  classes with probabilities given by  $\boldsymbol{\pi} \in \Delta^K$ , where  $\Delta^K$  denotes the  $K$ -simplex. We also assume that the  $K$ -th category corresponds to a special [MASK] token and let  $\mathbf{m} \in \mathcal{V}$  be the one-hot vector for this mask, i.e.,  $\mathbf{m}_K = 1$ . Additionally, let  $\mathbf{1} = \{1\}^K$  and  $\langle \mathbf{a}, \mathbf{b} \rangle$  and  $\mathbf{a} \odot \mathbf{b}$  respectively denote the dot and Hadamard products between two vectors  $\mathbf{a}$  and  $\mathbf{b}$ .

### 3.1 Interpolating Discrete Diffusion

We restrict our attention to forward processes  $q$  that interpolate between clean data  $\mathbf{x} \in \mathcal{V}$  and a target distribution  $\text{Cat}(\cdot; \boldsymbol{\pi})$ , forming a direct extension of Gaussian diffusion in (1). Let  $q$  define a sequence of increasingly noisy latent variables  $\mathbf{z}_t \in \mathcal{V}$ , where the time step  $t$  runs from  $t=0$  (least noisy) to  $t=1$  (most noisy). The marginal of  $\mathbf{z}_t$  conditioned on  $\mathbf{x}$  at time  $t$  is

$$q(\mathbf{z}_t|\mathbf{x}) = \text{Cat}(\mathbf{z}_t; \alpha_t \mathbf{x} + (1 - \alpha_t) \boldsymbol{\pi}), \quad (4)$$

where  $\alpha_t \in [0, 1]$  is a strictly decreasing function in  $t$ , with  $\alpha_0 \approx 1$  and  $\alpha_1 \approx 0$ ; see Suppl. E.1 for details. This implies transition probabilities  $q(\mathbf{z}_t|\mathbf{z}_s) = \text{Cat}(\mathbf{z}_t; \alpha_{t|s} \mathbf{z}_s + (1 - \alpha_{t|s}) \boldsymbol{\pi})$ , where  $\alpha_{t|s} = \alpha_t / \alpha_s$ . This indicates that during each diffusion step from  $s \rightarrow t$ , a fraction  $(1 - \alpha_{t|s})$  of the probability mass is transferred to the prior distribution  $\boldsymbol{\pi}$ . The reverse posterior is given as (see Suppl. 16 for details):

$$q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x}) = \text{Cat} \left( \mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{z}_t + (1 - \alpha_{t|s}) \mathbf{1} \boldsymbol{\pi}^\top \mathbf{z}_t] \odot [\alpha_s \mathbf{x} + (1 - \alpha_s) \boldsymbol{\pi}]}{\alpha_t \mathbf{z}_t^\top \mathbf{x} + (1 - \alpha_t) \mathbf{z}_t^\top \boldsymbol{\pi}} \right). \quad (5)$$

While (4) and (5) represent a special case of the more general diffusion processes proposed in D3PM [1], we show below that they yield a simplified variational lower bound objective and admit straightforward continuous time extensions.

并使用  $D_{\text{KL}}[\cdot]$  表示 Kullback–Leibler 散度，负 ELBO (NELBO) 等于 [54]:

$$\mathbb{E}_q \left[ \underbrace{-\log p_\theta(\mathbf{x}|\mathbf{z}_{t(0)})}_{\mathcal{L}_{\text{recons}}} + \underbrace{\sum_{i=1}^T D_{\text{KL}}[q(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)}, \mathbf{x}) \| p_\theta(\mathbf{z}_{s(i)}|\mathbf{z}_{t(i)})]}_{\mathcal{L}_{\text{diffusion}}} + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)}|\mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)})]}_{\mathcal{L}_{\text{prior}}} \right] \quad (2)$$

为了简洁起见，我们从下面的  $t(i)$  和  $s(i)$  中删除  $i$ ；一般来说， $s$  表示  $t$  之前的时间步。

## 2.2 离散扩散模型

扩散建模在离散数据中的应用可以分为两大类。首先是将离散结构嵌入连续空间中，然后对这些连续表示执行上面定义的高斯扩散[9,16,23,24,30,34,57]。与我们的方法更相关的是直接在离散结构上定义扩散过程的工作。D3PM [1] 引入了一个具有马尔可夫前向过程  $q(\mathbf{z}_t|\mathbf{z}_{t-1}) = \text{Cat}(\mathbf{z}_t; Q_t \mathbf{z}_{t-1})$  的框架，该过程由矩阵  $Q_t$  与  $T$  离散时间步长的乘法定义。这个过程会产生边际效应

$$q(\mathbf{z}_t|\mathbf{x}) = \text{Cat}(\mathbf{z}_t; \bar{Q}_t \mathbf{x}) = \text{Cat}(\mathbf{z}_t; Q_t \cdot Q_{t-1} \cdots Q_1 \mathbf{x}) \quad (3)$$

代表 (1) 的离散状态形式。将这种形式主义扩展到连续时间（如 (1) 中）依赖于连续时间马尔可夫链 (CTMC) 理论[5]。CTMC 框架又将扩散建模 [55] 的分数匹配观点推广到离散数据 [33, 59]。值得注意的是，SEDD [33] 将基于分数的方法与 ELBO 最大化联系起来，从而实现基于分数的模型的基于性能似然的训练。

## 3 简单的掩模扩散模型

虽然之前关于离散扩散的工作支持一般的前向过程（例如，D3PM 中的一般  $Q_t$ ），但吸收态（即掩蔽）扩散始终实现最佳性能 [1, 33]。在这项工作中，我们不支持一般的噪声过程，而是专注于掩蔽并导出严格的 Rao-Blackwellized 目标，这些目标优于一般方法并且不需要 CTMC 理论。在本节中，我们首先定义分类随机变量的扩散过程。后来在第二节。3.5，我们将此过程扩展到包含多个此类分类变量的序列。我们将我们的整体方法表示为掩蔽扩散语言模型 (MDLM)。

符号。我们将具有  $K$  类别的标量离散随机变量表示为“one-hot”列向量，并将  $\mathcal{V} \in \{\mathbf{x} \in \{0, 1\}^K : \sum_{i=1}^K \mathbf{x}_i = 1\}$  定义为所有此类向量的集合。将  $\text{Cat}(\cdot; \boldsymbol{\pi})$  定义为  $K$  类的分类分布，概率由  $\boldsymbol{\pi} \in \Delta^K$  给出，其中  $\Delta^K$  表示  $K$ -单纯形。我们还假设第  $K$  个类别对应于一个特殊的 [MASK] 标记，并令  $\mathbf{m} \in \mathcal{V}$  为该掩码的独热向量，即  $\mathbf{m}_K = 1$ 。此外，令  $\mathbf{1} = \{1\}^K$  和  $\langle \mathbf{a}, \mathbf{b} \rangle$  和  $\mathbf{a} \odot \mathbf{b}$  分别表示两个向量  $\mathbf{a}$  和  $\mathbf{b}$  之间的点积和 Hadamard 积。

### 3.1 插值离散扩散

我们将注意力限制在前向过程  $q$  上，该过程在干净数据  $\mathbf{x} \in \mathcal{V}$  和目标分布  $\text{Cat}(\cdot; \boldsymbol{\pi})$  之间进行插值，形成 (1) 中高斯扩散的直接扩展。让  $q$  定义一系列噪声逐渐增加的潜在变量  $\mathbf{z}_t \in \mathcal{V}$ ，其中时间步长  $t$  从  $t=0$ （噪声最小）到  $t=1$ （噪声最大）。在时间  $t$  以  $\mathbf{x}$  为条件的  $\mathbf{z}_t$  的边际为

$$q(\mathbf{z}_t|\mathbf{x}) = \text{Cat}(\mathbf{z}_t; \alpha_t \mathbf{x} + (1 - \alpha_t) \boldsymbol{\pi}), \quad (4)$$

其中  $\alpha_t \in [0, 1]$  是  $t$  中的严格递减函数，其中  $\alpha_0 \approx 1$  和  $\alpha_1 \approx 0$ ；见补充。E.1 详细信息。这意味着转移概率  $q(\mathbf{z}_t|\mathbf{z}_s) = \text{Cat}(\mathbf{z}_t; \alpha_{t|s} \mathbf{z}_s + (1 - \alpha_{t|s}) \boldsymbol{\pi})$ ，其中  $\alpha_{t|s} = \alpha_t / \alpha_s$ 。这表明在从  $s \rightarrow t$  开始的每个扩散步骤中，概率质量的一部分  $(1 - \alpha_{t|s})$  被转移到先验分布  $\boldsymbol{\pi}$ 。反向后验给出为（详细信息请参见补充材料 16）：

$$q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x}) = \text{Cat} \left( \mathbf{z}_s; \frac{[\alpha_{t|s} \mathbf{z}_t + (1 - \alpha_{t|s}) \mathbf{1} \boldsymbol{\pi}^\top \mathbf{z}_t] \odot [\alpha_s \mathbf{x} + (1 - \alpha_s) \boldsymbol{\pi}]}{\alpha_t \mathbf{z}_t^\top \mathbf{x} + (1 - \alpha_t) \mathbf{z}_t^\top \boldsymbol{\pi}} \right). \quad (5)$$

虽然 (4) 和 (5) 代表了 D3PM [1] 中提出的更一般扩散过程的特殊情况，但我们在下面表明它们产生了简化的变分下界目标并允许直接的连续时间扩展。

### 3.2 Masked Diffusion

Next, we focus on masking processes and derive a simple Rao-Blackwellized objective for this choice of  $q$ . This objective incurs lower variance during training and improves tightness.

#### 3.2.1 Forward Masking Process

In masked (i.e., absorbing state) diffusion, we set  $\pi = \mathbf{m}$ . At each noising step,  $t$ , the input  $\mathbf{x}$  transitions to a ‘masked’ state  $\mathbf{m}$  with some probability. If an input transitions to  $\mathbf{m}$  at any time  $t'$ , it will remain in this state for all  $t > t'$ :  $q(\mathbf{z}_t | \mathbf{z}_{t'} = \mathbf{m}) = \text{Cat}(\mathbf{z}_t; \mathbf{m})$ . At time  $T$ , all inputs are masked with probability 1.

The marginal of the forward process (4) is given by  $q(\mathbf{z}_t | \mathbf{x}) = \text{Cat}(\mathbf{z}_t; \alpha_t \mathbf{x} + (1 - \alpha_t) \mathbf{m})$ . Using properties of the masking process, the posterior  $q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x})$  simplifies (5); see Suppl. A.2:

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1 - \alpha_s) \mathbf{m} + (\alpha_s - \alpha_t) \mathbf{x}}{1 - \alpha_t}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases} \quad (6)$$

#### 3.2.2 Reverse Unmasking Process

The reverse process inverts the noise process defined by  $q$ . We consider both a finite number of steps  $T$ , as well as a continuous time model corresponding to  $T \rightarrow \infty$ . We begin with the discrete-time case for which the generative model is expressed as  $p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{z}_1) p_\theta(\mathbf{x} | \mathbf{z}_0) \prod_{i=1}^T p_\theta(\mathbf{z}_s | \mathbf{z}_t) d\mathbf{z}_{0:1}$ .

The optimal form for  $p_\theta(\mathbf{z}_s | \mathbf{z}_t)$  matches the true posterior in (6): this follows immediately from the definition of the diffusion objective in (2), which is a sum of terms of the form  $D_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) || p_\theta(\mathbf{z}_s | \mathbf{z}_t))$ . However, (6) is conditioned on  $\mathbf{x}$ , which we do not know. Therefore, we introduce a model  $\mathbf{x}_\theta(\mathbf{z}_t, t) : \mathcal{V} \times [0, 1] \rightarrow \Delta^K$  that approximates  $\mathbf{x}$  with a neural network. We can also omit explicit dependence of  $\mathbf{x}_\theta$  on time  $t$ , which simplifies sampling, yielding a 2x inference speed-up (see Suppl. E.2).

#### 3.2.3 SUBS Parameterization

The specific parameterization for  $p_\theta(\mathbf{z}_s | \mathbf{z}_t)$  that we use is

$$p_\theta(\mathbf{z}_s | \mathbf{z}_t) = q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t)) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t), & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1 - \alpha_s) \mathbf{m} + (\alpha_s - \alpha_t) \mathbf{x}_\theta(\mathbf{z}_t, t)}{1 - \alpha_t}\right), & \mathbf{z}_t = \mathbf{m}, \end{cases} \quad (7)$$

Furthermore, we induce 2 key properties of the absorbing state diffusion process into our denoising model,  $\mathbf{x}_\theta(\mathbf{z}_t, t)$ : an unmasked token remains unchanged during reverse diffusion, and the clean input is never masked. We implement these as substitutions to the output of  $\mathbf{x}_\theta(\mathbf{z}_t, t)$ , hence we call our parameterization SUBS.

**Zero Masking Probabilities** First, notice that by definition,  $\langle \mathbf{x}, \mathbf{m} \rangle = 0$ . For this reason, we design the denoising network such that  $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$ , i.e., we substitute the logit index corresponding to the [MASK] token with  $-\infty$ .

**Carry-Over Unmasking** Second, if  $\mathbf{z}_t$  is unmasked, then we desire  $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{z}_t$ , i.e., unmasked latents are ‘carried over’. We accomplish this by substituting the output of our network to simply copy unmasked inputs.

In Suppl. B.1, we show that “Zero Masking Probabilities” property simplifies the D3PM’s NELBO (39) to (41), and “Carry-Over Unmasking” further simplifies (41) to (43) whose continuous time equivalent is the simplified NELBO (10). Table 8 shows that each simplification leads to an improved likelihood.

### 3.3 Rao-Blackwellized Likelihood Bounds

Recall from (2) that the diffusion training objective has the form  $\mathcal{L}_{\text{recons}} + \mathcal{L}_{\text{diffusion}} + \mathcal{L}_{\text{prior}}$ . For the simplified reverse process in (7), the discrete-time diffusion loss for finite  $T$  simplifies to (Suppl. B.1.3):

$$\mathcal{L}_{\text{diffusion}} = \sum_{i=1}^T \mathbb{E}_q [D_{\text{KL}}(q(\mathbf{z}_{s(i)} | \mathbf{z}_{t(i)}, \mathbf{x}) || p_\theta(\mathbf{z}_{s(i)} | \mathbf{z}_{t(i)}))] = \sum_{i=1}^T \mathbb{E}_q \left[ \frac{\alpha_{t(i)} - \alpha_{s(i)}}{1 - \alpha_{t(i)}} \log \langle \mathbf{x}_\theta(\mathbf{z}_{t(i)}), \mathbf{x} \rangle \right] \quad (8)$$

### 3.2 掩蔽扩散

接下来，我们关注掩蔽过程，并针对  $q$  的选择导出一个简单的 Rao-Blackwellized 目标。该目标在训练期间会产生较低的方差并提高紧密度。

#### 3.2.1 前向掩蔽过程

在掩蔽（即吸收态）扩散中，我们设置  $\pi = \mathbf{m}$ 。在每个噪声步骤  $t$  中，输入  $\mathbf{x}$  以一定的概率转变为“屏蔽”状态  $\mathbf{m}$ 。如果输入在任何时间  $t'$  转换到  $\mathbf{m}$ ，则所有  $t > t'$  都将保持此状态： $q(\mathbf{z}_t | \mathbf{z}_{t'} = \mathbf{m}) = \text{Cat}(\mathbf{z}_t; \mathbf{m})$ 。在时间  $T$ ，所有输入都以概率 1 进行屏蔽。

前向过程 (4) 的边际由  $q(\mathbf{z}_t | \mathbf{x}) = \text{Cat}(\mathbf{z}_t; \alpha_t \mathbf{x} + (1 - \alpha_t) \mathbf{m})$  给出。使用掩蔽过程的属性，后验  $q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x})$  简化 (5)；见补充。A.2:

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1 - \alpha_s) \mathbf{m} + (\alpha_s - \alpha_t) \mathbf{x}}{1 - \alpha_t}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases} \quad (6)$$

#### 3.2.2 反向解密过程

相反的过程反转由  $q$  定义的噪声过程。我们考虑有限数量的步骤  $T$  以及对应于  $T \rightarrow \infty$  的连续时间模型。我们从离散时间情况开始，生成模型表示为

$$p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{z}_1) p_\theta(\mathbf{x} | \mathbf{z}_0) \prod_{i=1}^T p_\theta(\mathbf{z}_s | \mathbf{z}_t) d\mathbf{z}_{0:1}.$$

$p_\theta(\mathbf{z}_s | \mathbf{z}_t)$  的最佳形式与 (6) 中的真实后验匹配：这直接来自 (2) 中扩散目标的定义，它是  $D_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$  形式的项之和。然而，(6) 的条件是  $\mathbf{x}$ ，我们不知道。因此，我们引入一个模型  $\mathbf{x}_\theta(\mathbf{z}_t, t) : \mathcal{V} \times [0, 1] \rightarrow \Delta^K$ ，用神经网络逼近  $\mathbf{x}$ 。我们还可以省略  $\mathbf{x}_\theta$  对时间  $t$  的显式依赖，这简化了采样，产生 2 倍的推理加速（参见补充材料 E.2）。

#### 3.2.3 SUBS 参数化

我们使用的  $p_\theta(\mathbf{z}_s | \mathbf{z}_t)$  的具体参数化是

$$p_\theta(\mathbf{z}_s | \mathbf{z}_t) = q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t)) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t), & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1 - \alpha_s) \mathbf{m} + (\alpha_s - \alpha_t) \mathbf{x}_\theta(\mathbf{z}_t, t)}{1 - \alpha_t}\right), & \mathbf{z}_t = \mathbf{m}, \end{cases} \quad (7)$$

此外，我们将吸收态扩散过程的 2 个关键属性引入到我们的去噪模型  $\mathbf{x}_\theta(\mathbf{z}_t, t)$  中：未屏蔽的标记在反向扩散过程中保持不变，并且干净的输入永远不会被屏蔽。我们将它们实现为对  $\mathbf{x}_\theta(\mathbf{z}_t, t)$  输出的替换，因此我们将参数化称为 SUBS。

零掩蔽概率首先，请注意，根据定义， $\langle \mathbf{x}, \mathbf{m} \rangle = 0$ 。因此，我们设计去噪网络，使得  $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$ ，即，我们用  $-\infty$  替换与 [MASK] 标记对应的 logit 索引。

其次，如果  $\mathbf{z}_t$  未被屏蔽，那么我们希望  $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{z}_t$ ，即未屏蔽的潜伏被“结转”。我们通过将网络的输出替换为简单地复制未屏蔽的输入来实现此目的。

在补充中。B.1，我们证明“零掩蔽概率”属性将 D3PM 的 NELBO (39) 简化为 (41)，而“结转解掩蔽”进一步简化 (41) 到 (43)，其连续时间等效性是简化的 NELBO (10)。表 8 显示每次简化都会提高可能性。

### 3.3 Rao-Blackwellized 似然界

回想一下 (2)，扩散训练目标的形式为  $\mathcal{L}_{\text{recons}} + \mathcal{L}_{\text{diffusion}} + \mathcal{L}_{\text{prior}}$  对于 (7) 中的简化逆过程，有限  $T$  的离散时间扩散损失简化为（补充材料 B.1.3）：

$$\mathcal{L}_{\text{diffusion}} = \sum_{i=1}^T \mathbb{E}_q [D_{\text{KL}}(q(\mathbf{z}_{s(i)} | \mathbf{z}_{t(i)}, \mathbf{x}) \| p_\theta(\mathbf{z}_{s(i)} | \mathbf{z}_{t(i)}))] = \sum_{i=1}^T \mathbb{E}_q \left[ \frac{\alpha_{t(i)} - \alpha_{s(i)}}{1 - \alpha_{t(i)}} \log \langle \mathbf{x}_\theta(\mathbf{z}_{t(i)}), \mathbf{x} \rangle \right] \quad (8)$$

Note that this objective is simpler and more well-behaved than the expression one would obtain for  $D_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})||p_\theta(\mathbf{z}_s|\mathbf{z}_t))$  under the parameterization induced by using  $p_\theta(\mathbf{z}_s|\mathbf{z}_t) = q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t))$  from (5), which is similar to what is used by D3PM [1] (see Suppl. A.2.4):

$$\left[ \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s) (\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t) (\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \quad (9)$$

We refer to the process of obtaining (8) in lieu of (9) as a form of Rao-Blackwellization. Specifically, we analytically compute expectations such as  $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$  in order to simplify objective (9) to obtain (8). Without analytical simplifications, a model must learn  $\theta$  such that  $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$  holds. Unlike in regular Rao-Blackwellization, simplifications are possible because of modeling choices for  $\mathbf{x}_\theta(\mathbf{z}_t, t)$  (zero masking probabilities and carry-over unmasking). In that sense, our approach has similarities to graphical modeling, where incorporating conditional independencies into  $p_\theta$  sets certain log-likelihood terms to zero. However, our approach also empirically helps reduce variance, hence we refer to it as Rao-Blackwellization, somewhat abusing the usual terminology.

### 3.4 Continuous-Time Likelihood Bounds

Previous works have shown empirically and mathematically that increasing the number of steps  $T$  yields a tighter approximation to the ELBO [29]. Following a similar argument, we form an continuous extension of (8) by taking  $T \rightarrow \infty$  (see Suppl. B.2), which yields the following NELBO,  $\mathcal{L}_{\text{NELBO}}^\infty$ :

$$\mathcal{L}_{\text{NELBO}}^\infty = \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle dt \quad (10)$$

**Invariance to the noise schedule** The function  $\alpha_t$  is invertible due to the monotonicity assumption in Sec. 3.1, and so we can perform the following change of variables in (10):  $\gamma \equiv \log(1 - \alpha_t)$ . Thus, the diffusion loss can be equivalently expressed as  $\mathcal{L}_{\text{NELBO}}^\infty = -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \mathbf{x}_\theta(\mathbf{z}_\gamma, \gamma), \mathbf{x} \rangle d\gamma$ ; see Suppl. E.1.1 for details. This new formulation demonstrates that the diffusion loss is invariant to the functional form of  $\alpha_t$ , which we verify empirically in Suppl. E.1.

### 3.5 Masked Diffusion Language Models

Next, we apply masked diffusion to language modeling over sequences  $\mathbf{x}^{1:L}$  of  $L$  tokens, with  $\mathbf{x}^\ell$  denoting the  $\ell$ -th token. We make the assumption that the forward noising process is applied independently across a sequence and that, conditioned on a sequence of latents  $\mathbf{z}_t^{1:L}$ , the denoising process factorizes independently across tokens, i.e.,  $p_\theta(\mathbf{z}_s^{1:L} | \mathbf{z}_t^{1:L}) = \prod_{\ell=1}^L p_\theta(\mathbf{z}_s^\ell | \mathbf{z}_t^{1:L})$ . Thus, we use a single model to compute  $\mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t)$  for each  $\ell$  from a masked sequence  $\mathbf{z}_t$ , optimizing:

$$\mathcal{L}_{\text{NELBO}}^\infty = \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1 - \alpha_t} \sum_{\ell} \log \langle \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \mathbf{x}^\ell \rangle dt \quad (11)$$

Interestingly, our objective has a simple form: it is the weighted average of masked language modeling (MLM) losses [15]. Thus our work establishes a connection between generative diffusion models and encoder-only BERT models. Our objective enables principled selection of a (randomized) masking rate, and also endows BERT-style models with principled generation capabilities; see Sec. 6. The full training algorithm is provided in Suppl. B.3.

**Note:** Although (11) imposes a loss on all tokens, unmasked tokens don't contribute to the loss, as they are copied over by the denoising network due to "carry-over unmasking" (Sec. 3.2.3), effectively reducing  $\log \langle \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \mathbf{x}^\ell \rangle$  to zero.

#### 3.5.1 Training Considerations for Masked Diffusion

One of the key contributions of our work is a well-engineered implementation of masked diffusion models. Our experiments demonstrate that these improvements greatly boost performance even for methods previously thought to perform poorly, e.g., Austin et al. [1]. Below we briefly summarize these implementation details. First, we find that tokenization is critical to performance. Small vocabularies, such as the 8k vocabulary in Austin et al. [1], result in longer-range dependencies that decrease the performance of both diffusion and AR models. Additionally, by focusing on masked diffusion, we

请注意，这个目标比在使用 (5) 中的  $p_\theta(\mathbf{z}_s|\mathbf{z}_t) = q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t))$  诱导的参数化下为  $D_{\text{KL}}(q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})\|p_\theta(\mathbf{z}_s|\mathbf{z}_t))$  获得的表达式更简单且表现更好，这类似于 D3PM [1] 使用的内容（参见补充材料 A.2.4）：

$$\left[ \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s) (\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t) (\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \quad (9)$$

我们将获得 (8) 代替 (9) 的过程称为 Rao-Blackwellization 的一种形式。具体来说，我们分析计算诸如  $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$  之类的期望，以便简化目标 (9) 以获得 (8)。如果没有分析简化，模型必须学习  $\theta$  以使  $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$  成立。与常规的 Rao-Blackwellization 不同，由于对  $\mathbf{x}_\theta(\mathbf{z}_t, t)$  (零掩码概率和遗留去掩码) 的建模选择，简化是可能的。从这个意义上说，我们的方法与图形建模有相似之处，其中将条件独立性合并到  $p_\theta$  中将某些对数似然项设置为零。然而，我们的方法在经验上也有助于减少方差，因此我们将其称为 Rao-Blackwellization，有点滥用常用术语。

### 3.4 连续时间似然界

先前的工作已经从经验和数学角度表明，增加步数  $T$  会产生更严格的 ELBO 近似值 [29]。遵循类似的论证，我们通过采取  $T \rightarrow \infty$  (参见补充材料来形成 (8) 的连续扩展。B.2)，产生以下 NELBO， $\mathcal{L}_{\text{NELBO}}^\infty$ ：

$$\mathcal{L}_{\text{NELBO}}^\infty = \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle dt \quad (10)$$

噪声表的不变性 由于第 2 节中的单调性假设，函数  $\alpha_t$  是可逆的。3.1，因此我们可以对 (10) 中的变量进行以下更改： $\gamma \equiv \log(1 - \alpha_t)$ 。因此，扩散损失可以等价地表示为  $\mathcal{L}_{\text{NELBO}}^\infty = -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \mathbf{x}_\theta(\mathbf{z}_\gamma, \gamma), \mathbf{x} \rangle d\gamma$ ；见补充。E.1.1 详细信息。这个新的公式表明，扩散损失对于  $\alpha_t$  的函数形式是不变的，我们在补充中凭经验验证了这一点。E.1。

### 3.5 掩蔽扩散语言模型

接下来，我们将掩码扩散应用于对  $L$  标记的序列  $\mathbf{x}^{1:L}$  的语言建模，其中  $\mathbf{x}^\ell$  表示第  $\ell$  个标记。我们假设前向噪声过程独立地应用于整个序列，并且以潜在序列  $\mathbf{z}_t^{1:L}$  为条件，去噪过程独立地跨标记进行因式分解，即  $p_\theta(\mathbf{z}_s^{1:L} | \mathbf{z}_t^{1:L}) = \prod_{\ell=1}^L p_\theta(\mathbf{z}_s^\ell | \mathbf{z}_t^{1:L})$ 。因此，我们使用单个模型来计算来自屏蔽序列  $\mathbf{z}_t$  的每个  $\ell$  的  $\mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t)$ ，优化：

$$\mathcal{L}_{\text{NELBO}}^\infty = \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1 - \alpha_t} \sum_{\ell} \log \langle \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \mathbf{x}^\ell \rangle dt \quad (11)$$

有趣的是，我们的目标有一个简单的形式：它是掩码语言建模 (MLM) 损失的加权平均值 [15]。因此，我们的工作生成扩散模型和仅编码器的 BERT 模型之间建立了联系。我们的目标能够原则性地选择 (随机) 掩蔽率，并且还赋予 BERT 式模型原则性的生成能力；见第 2 节。6. 补充材料中提供了完整的训练算法。B.3。

注意：虽然 (11) 对所有令牌造成了损失，但未屏蔽的令牌不会造成损失，因为它们由于“结转未屏蔽” (第 3.2.3 节) 而被去噪网络复制，有效地将  $\log \langle \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \mathbf{x}^\ell \rangle$  减少到零。

#### 3.5.1 掩蔽扩散的训练注意事项

我们工作的主要贡献之一是精心设计的掩蔽扩散模型的实现。我们的实验表明，这些改进极大地提高了性能，即使对于以前被认为性能不佳的方法 (例如 Austin 等人) 也是如此。[1]。下面我们简单总结一下这些实现细节。首先，我们发现标记化对于性能至关重要。小词汇表，例如 Austin 等人中的 8k 词汇表。[1]，导致较长距离的依赖性，从而降低扩散模型和 AR 模型的性能。此外，通过关注掩模扩散，我们

are able to provide a numerically stable implementation of the objective function. Namely, since previous formulations of discrete diffusion were constructed to accommodate a wide range of limiting distributions [1], the objective was implemented by materializing the full transition matrices  $\bar{Q}_t$  and posterior probabilities. In contrast, we evaluate  $D_{\text{KL}}[q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) || p_\theta(\mathbf{z}_s | \mathbf{z}_t)]$  by examining only the masked token indices rather than comparing the full true and approximate posterior distributions.

Furthermore, we modernize the architecture for the denoising network relative to D3PM [1]. In lieu of the T5 architecture used in D3PM, we use the diffusion transformer (DiT) introduced in Peebles & Xie [42], which integrates time step conditioning into a standard encoder-only transformer [62] and uses rotary positional embeddings [58]. In addition, we implement a low-discrepancy sampler that reduces the variance of the ELBO, similar to Kingma et al. [29] and draws correlated samples  $t_i$  rather than performing i.i.d. sampling.

## 4 Inference and Sampling in Masked Diffusion Language Models

### 4.1 Efficient Ancestral Sampling

To generate a sequence of length  $L$ , the reverse diffusion process starts with the sequence  $\mathbf{z}_{t=1}^{1:L}$  where  $\mathbf{z}_{t=1}^\ell = \mathbf{m}$ , for all  $\ell \in \{1, \dots, L\}$ . Then the subsequent latents,  $\mathbf{z}_t^{1:L}$  are generated by discretizing the reverse diffusion process with some finite  $T$ . Given  $\mathbf{z}_t^{1:L}$ , we construct  $\mathbf{z}_s^{1:L}$  by sampling each token  $\mathbf{z}_s^\ell$  independently from the distribution  $p_\theta(\mathbf{z}_s^\ell | \mathbf{z}_t^{1:L})$  given in (7).

Note that in the reverse process, unmasked tokens remain unchanged. Thus, if no new tokens in  $\mathbf{z}_s^{1:L}$  become unmasked (which can occur often in early denoising stages for large  $T$ ), then  $\mathbf{z}_s^{1:L} = \mathbf{z}_t^{1:L}$ . Additionally if the denoising model,  $\mathbf{x}_\theta(\mathbf{z}_t^{1:L})$  is not conditioned on time, then we can simply draw a new sample from  $p_\theta(\mathbf{z}_{s-1/T}^{1:L} | \mathbf{z}_s^{1:L})$  using the previously computed and cached value  $\mathbf{x}_\theta(\mathbf{z}_t^{1:L})$ . This means we have effectively “skipped” over the time step  $s$ , saving a function call to the denoising network. Note that SEDD [33] does not support this caching because the denoising network models time-dependent rates, which requires conditioning on time.

### 4.2 Semi-Autoregressive Masked Diffusion Language Models

Our method also admits an effective semi-autoregressive (SAR) decoding method that allows the model to generate sequences of arbitrary length [24, 52, 53]. Let  $\tilde{\mathbf{x}}^{1:L}$  represent the output from sampling a sequence of  $L$  tokens using the reverse diffusion process described above. To generate additional  $L' < L$  tokens, we propose a generation algorithm in which the latter  $L - L'$  tokens  $\tilde{\mathbf{x}}^{L':L}$  are used as a prefix for an additional round of generation. Given the carry-over unmasking described in Sec. 3.2.3, these prefix tokens will simply be copied over at each decoding step. The remaining tokens are generated as above with  $\mathbf{z}_s^\ell \sim p_\theta(\mathbf{z}_s^\ell | \mathbf{z}_t^{L:L+L'})$  for all  $\ell \in \{L+1, \dots, L+L'\}$ , with  $\mathbf{z}_{t=1}^{L-L':L}$  initialized to  $\tilde{\mathbf{x}}^{L-L':L}$  as opposed to being initialized as masked tokens  $\mathbf{m}$ . At the end of this process, we have produced  $L+L'$  tokens  $\text{concat}[\tilde{\mathbf{x}}^{1:L}, \tilde{\mathbf{x}}^{L+1:L+L'}]$ , where  $\text{concat}[\cdot]$  denotes concatenation along the sequence length dimension. This process can repeat indefinitely, with the prefix shifted for every new round of generation.

## 5 Experiments

### 5.1 Masked Diffusion Language Models

**Experimental Setup** We evaluate MDLM as a generative model of language and as a representation model via fine-tuning on downstream tasks.

For language modeling likelihood evaluation, we conduct experiments on two datasets: The One Billion Words Dataset (LM1B; [8]) and OpenWebText (OWT; [18]). We use the `bert-base-uncased` tokenizer for LM1B, and report perplexities on the test split. Models have a context size of 128. For OWT, which does not have a pre-defined split, we reserve the last 100K documents as a held-out validation set and report perplexities on this set. We use the GPT2 tokenizer [45] for OWT. Models have a context size of 1,024. We utilize the transformer architecture from Lou et al. [33], which augments the diffusion transformer [42] with rotary embeddings [58]. MDLM was trained for 1M or 10M steps (corresponding to 33B, 327B tokens, respectively) on LM1B and 1M steps on OWT (which corresponds to 262B tokens). The corresponding AR baseline was trained for half the number of steps

能够提供目标函数的数值稳定实现。也就是说，由于先前的离散扩散公式是为了适应广泛的极限分布[1]而构建的，因此该目标是通过具体化完整的转移矩阵 $\bar{Q}_t$ 和后验概率来实现的。相反，我们通过仅检查屏蔽标记索引而不是比较完整的真实分布和近似后验分布来评估  $D_{KL}[q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) || p_\theta(\mathbf{z}_s | \mathbf{z}_t)]$ 。

此外，我们相对于 D3PM [1] 对去噪网络的架构进行了现代化改造。我们使用 Peebles & Xie [42] 中引入的扩散变压器 (DiT) 来代替 D3PM 中使用的 T5 架构，它将时间步调节集成到标准的仅编码器变压器 [62] 中，并使用旋转位置嵌入 [58]。此外，我们实现了一个低差异采样器，可以减少 ELBO 的方差，类似于 Kingma 等人。[29]并绘制相关样本 $t_i$ 而不是执行独立同分布。采样。

## 4 掩蔽扩散语言模型中的推理和采样

### 4.1 高效祖先采样

为了生成长度为  $L$  的序列，对于所有  $\ell \in \{1, \dots, L\}$ ，反向扩散过程从序列  $\mathbf{z}_{t=1}^{1:L}$ （其中  $\mathbf{z}_{t=1}^\ell = \mathbf{m}$ ）开始。然后，通过使用一些有限的  $T$  离散化反向扩散过程来生成后续潜伏  $\mathbf{z}_t^{1:L}$  给定  $\mathbf{z}_t^{1:L}$ ，我们通过独立于 (7) 中给出的分布  $p_\theta(\mathbf{z}_s^\ell | \mathbf{z}_t^{1:L})$  采样每个标记  $\mathbf{z}_s^\ell$  来构造  $\mathbf{z}_s^{1:L}$ 。

请注意，在相反的过程中，未屏蔽的令牌保持不变。因此，如果  $\mathbf{z}_s^{1:L}$  中没有新的标记被揭露（这通常发生在大  $T$  的早期去噪阶段），则  $\mathbf{z}_s^{1:L} = \mathbf{z}_t^{1:L}$ 。此外，如果去噪模型  $\mathbf{x}_\theta(\mathbf{z}_t^{1:L})$  不以时间为条件，那么我们可以简单地使用之前计算和缓存的值  $\mathbf{x}_\theta(\mathbf{z}_t^{1:L})$  从  $p_\theta(\mathbf{z}_{s-1/T}^{1:L} | \mathbf{z}_s^{1:L})$  中提取新样本。这意味着我们实际上“跳过”了时间步  $s$ ，保存了对去噪网络的函数调用。请注意，SEDD [33] 不支持这种缓存，因为去噪网络对时间相关的速率进行建模，这需要对时间进行调节。

### 4.2 半自回归掩蔽扩散语言模型

我们的方法还采用有效的半自回归 (SAR) 解码方法，允许模型生成任意长度的序列[24,52,53]。让  $\tilde{\mathbf{x}}^{1:L}$  表示使用上述反向扩散过程对  $L$  标记序列进行采样的输出。为了生成额外的  $L' < L$  令牌，我们提出了一种生成算法，其中后面的  $L-L'$  令牌  $\tilde{\mathbf{x}}^{L':L}$  用作附加一轮生成的前缀。鉴于第 2 节中描述的遗留揭露。3.2.3 中，这些前缀标记将在每个解码步骤中简单地复制。其余标记按照上面的方式为所有  $\ell \in \{L+1, \dots, L+L'\}$ ，生成  $\mathbf{z}_s^\ell \sim p_\theta(\mathbf{z}_s^\ell | \mathbf{z}_t^{L:L+L'})$ ，其中  $\mathbf{z}_{t=1}^{L-L':L}$  初始化为  $\tilde{\mathbf{x}}^{L-L':L}$ ，而不是初始化为掩码标记  $\mathbf{m}$ 。在此过程结束时，我们生成了  $L+L'$  标记  $\text{concat}[\tilde{\mathbf{x}}^{1:L}, \tilde{\mathbf{x}}^{L+1:L+L'}]$ ，其中  $\text{concat}[\cdot]$  表示沿序列长度维度的串联。这个过程可以无限地重复，每次新一轮的生成都会改变前缀。

## 5 实验

### 5.1 掩蔽扩散语言模型

实验设置我们通过对下游任务进行微调来将 MDLM 评估为语言生成模型和表示模型。

对于语言建模可能性评估，我们在两个数据集上进行实验：十亿字数据集 (LM1B; [8]) 和 OpenWebText (OWT; [18])。我们对 LM1B 使用 bert-base-uncased 分词器，并报告测试分割的困惑。模型的上下文大小为 128。对于没有预定义分割的 OWT，我们保留最后 100K 文档作为保留验证集并报告该集的困惑。我们使用 GPT2 分词器 [45] 进行 OWT。模型的上下文大小为 1,024。我们利用 Lou 等人的变压器架构。[33]，它通过旋转嵌入[58]增强了扩散变压器[42]。MDLM 在 LM1B 上接受 1M 或 10M 步骤（分别对应于 33B、327B 令牌），在 OWT 上接受 1M 步骤（对应于 262B 令牌）。相应的 AR 基线训练了一半的步数

Table 1: Test perplexities (PPL; ↓) on LM1B. <sup>†</sup>Reported in He et al. [26]. Best diffusion value is bolded.

|                                       |                                | Parameters | PPL (↓)        |
|---------------------------------------|--------------------------------|------------|----------------|
| <i>Autoregressive</i>                 | Transformer-X Base [13]        | 0.46B      | 23.5           |
|                                       | OmniNet <sub>T</sub> [61]      | 100M       | 21.5           |
| <i>Diffusion</i>                      | BERT-Mouth [64] <sup>†</sup>   | 110M       | ≤142.89        |
|                                       | D3PM (absorb) [1]              | 70M        | ≤76.90         |
|                                       | Diffusion-LM [30] <sup>†</sup> | 80M        | ≤118.62        |
|                                       | DiffusionBert [26]             | 110M       | ≤63.78         |
|                                       | SEDD [33] (33B tokens)         | 110M       | ≤32.79         |
| <i>Autoregressive<br/>(Retrained)</i> | Transformer (33B tokens)       | 110M       | 22.32          |
|                                       | Transformer (327B tokens)      |            | 20.86          |
| <i>Diffusion<br/>(Ours)</i>           | MDLM (33B tokens)              | 110M       | ≤27.04         |
|                                       | MDLM (327B tokens)             |            | ≤ <b>23.00</b> |

to ensure similar number of tokens seen (details in Suppl. D.2). Full hyperparameters are given in Suppl. D.4. On OWT, we train with and without time step conditioning.

For representation learning, we pre-train models on the C4 dataset [46], then fine-tune and evaluate models on the GLUE benchmark [65]. Models have a context size of 128. We use the bert-base-uncased tokenizer for the representation learning experiments. We utilize the MosaicBERT architecture from Portes et al. [43], an extension of the original BERT architecture [15]. We pre-train a bidirectional MosaicBERT using an MLM objective for 37B tokens of C4, as well as a causal variant on the same data. We further fine-tune MosaicBERT model using the MDLM for 327M tokens, less than 1% of the pre-training data. We provide the full hyperparameters in Suppl. D.6.

**Likelihood Evaluation** On LM1B, MDLM outperforms all previous diffusion methods (Table 1). Compared to the SEDD baseline reported by Lou et al. [33], trained for 33B tokens, MDLM, which we train for the same amount, achieves a 17% improvement on the perplexity bound. Finally, MDLM gets within 14% of an AR baseline and continues to improve with more training. We see the same trend for models trained on OWT, a larger dataset, shown in Table 2 – MDLM outperforms prior diffusion methods, closing the gap towards AR models. In Table 12 we find that models trained with and without time conditioning attain similar perplexities on OWT. Additionally, Figure 3 demonstrates the reduced variance we achieve from our objective, when compared to previous masked diffusion models such as SEDD [33].

**Zero-Shot Likelihood Evaluation** We also explore models’ ability to generalize by taking models trained on OWT and evaluating how well they model unseen datasets. We compare the perplexities of our MDLM with SEDD [1] and an AR Transformer language model. Our zero-shot datasets include the validation splits of Penn Tree Bank (PTB; [36]), Wikitext [38], LM1B, Lambada [41], AG News [68], and Scientific Papers (Pubmed and Arxiv subsets; [10]). Full experimental details are available in Suppl. D.4.

MDLM consistently outperforms the SEDD diffusion parameterization. In some cases, e.g., for Lambada and Scientific Papers, MDLM attains better perplexity than AR. We hypothesize that these datasets are farther from OWT, and that diffusion models may be more robust to out-of-domain evaluation due to the unmasking-based objective.

**Downstream Task Evaluation** We find that BERT fine-tuned with MDLM to be a generative model results in strong perplexities while preserving performance on downstream tasks. On the C4 validation set, the AR model attains perplexity (PPL) of 22, the pre-trained BERT attains a PPL upper bound of 78 (evaluated using the MDLM variational bound), and BERT + MDLM-FT attains a PPL upper bound of 35. In Table 4, we further find that BERT + MDLM fine-tuning has no degradation in downstream

Table 2: Test perplexities (PPL; ↓) on OWT for models trained for 262B tokens. <sup>†</sup> denotes retrained models.

|                   | PPL (↓)        |
|-------------------|----------------|
| AR <sup>†</sup>   | 17.54          |
| SEDD <sup>†</sup> | ≤24.10         |
| MDLM (Ours)       | ≤ <b>23.21</b> |

表 1: LM1B 上的测试困惑度 (PPL;  $\downarrow$ )。†He 等人报道。[26]。最佳扩散值以粗体显示。

|                                   |                                | Parameters | PPL ( $\downarrow$ )  |
|-----------------------------------|--------------------------------|------------|-----------------------|
| <i>Autoregressive</i>             | Transformer-X Base [13]        | 0.46B      | 23.5                  |
|                                   | OmniNet <sub>T</sub> [61]      | 100M       | 21.5                  |
| <i>Diffusion</i>                  | BERT-Mouth [64] <sup>†</sup>   | 110M       | $\leq 142.89$         |
|                                   | D3PM (absorb) [1]              | 70M        | $\leq 76.90$          |
|                                   | Diffusion-LM [30] <sup>†</sup> | 80M        | $\leq 118.62$         |
|                                   | DiffusionBert [26]             | 110M       | $\leq 63.78$          |
|                                   | SEDD [33] (33B tokens)         | 110M       | $\leq 32.79$          |
| <i>Autoregressive (Retrained)</i> | Transformer (33B tokens)       | 110M       | 22.32                 |
|                                   | Transformer (327B tokens)      |            | 20.86                 |
| <i>Diffusion (Ours)</i>           | MDLM (33B tokens)              | 110M       | $\leq 27.04$          |
|                                   | MDLM (327B tokens)             |            | $\leq \mathbf{23.00}$ |

确保看到的令牌数量相似（详细信息见补充材料 D.2）。完整的超参数在补充中给出。D.4。在 OWT 上，我们训练有或没有时间步调节。

对于表示学习，我们在 C4 数据集 [46] 上预训练模型，然后在 GLUE 基准 [65] 上微调和评估模型。模型的上下文大小为 128。我们使用 bert-base-uncased 分词器进行表示学习实验。我们利用 Portes 等人的 MosaicBERT 架构。[43]，原始 BERT 架构的扩展 [15]。我们使用 C4 的 37B 代币的 MLM 目标以及相同数据的因果变体来预训练双向 MosaicBERT。我们使用 MDLM 进一步微调 MosaicBERT 模型，处理 3.27 亿个令牌，不到预训练数据的 1%。我们在补充中提供了完整的超参数。D.6。

可能性评估 在 LM1B 上，MDLM 优于所有先前的扩散方法（表 1）。与 Lou 等人报告的 SEDD 基线相比。[33]，针对 33B 令牌进行训练，我们训练相同数量的 MDLM，在困惑度界限上实现了 17% 的改进。最后，MDLM 的误差在 AR 基线的 14% 以内，并且通过更多训练继续改进。我们看到在 OWT（一个更大的数据集）上训练的模型也有相同的趋势，如表 2 所示 - MDLM 优于先前的扩散方法，缩小了与 AR 模型的差距。在表 12 中，我们发现经过时间调节和不经时间调节训练的模型在 OWT 上达到了相似的困惑度。此外，图 3 展示了与之前的掩蔽扩散模型（例如 SEDD [33]）相比，我们从目标中实现的方差减少。

零样本自然评估 我们还通过采用 OWT 训练的模型并评估它们对未见过的数据集建模的效果来探索模型的泛化能力。我们将 MDLM 与 SEDD [1] 和 AR Transformer 语言模型的困惑度进行比较。我们的零样本数据集包括 Penn Tree Bank (PTB; [36])、WikiText [38]、LM1B、Lambada [41]、AG News [68] 和科学论文 (Pubmed 和 Arxiv 子集; [10]) 的验证分割。完整的实验细节可在补充材料中找到。D.4。

MDLM 始终优于 SEDD 扩散参数化。在某些情况下，例如对于 Lambada 和科学论文，MDLM 比 AR 获得更好的困惑度。我们假设这些数据集更远

来自 OWT，并且由于基于揭露的目标，扩散模型对于域外评估可能更加稳健。

表 2: 针对 262B 令牌训练的模型在 OWT 上测试困惑度 (PPL;  $\downarrow$ )。†表示重新训练的模型。

|                   | PPL ( $\downarrow$ )  |
|-------------------|-----------------------|
| AR <sup>†</sup>   | 17.54                 |
| SEDD <sup>†</sup> | $\leq 24.10$          |
| MDLM (Ours)       | $\leq \mathbf{23.21}$ |

下游任务评估 我们发现 BERT 与 MDLM 一起微调为生成模型，会导致强烈的困惑，同时保留下游任务的性能。在 C4 验证集上，AR 模型的困惑度 (PPL) 为 22，预训练的 BERT 的 PPL 上限为 78（使用 MDLM 变分界进行评估），BERT + MDLM-FT 的 PPL 上限为 35。在表 4 中，我们进一步发现 BERT + MDLM 微调在下游没有退化

Table 3: Zero-shot perplexities ( $\downarrow$ ) of models trained for 524B tokens on OWT. All perplexities for diffusion models are upper bounds.

|                  | PTB          | Wikitext     | LM1B         | Lambada      | AG News      | Pubmed       | Arxiv        |
|------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| AR (Retrained)   | <b>82.05</b> | <b>25.75</b> | <b>51.25</b> | 51.28        | <b>52.09</b> | 49.01        | 41.73        |
| SEDD (Retrained) | 100.09       | 34.28        | 68.20        | 49.86        | 62.09        | 44.53        | 38.48        |
| MDLM (Ours)      | 95.26        | 32.83        | 67.01        | <b>47.52</b> | 61.15        | <b>41.89</b> | <b>37.37</b> |

Table 4: GLUE evaluation results. Evaluation measures ( $\uparrow$ ) are F1 score for QQP and MRPC, Spearman correlations for STS-B, and accuracy for the rest. For MNLI, we report match/mismatch accuracies.

|          | MNLI<br>(m/mm)     | QQP          | QNLI         | SST-2        | COLA         | STS-B        | MRPC         | RTE          | Avg          |
|----------|--------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| AR       | 80.94/80.78        | 86.98        | 86.16        | 90.14        | 33.43        | 84.32        | 83.88        | 47.29        | 74.88        |
| BERT     | 84.43/85.35        | 88.41        | <b>90.46</b> | <b>92.20</b> | 54.81        | <b>88.41</b> | 89.16        | 61.37        | 81.62        |
| +MDLM-FT | <b>84.76/85.07</b> | <b>88.49</b> | 90.30        | <b>92.20</b> | <b>57.69</b> | 87.48        | <b>90.53</b> | <b>62.09</b> | <b>82.06</b> |

GLUE performance compared to the BERT initialization. While the perplexity of our method is higher than the AR baseline, the downstream task performance is significantly better.

**Semi-Autoregressive Modeling** To test the SAR decoding algorithm presented in Sec. 4.2, we compare to SSD-LM [24] a diffusion model that was designed to generate blocks of text autoregressively. We generate 200 sequences of length 2048 tokens on a single 3090 GPU and evaluate generative perplexity under a pre-trained GPT-2 [45] model. The SSD-LM sequences are generated using blocks of 25 tokens (as implemented in their pre-trained model) and the MDLM sequences are generated using  $L' = 512$ . In Table 5, we find that in addition to achieving better generative perplexity, MDLM enables  $\sim 25$ -30x faster SAR decoding relative to SSD-LM.

Table 5: Semi-AR generative perplexity (Gen. PPL;  $\downarrow$ ) for sequences of 2048 tokens.

|             | Gen. PPL ( $\downarrow$ ) | Sec/Seq ( $\downarrow$ ) |
|-------------|---------------------------|--------------------------|
| SSD-LM      | 35.43                     | 2473.9                   |
| MDLM (Ours) | <b>27.18</b>              | <b>89.3</b>              |

## 5.2 Masked Diffusion DNA Models

We also explore applications to the generative modeling of biological sequences [14, 47] using a state space model (SSM) backbone [22]. Namely, we build on the recently-proposed Caduceus DNA language model [50], which uses as a backbone the data-dependent SSM Mamba block [21].

**Experimental Setup** We pre-train the encoder-only Caduceus [50], which is an MLM, on the HG38 human reference genome [11] and perform fine-tuning using our diffusion parameterization. We use a context length of 1024 tokens and follow Schiff et al. [50] for the experimental setup, other than learning rate which was reduced to  $1e-3$ . See Suppl. D.7 for full experimental details. We assess both generative performance using perplexity and downstream performance on Genomics Benchmarks [20] across language diffusion paradigms and AR models.

**Generative Performance** We fine-tune the Caduceus MLM across diffusion parameterizations and compare perplexities against AR models. We report perplexity values in Table 6. MDLM outperforms all other diffusion language modeling schemes.

**Downstream Task Fine-tuning** We perform downstream evaluation with the Genomics Benchmarks [20], a recently proposed benchmark with eight regulatory element classification tasks. As shown in Table 7, our generative fine-tuning paradigm preserves or improves upon downstream performance from MLM pre-training. Absorbing-state diffusion methods outperform Plaid across tasks except for the simplest task Human vs. Worm, where all methods have roughly the same performance. For tasks where the input is a biased subsample of the full genome, we observe that the correlation between perplexity and downstream performance is weaker; see Suppl. D.7.

表 3: 在 OWT 上针对 524B 令牌训练的模型的零样本困惑度 ( $\downarrow$ )。扩散模型的所有困惑度都是上限。

|                  | PTB          | Wikitext     | LM1B         | Lambada      | AG News      | Pubmed       | Arxiv        |
|------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| AR (Retrained)   | <b>82.05</b> | <b>25.75</b> | <b>51.25</b> | 51.28        | <b>52.09</b> | 49.01        | 41.73        |
| SEDD (Retrained) | 100.09       | 34.28        | 68.20        | 49.86        | 62.09        | 44.53        | 38.48        |
| MDLM (Ours)      | 95.26        | 32.83        | 67.01        | <b>47.52</b> | 61.15        | <b>41.89</b> | <b>37.37</b> |

表 4: GLUE 评估结果。评估指标 ( $\uparrow$ ) 是 QQP 和 MRPC 的 F1 分数、STS-B 的 Spearman 相关性以及其余的准确性。对于 MNLI，我们报告匹配/不匹配准确度。

|          | MNLI<br>(m/mm)     | QQP          | QNLI         | SST-2        | COLA         | STS-B        | MRPC         | RTE          | Avg          |
|----------|--------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| AR       | 80.94/80.78        | 86.98        | 86.16        | 90.14        | 33.43        | 84.32        | 83.88        | 47.29        | 74.88        |
| BERT     | 84.43/85.35        | 88.41        | <b>90.46</b> | <b>92.20</b> | 54.81        | <b>88.41</b> | 89.16        | 61.37        | 81.62        |
| +MDLM-FT | <b>84.76/85.07</b> | <b>88.49</b> | 90.30        | <b>92.20</b> | <b>57.69</b> | 87.48        | <b>90.53</b> | <b>62.09</b> | <b>82.06</b> |

GLUE 性能与 BERT 初始化相比。虽然我们方法的困惑度高于 AR 基线，但下游任务性能明显更好。

半自回归建模 测试第 2 节中提出的 SAR 解码算法。4.2，我们将 SSD-LM [24] 进行比较，这是一种旨在自回归生成文本块的扩散模型。我们在单个 3090 GPU 上生成 200 个长度为 2048 个标记的序列，并在预训练的 GPT-2 [45] 模型下评估生成困惑度。SSD-LM 序列是使用以下块生成的

表 5: 2048 个标记序列的半 AR 生成困惑度 (Gen. PPL;  $\downarrow$ )。

|             | Gen. PPL ( $\downarrow$ ) | Sec/Seq ( $\downarrow$ ) |
|-------------|---------------------------|--------------------------|
| SSD-LM      | 35.43                     | 2473.9                   |
| MDLM (Ours) | <b>27.18</b>              | <b>89.3</b>              |

25 个令牌（在其预训练模型中实现）和 MDLM 序列是使用  $L' = 512$  生成的。在表 5 中，我们发现除了实现更好的生成困惑度之外，MDLM 还使  $\sim$  SAR 解码速度比 SSD-LM 快 25-30 倍。

## 5.2 掩蔽扩散 DNA 模型

我们还探索了使用状态空间模型 (SSM) 主干 [22] 的生物序列生成模型 [14, 47] 的应用。也就是说，我们建立在最近提出的 Caduceus DNA 语言模型 [50] 的基础上，该模型使用数据依赖的 SSM Mamba 块 [21] 作为主干。

实验设置我们在 HG38 人类参考基因组 [11] 上预训练仅编码器的 Caduceus [50]（一种 MLM），并使用我们的扩散参数化进行微调。我们使用 1024 个标记的上下文长度并遵循 Schiff 等人的方法。[50] 对于实验设置，除了学习率降低到  $1e-3$  之外。参见补充。D.7 了解完整的实验细节。我们使用困惑度评估生成性能，并评估跨语言扩散范式 and AR 模型的基因组学基准 [20] 的下游性能。

生成性能 我们对扩散参数化的 Caduceus MLM 进行了微调，并将困惑度与 AR 模型进行了比较。我们在表 6 中报告了困惑度值。MDLM 优于所有其他扩散语言建模方案。

下游任务微调我们使用基因组学基准[20]进行下游评估，这是最近提出的包含八个监管要素分类任务的基准。如表 7 所示，我们的生成微调范例保留或改进了 MLM 预训练的下游性能。除了最简单的人类与蠕虫任务外，吸收状态扩散方法在所有任务中都优于 Plaid，其中所有方法的性能大致相同。对于输入是全基因组有偏差子样本的任务，我们观察到困惑度和下游性能之间的相关性较弱；见补充。D.7。

Table 6: Test perplexities (PPL;  $\downarrow$ ) of generative fine-tuning of the Caduceus MLM [50] on the HG38 reference genome. Best diffusion model values are bolded. Error bars indicate the difference between the maximum and minimum values across 5 random seeds used for fine-tuning.

|                                   |          | Params | PPL ( $\downarrow$ )           |
|-----------------------------------|----------|--------|--------------------------------|
| <i>Autoregressive (Retrained)</i> | Mamba    | 465K   | $3.067 \pm .010$               |
|                                   | HyenaDNA | 433K   | $3.153 \pm .001$               |
| <i>Diffusion (Retrained)</i>      | Plaid    | 507K   | $\leq 3.240 \pm .005$          |
|                                   | SEDD     | 467K   | $\leq 3.216 \pm .003$          |
| <i>Diffusion (Ours)</i>           | MDLM     | 467K   | $\leq \mathbf{3.199} \pm .010$ |

Table 7: Genomic Benchmarks. Top-1 accuracy ( $\uparrow$ ) across 5-fold cross-validation (CV) for a pre-trained AR Mamba, and a pre-trained Caduceus model fine-tuned with different diffusion parameterizations. The best values per task are bolded and the second best are italicized. Error bars indicate the difference between the maximum and minimum values across 5 random seeds used for CV.

| Model<br>Fine-Tuning Objective<br>(Parameter Count) | Mamba<br>AR<br>(465K) | Caduceus<br>MLM<br>(467K)    | Caduceus<br>Plaid<br>(507k)  | Caduceus<br>SEDD<br>(467k)   | Caduceus<br>MDLM (ours)<br>(467k) |
|-----------------------------------------------------|-----------------------|------------------------------|------------------------------|------------------------------|-----------------------------------|
| Mouse Enhancers                                     | 0.763 ( $\pm 0.008$ ) | <b>0.810</b> ( $\pm 0.016$ ) | 0.745 ( $\pm 0.079$ )        | 0.784 ( $\pm 0.058$ )        | <i>0.795</i> ( $\pm 0.029$ )      |
| Coding vs. Intergenic                               | 0.897 ( $\pm 0.004$ ) | <b>0.913</b> ( $\pm 0.003$ ) | <i>0.908</i> ( $\pm 0.003$ ) | <b>0.913</b> ( $\pm 0.005$ ) | <b>0.913</b> ( $\pm 0.003$ )      |
| Human vs. Worm                                      | 0.967 ( $\pm 0.002$ ) | <i>0.970</i> ( $\pm 0.002$ ) | <b>0.971</b> ( $\pm 0.001$ ) | <i>0.970</i> ( $\pm 0.003$ ) | <i>0.970</i> ( $\pm 0.003$ )      |
| Human Enhancers Cohn                                | 0.734 ( $\pm 0.027$ ) | 0.737 ( $\pm 0.001$ )        | <i>0.743</i> ( $\pm 0.010$ ) | <b>0.746</b> ( $\pm 0.015$ ) | <i>0.743</i> ( $\pm 0.016$ )      |
| Human Enhancer Ensembl                              | 0.856 ( $\pm 0.003$ ) | <b>0.907</b> ( $\pm 0.000$ ) | 0.885 ( $\pm 0.003$ )        | <i>0.905</i> ( $\pm 0.006$ ) | 0.899 ( $\pm 0.004$ )             |
| Human Regulatory                                    | 0.861 ( $\pm 0.008$ ) | <b>0.874</b> ( $\pm 0.003$ ) | <i>0.868</i> ( $\pm 0.010$ ) | 0.828 ( $\pm 0.037$ )        | <i>0.868</i> ( $\pm 0.004$ )      |
| Human OCR Ensembl                                   | 0.806 ( $\pm 0.005$ ) | <i>0.821</i> ( $\pm 0.000$ ) | 0.820 ( $\pm 0.004$ )        | 0.816 ( $\pm 0.008$ )        | <b>0.823</b> ( $\pm 0.008$ )      |
| Human NonTATA Promoters                             | 0.926 ( $\pm 0.008$ ) | <i>0.935</i> ( $\pm 0.014$ ) | <i>0.935</i> ( $\pm 0.007$ ) | <i>0.935</i> ( $\pm 0.014$ ) | <b>0.940</b> ( $\pm 0.007$ )      |

### 5.3 Ablation Analysis

In Table 8, we can see the effect of our streamlined masked diffusion implementation. The improvements described in Sec. 3.5.1 allow us to greatly reduce perplexity of previously discounted models, such as D3PM (see the bottom row of this table, which is mathematically equivalent to the D3PM formulation). While most works assumed that D3PM achieves mediocre log-likelihoods, we show that is incorrect: our re-implementation almost matches state-of-the-art score-based methods. This introduces a new strong baseline that opens new research opportunities. Additionally, in Table 8, we ablate different components of MDLM. We observe that the perplexity for MDLM trained with a discrete  $T=1000$  marginally worsens by 0.1 compared to MDLM trained in continuous time. Additionally, removing the “carry over” operation from the SUBS parameterization increases the perplexity by 1.5 points. However, further removing the “zero masking” operation does not lead to any meaningful change in perplexity. We provide further ablations for the continuous time formulation in the Appendix, showing in Table 11 that for a pre-trained model, at inference, increasing  $T$  yields better likelihoods.

Table 8: Test perplexities (PPL;  $\downarrow$ ) for MDLM ablations on LM1B. For the discrete-time models, we use  $T = 1000$ . Standard deviation is measured over 5 seeds during evaluation.

|                          | PPL ( $\leq$ )         |
|--------------------------|------------------------|
| MDLM (47)                | <b>27.04</b> $\pm$ .01 |
| w/o continuous time (43) | 27.19 $\pm$ .07        |
| & w/o carry-over (41)    | 28.56 $\pm$ .15        |
| & w/o zero masking (39)  | 28.51 $\pm$ .15        |

## 6 Related Work

**Comparison to D3PM** Masked diffusion is a strict subset of D3PM [1]; setting  $Q_{t|s} = \alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1m}^\top$  in their framework yields our forward diffusion. We improve over D3PM in three ways: (1) we adopt the SUBS parameterization; (2) this allows us to derive a simplified objective that analytically simplifies certain expectations to zero; (3) we adopt well-engineered training recipes that improve performance. Both (1) and (2) are possible because we focus on masking instead of developing a general discrete diffusion framework. Surprisingly, (3) has the largest contribution to performance.

表 6: Caduceus MLM [50] 在 HG38 参考基因组上的生成微调的测试困惑 (PPL; ↓)。最佳扩散模型值以粗体显示。误差线表示用于微调的 5 个随机种子的最大值和最小值之间的差异。

|                                   |          | Params | PPL (↓)               |
|-----------------------------------|----------|--------|-----------------------|
| <i>Autoregressive (Retrained)</i> | Mamba    | 465K   | 3.067 ± .010          |
|                                   | HyenaDNA | 433K   | 3.153 ± .001          |
| <i>Diffusion (Retrained)</i>      | Plaid    | 507K   | ≤ 3.240 ± .005        |
|                                   | SEDD     | 467K   | ≤ 3.216 ± .003        |
| <i>Diffusion (Ours)</i>           | MDLM     | 467K   | ≤ <b>3.199</b> ± .010 |

表 7: 基因组基准。预训练的 AR Mamba 和使用不同扩散参数进行微调的预训练的 Caduceus 模型的 5 倍交叉验证 (CV) 的 Top-1 准确度 (↑)。每个任务的最佳值以粗体显示，次佳值以斜体显示。误差线表示用于 CV 的 5 个随机种子的最大值和最小值之间的差异。

| Model Fine-Tuning Objective (Parameter Count) | Mamba AR (465K) | Caduceus MLM (467K)   | Caduceus Plaid (507k) | Caduceus SEDD (467k)  | Caduceus MDLM (ours) (467k) |
|-----------------------------------------------|-----------------|-----------------------|-----------------------|-----------------------|-----------------------------|
| Mouse Enhancers                               | 0.763 {±0.008}  | <b>0.810</b> {±0.016} | 0.745 {±0.079}        | 0.784 {±0.058}        | <i>0.795</i> {±0.029}       |
| Coding vs. Intergenic                         | 0.897 {±0.004}  | <b>0.913</b> {±0.003} | <i>0.908</i> {±0.003} | <b>0.913</b> {±0.005} | <b>0.913</b> {±0.003}       |
| Human vs. Worm                                | 0.967 {±0.002}  | <i>0.970</i> {±0.002} | <b>0.971</b> {±0.001} | <i>0.970</i> {±0.003} | <i>0.970</i> {±0.003}       |
| Human Enhancers Cohn                          | 0.734 {±0.027}  | 0.737 {±0.001}        | <i>0.743</i> {±0.010} | <b>0.746</b> {±0.015} | <i>0.743</i> {±0.016}       |
| Human Enhancer Ensembl                        | 0.856 {±0.003}  | <b>0.907</b> {±0.000} | 0.885 {±0.003}        | <i>0.905</i> {±0.006} | 0.899 {±0.004}              |
| Human Regulatory                              | 0.861 {±0.008}  | <b>0.874</b> {±0.003} | <i>0.868</i> {±0.010} | 0.828 {±0.037}        | <i>0.868</i> {±0.004}       |
| Human OCR Ensembl                             | 0.806 {±0.005}  | <i>0.821</i> {±0.000} | 0.820 {±0.004}        | 0.816 {±0.008}        | <b>0.823</b> {±0.008}       |
| Human NonTATA Promoters                       | 0.926 {±0.008}  | <i>0.935</i> {±0.014} | <i>0.935</i> {±0.007} | <i>0.935</i> {±0.014} | <b>0.940</b> {±0.007}       |

### 5.3 消融分析

在表 8 中，我们可以看到简化的掩模扩散实现的效果。第 2 节中描述的改进。3.5.1 使我们能够大大减少以前折扣模型的复杂性，例如 D3PM（参见该表的底行，它在数学上相当于 D3PM 公式）。虽然大多数工作假设 D3PM 实现平庸的对数似然，但我们证明这是不正确的：我们的重新实现几乎与最先进的基于分数的方法相匹配。这引入了一个新的强有力的基线，开辟了新的研究机会。此外，在表 8 中，我们消融了

表 8: LM1B 上 MDLM 消融的测试困惑度 (PPL; ↓)。对于离散时间模型，我们使用  $T = 1000$ 。在评估过程中测量 5 个种子的标准偏差。

|                          | PPL (≤)            |
|--------------------------|--------------------|
| MDLM (47)                | <b>27.04</b> ± .01 |
| w/o continuous time (43) | 27.19 ± .07        |
| & w/o carry-over (41)    | 28.56 ± .15        |
| & w/o zero masking (39)  | 28.51 ± .15        |

MDLM 的不同组件。我们观察到，与连续时间训练的 MDLM 相比，使用离散  $T = 1000$  训练的 MDLM 的困惑度略微恶化了 0.1。此外，从 SUBS 参数化中删除“结转”操作会使困惑度增加 1.5 个点。然而，进一步删除“零掩蔽”操作不会导致困惑度发生任何有意义的变化。我们在附录中提供了对连续时间公式的进一步消融，如表 11 所示，对于预训练模型，在推理时，增加  $T$  会产生更好的可能性。

## 6 相关工作

与 D3PM 的比较 掩模扩散是 D3PM 的严格子集 [1]；在他们的框架中设置  $Q_{t|s} = \alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1m}^\top$  会产生我们的前向扩散。我们在三个方面对 D3PM 进行了改进：（1）我们采用 SUBS 参数化；（2）这使我们能够得出一个简化的目标，通过分析将某些期望简化为零；（3）我们采用精心设计的培训方案来提高绩效。（1）和（2）都是可能的，因为我们专注于掩蔽而不是开发通用的离散扩散框架。令人惊讶的是，（3）对性能的贡献最大。

**Comparison to CTMC** Most implementations of diffusion work best in continuous time. However, extending D3PM in this way requires computing the limit of the product of an infinite number of matrices  $Q_T \cdot Q_{T-1} \cdots Q_t$  as  $T \rightarrow \infty$ , which requires advanced CTMC theory [5]. Our work describes simple continuous-time formulations for the most common noise processes (e.g., masking and uniform  $\pi$ ), thus helping make an important part of the literature more accessible. In Suppl. C, we show that our results are compatible with CTMC, using the rate forward matrix  $R_t = \frac{\alpha'_t}{\alpha_t}(\mathbf{I} - \mathbf{1}\mathbf{m}^\top)$  and the reverse rate  $\tilde{R}_t(\mathbf{y}', \mathbf{y})$  for the transition  $\mathbf{y} \rightarrow \mathbf{y}'$ , where  $\mathbf{y}, \mathbf{y}' \in \mathcal{V}$ :

$$\tilde{R}_t(\mathbf{y}', \mathbf{y}) = -\frac{\alpha'_t}{1 - \alpha_t} [\mathbf{y}']^\top [\mathbf{x}_\theta(\mathbf{y}, t) - \mathbf{m}] \langle \mathbf{y}, \mathbf{m} \rangle \quad (12)$$

**Comparison to Score Estimation** Score-based approaches to diffusion [55] extend to discrete states, although they typically further build upon advanced CTMC theory. In particular, SEDD [33] optimizes an ELBO<sup>3</sup> that is a function of the score model, obtaining state-of-the-art log-likelihoods among diffusion models. Our approach, however, is much simpler and does not require advanced theory. Furthermore, we can extract the score for MDLM (76), as demonstrated in Suppl. C.3, making it compatible with various techniques designed for score-based algorithms, such as samplers [5], score parameterization [33], efficient designs of the denoising network [59], guidance techniques, and more.

**Comparison to BERT** Our work provides a principled way of making BERT generative when trained with randomized masking rates. Previous work on generating from BERT used Gibbs sampling or ad-hoc methods [17, 32, 64]. The connection between BERT and diffusion was first made by Austin et al. [1]: their objective effectively involves unmasking. He et al. [26] additionally starts training from a pretrained BERT. However, both works use an objective that is similar to (9), which is less numerically stable than our objective (see Section 3.5.1). Austin et al. [1] mention in their appendix that their ELBO simplifies to a weighted masking (MLM) loss similar to (8), but it uses a more complex formula for the weights and is limited to the discrete time setting unlike our work. Furthermore, they do not train with that objective. Our work derives a simpler expression for the average of MLM losses, implements it, and obtains better likelihoods.

**Comparison to Latent Diffusion LMs** In contrast to this work, which defines diffusion over discrete structures, Plaid [23] and Diffusion LM [30] define a Gaussian diffusion process over word embeddings. Zhang et al. [67] and Hu et al. [28] extend this approach to flow matching over word embeddings, enabling the design of faster samplers. Discrete Flow Matching (DFM) [6] applies flow matching to discrete structures, using a cross-entropy loss as their training objective:  $-\mathbb{E}_{q,t} \log p_\theta(\mathbf{x}^{1:L} | \mathbf{z}_t^{1:L})$ . Similar to Chang et al. [7], DFM’s objective, while effective, is not weighted to serve as a proper ELBO. In MDLM, however, we derive a tight, principled lower bound on the log-likelihood.

**Concurrent Works** Concurrent to our work, Shi et al. [51] and Ou et al. [40] derive a similar simplified objective for masked diffusion processes. While Ou et al. [40] start from a score matching perspective, we tackle this problem from a variational lens similar to Shi et al. [51]. Similar to Ou et al. [40], we formulate efficient samplers in Section 4.1 by leveraging a time-independent denoising network.

A key differentiation between our work and that of Shi et al. [51], Ou et al. [40] is the semi-autoregressive decoding method we present in Section 4.2. While [51, 40] are restricted to sample sequences of a fixed length, we propose samplers to generate arbitrary lengths of text like a traditional language model. Furthermore, we establish the connection between our simplified objective and the masked language modeling (MLM) objective. As a result, we endow BERT-style models with principled generation capabilities while maintaining representation learning capabilities. Whereas [51, 40] only evaluate on NLP datasets, we show that masked diffusion is also effective in modeling biological sequences.

## 7 Conclusion

In this work, we explore masked diffusion. With a well-engineered implementation that supports a simple variational objective, we attain state-of-the-art diffusion perplexities on language benchmarks and demonstrate how to efficiently convert BERT-style encoders into generative models. Given we are working on language modeling, we carry any of the inherent risks and opportunities that come with this line of research.

<sup>3</sup>Lou et al. [33] mention that their ELBO can be derived from Benton et al. [4], Hanson [25] but do not provide an explicit derivation. To address this, we present a rigorous derivation in Suppl. C.2 using CTMC theory.

与 CTMC 的比较 大多数扩散实现在连续时间内效果最佳。然而，以这种方式扩展 D3PM 需要计算无限多个矩阵  $Q_T \cdot Q_{T-1} \cdots Q_t$  的乘积为  $T \rightarrow \infty$  的极限，这需要先进的 CTMC 理论[5]。我们的工作描述了最常见噪声过程的简单连续时间公式（例如，掩蔽和均匀  $\pi$ ），从而有助于使文献的重要部分更容易理解。在补充中，C，我们证明我们的结果与 CTMC 兼容，使用速率前向矩阵  $R_t = \frac{\alpha'_t}{\alpha_t}(\mathbf{I} - \mathbf{1}\mathbf{m}^\top)$  和反向速率  $\tilde{R}_t(\mathbf{y}', \mathbf{y})$  进行转换  $\mathbf{y} \rightarrow \mathbf{y}'$ ，其中  $\mathbf{y}, \mathbf{y}' \in \mathcal{V}$ :

$$\tilde{R}_t(\mathbf{y}', \mathbf{y}) = -\frac{\alpha'_t}{1 - \alpha_t} [\mathbf{y}']^\top [\mathbf{x}_\theta(\mathbf{y}, t) - \mathbf{m}] \langle \mathbf{y}, \mathbf{m} \rangle \quad (12)$$

与分数估计的比较 基于分数的扩散方法 [55] 扩展到离散状态，尽管它们通常进一步建立在先进的 CTMC 理论之上。特别是，SEDD [33] 优化了作为分数模型函数的 ELBO<sup>3</sup>，获得了扩散模型中最先进的对数似然。然而，我们的方法要简单得多，不需要先进的理论。此外，我们可以提取 MDLM 的分数 (76)，如补充材料中所示。C.3，使其与为基于分数的算法设计的各种技术兼容，例如采样器[5]、分数参数化[33]、去噪网络[59]的高效设计、引导技术等等。

与 BERT 的比较 我们的工作提供了一种在使用随机掩蔽率进行训练时使 BERT 生成的原则方法。之前从 BERT 生成的工作使用吉布斯采样或临时方法 [17, 32, 64]。Austin 等人首先将 BERT 与扩散联系起来。[1]：他们的目标实际上涉及揭露真相。他等人。[26] 另外从预训练的 BERT 开始训练。然而，这两项工作都使用类似于 (9) 的目标，它在数值上不如我们的目标稳定（参见第 3.5.1 节）。奥斯汀等人。[1] 在其附录中提到，他们的 ELBO 简化为类似于 (8) 的加权掩蔽 (MLM) 损失，但我们的工作不同，它使用更复杂的权重公式，并且仅限于离散时间设置。此外，他们的训练并不是为了这个目标。我们的工作推导了传销平均损失的更简单的表达式，实现了它，并获得了更好的可能性。

与潜在扩散 LM 的比较 与定义离散结构上的扩散的这项工作相反，Plaid [23] 和扩散 LM [30] 定义了词嵌入上的高斯扩散过程。张等人。[67] 和胡等人。[28] 将这种方法扩展到词嵌入上的流匹配，从而能够设计更快的采样器。离散流匹配 (DFM) [6] 将流匹配应用于离散结构，使用交叉熵损失作为训练目标： $-\mathbb{E}_{q,t} \log p_\theta(\mathbf{x}^{1:L} | \mathbf{z}_t^{1:L})$ 。与张等人类似。[7]，DFM 的目标虽然有效，但并未加权作为适当的 ELBO。然而，在 MDLM 中，我们得出了一个严格的、有原则的对数似然下限。

并行工作与我们的工作同时进行，Shi 等人。[51] 和 Ou 等人。[40] 导出了掩蔽扩散过程的类似简化目标。而 Ou 等人。[40] 从分数匹配的角度出发，我们从类似于 Shi 等人的变分镜头来解决这个问题。[51]。与 Ou 等人类似。[40]，我们在 4.1 节中利用时间无关的去噪网络制定了高效的采样器。

我们的工作与 Shi 等人的工作之间的一个关键区别。[51]，欧等人。[40] 是我们在 4.2 节中提出的半自回归解码方法。虽然 [51, 40] 仅限于固定长度的样本序列，但我们建议采样器像传统语言模型一样生成任意长度的文本。此外，我们在简化目标和掩码语言建模 (MLM) 目标之间建立了联系。因此，我们赋予 BERT 式模型有原则的生成能力，同时保持表示学习能力。虽然 [51, 40] 仅在 NLP 数据集上进行评估，但我们表明掩蔽扩散在生物序列建模中也很有效。

## 7 结论

在这项工作中，我们探索了掩蔽扩散。通过精心设计的支持简单变分目标的实现，我们在语言基准上获得了最先进的扩散困惑，并演示了如何有效地将 BERT 式编码器转换为生成模型。鉴于我们正在研究语言建模，我们承担着这一研究领域带来的任何固有风险和机遇。

<sup>3</sup>Lou et al. [33] mention that their ELBO can be derived from Benton et al. [4], Hanson [25] but do not provide an explicit derivation. To address this, we present a rigorous derivation in Suppl. C.2 using CTMC theory.

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# Appendices

## Appendix A Discrete time ELBO

This section is organized as follows: First, we derive the expressions for the true posterior and the approximate posterior as outlined in Suppl. A.1. We then simplify these expressions specifically for the case of absorbing state diffusion in Suppl. A.2. Finally, we derive the expression for the ELBO for absorbing state diffusion in Suppl. A.2.3.

### A.1 Generic case

Given the state transition matrix  $Q_t$ , prior  $\pi$ , and the latent variables  $\mathbf{z}_s$  and  $\mathbf{z}_t$ , where  $s < t$ , let

$$Q_{t|s} = \alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1} \pi^\top. \quad (13)$$

#### A.1.1 $q(\mathbf{z}_t | \mathbf{z}_s)$

Thus, the marginals in (3) correspond to the following forward process:

$$\begin{aligned}
q(\mathbf{z}_t | \mathbf{z}_s) &= \text{Cat}(\mathbf{z}_t; Q_{t|s}^\top \mathbf{z}_s) \\
&= \text{Cat}(\mathbf{z}_t; [\alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1} \pi^\top]^\top \mathbf{z}_s) \\
&= \text{Cat}(\mathbf{z}_t; \alpha_{t|s} \mathbf{z}_s + (1 - \alpha_{t|s}) \pi \mathbf{1}^\top \mathbf{z}_s) && \because \mathbf{1}^\top \mathbf{z}_s = 1 \\
&= \text{Cat}(\mathbf{z}_t; \alpha_{t|s} \mathbf{z}_s + (1 - \alpha_{t|s}) \pi).
\end{aligned} \quad (14)$$

The above equation indicates that during each diffusion step from  $s \rightarrow t$ , a fraction  $(1 - \alpha_{t|s})$  of the probability mass is transferred to the prior distribution  $\pi$ .



### A.1.2 $q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})$

Austin et al. [1] show that the posterior corresponding to (14) is given as follows:

$$q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x}) = \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{z}_t \odot Q_s^\top \mathbf{x}}{\mathbf{z}_t^\top Q_t^\top \mathbf{x}}\right), \quad (15)$$

which we simplify to the following:

$$\begin{aligned} q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x}) &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{I} + (1-\alpha_{t|s})\mathbf{1}\pi^\top]\mathbf{z}_t \odot [\alpha_s\mathbf{I} + (1-\alpha_s)\mathbf{1}\pi^\top]^\top \mathbf{x}}{\mathbf{z}_t^\top [\alpha_t\mathbf{I} + (1-\alpha_t)\mathbf{1}\pi^\top]^\top \mathbf{x}}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t + (1-\alpha_{t|s})\mathbf{1}\pi^\top \mathbf{z}_t] \odot [\alpha_s\mathbf{x} + (1-\alpha_s)\pi]}{\mathbf{z}_t^\top [\alpha_t\mathbf{x} + (1-\alpha_t)\pi \mathbf{1}^\top \mathbf{x}]}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t + (1-\alpha_{t|s})\mathbf{1}\pi^\top \mathbf{z}_t] \odot [\alpha_s\mathbf{x} + (1-\alpha_s)\pi]}{\alpha_t \mathbf{z}_t^\top \mathbf{x} + (1-\alpha_t)\mathbf{z}_t^\top \pi}\right). \quad \because \mathbf{1}^\top \mathbf{x} = 1 \end{aligned} \quad (16)$$

### A.1.3 $p_\theta(\mathbf{z}_s|\mathbf{z}_t)$

Austin et al. [1] approximate the reverse process in the following manner:

$$p_\theta(\mathbf{z}_s|\mathbf{z}_t) = q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t)) = \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{z}_t \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{\mathbf{z}_t^\top Q_t^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right). \quad (17)$$

where  $\mathbf{x}_\theta(\mathbf{z}_t, t) : \mathcal{V} \times [0, 1] \rightarrow \Delta^K$  is an approximation for  $\mathbf{x}$ .

## A.2 Absorbing state

For the absorbing state diffusion process we have  $\pi = \mathbf{m}$ .

### A.2.1 $q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})$

Since,  $\mathbf{z}_t \in \{\mathbf{x}, \mathbf{m}\}$ , takes only 2 values we consider the separate cases:  $\mathbf{z}_t = \mathbf{x}$  and  $\mathbf{z}_t = \mathbf{m}$ .

**Case 1.** Consider the case  $\mathbf{z}_t = \mathbf{x}$  i.e.  $\mathbf{z}_t$  is unmasked. From (16), we have the following:

$$\begin{aligned} q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{x}, \mathbf{x}) &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{x} + (1-\alpha_{t|s})\mathbf{1}\mathbf{m}^\top \mathbf{x}] \odot [\alpha_s\mathbf{x} + (1-\alpha_s)\mathbf{m}]}{\alpha_t \mathbf{x}^\top \mathbf{x} + (1-\alpha_t)\mathbf{x}^\top \mathbf{m}}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{x}] \odot [\alpha_s\mathbf{x} + (1-\alpha_s)\mathbf{m}]}{\alpha_t}\right) \quad \because \mathbf{x}^\top \mathbf{m} = 0 \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{\alpha_t \mathbf{x}}{\alpha_t}\right) \quad \because \mathbf{x} \odot \mathbf{m} = \mathbf{0} \text{ and } \alpha_t = \alpha_{t|s}\alpha_s \\ &= \text{Cat}(\mathbf{z}_s; \mathbf{x}) \quad \because \alpha_t = \alpha_{t|s}\alpha_s \end{aligned} \quad (18)$$

Thus, we have the following:

$$q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{x}, \mathbf{x}) = \text{Cat}(\mathbf{z}_s; \mathbf{x}). \quad (19)$$

**Case 2.** Consider the case  $\mathbf{z}_t = \mathbf{m}$ . By substituting  $\mathbf{z}_t = \mathbf{m}$  and  $\pi = \mathbf{m}$  in (16),  $q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})$  simplifies to the following:

$$\begin{aligned} q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}, \mathbf{x}) &= \text{Cat}\left(\mathbf{z}_s; \frac{(\alpha_{t|s}\mathbf{m} + (1-\alpha_{t|s})\mathbf{1}) \odot (\alpha_s\mathbf{x} + (1-\alpha_s)\mathbf{m})}{(1-\alpha_t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{(\alpha_{t|s}(1-\alpha_s)\mathbf{m} + (1-\alpha_{t|s})(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x})}{(1-\alpha_t)}\right) \end{aligned}$$

### A.1.2 $q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})$

奥斯汀等人。 [1]表明对应于(14)的后验为

如下：

$$q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x}) = \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{z}_t \odot Q_s^\top \mathbf{x}}{\mathbf{z}_t^\top Q_t^\top \mathbf{x}}\right), \quad (15)$$

我们将其简化为：

$$\begin{aligned} q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x}) &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{I} + (1-\alpha_{t|s})\mathbf{1}\pi^\top]\mathbf{z}_t \odot [\alpha_s\mathbf{I} + (1-\alpha_s)\mathbf{1}\pi^\top]^\top \mathbf{x}}{\mathbf{z}_t^\top [\alpha_t\mathbf{I} + (1-\alpha_t)\mathbf{1}\pi^\top]^\top \mathbf{x}}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t + (1-\alpha_{t|s})\mathbf{1}\pi^\top \mathbf{z}_t] \odot [\alpha_s\mathbf{x} + (1-\alpha_s)\pi]}{\mathbf{z}_t^\top [\alpha_t\mathbf{x} + (1-\alpha_t)\pi \mathbf{1}^\top \mathbf{x}]}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t + (1-\alpha_{t|s})\mathbf{1}\pi^\top \mathbf{z}_t] \odot [\alpha_s\mathbf{x} + (1-\alpha_s)\pi]}{\alpha_t \mathbf{z}_t^\top \mathbf{x} + (1-\alpha_t)\mathbf{z}_t^\top \pi}\right). \end{aligned} \quad \because \mathbf{1}^\top \mathbf{x} = 1 \quad (16)$$

### A.1.3 $p_\theta(\mathbf{z}_s|\mathbf{z}_t)$

奥斯汀等人。 [1]按以下方式近似逆过程：

$$p_\theta(\mathbf{z}_s|\mathbf{z}_t) = q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x} = \mathbf{x}_\theta(\mathbf{z}_t, t)) = \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{z}_t \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{\mathbf{z}_t^\top Q_t^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right). \quad (17)$$

其中  $\mathbf{x}_\theta(\mathbf{z}_t, t) : \mathcal{V} \times [0, 1] \rightarrow \Delta^K$  是  $\mathbf{x}$  的近似值。

## A.2 吸收状态

对于吸收态扩散过程，我们有  $\pi = \mathbf{m}$ 。

### A.2.1 $q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})$

由于  $\mathbf{z}_t \in \{\mathbf{x}, \mathbf{m}\}$  仅采用 2 个值，我们考虑不同的情况： $\mathbf{z}_t = \mathbf{x}$  和  $\mathbf{z}_t = \mathbf{m}$ 。

情况 1. 考虑情况  $\mathbf{z}_t = \mathbf{x}$ ，即  $\mathbf{z}_t$  未被屏蔽。由(16)，我们有以下结论：

$$\begin{aligned} q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{x}, \mathbf{x}) &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{x} + (1-\alpha_{t|s})\mathbf{1}\mathbf{m}^\top \mathbf{x}] \odot [\alpha_s\mathbf{x} + (1-\alpha_s)\mathbf{m}]}{\alpha_t \mathbf{x}^\top \mathbf{x} + (1-\alpha_t)\mathbf{x}^\top \mathbf{m}}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{x}] \odot [\alpha_s\mathbf{x} + (1-\alpha_s)\mathbf{m}]}{\alpha_t}\right) \quad \because \mathbf{x}^\top \mathbf{m} = 0 \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{\alpha_t \mathbf{x}}{\alpha_t}\right) \quad \because \mathbf{x} \odot \mathbf{m} = \mathbf{0} \text{ 和 } \alpha_t = \alpha_{t|s}\alpha_s \\ &= \text{Cat}(\mathbf{z}_s; \mathbf{x}) \quad \because \alpha_t = \alpha_{t|s}\alpha_s \end{aligned} \quad (18)$$

因此，我们有以下内容：

$$q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{x}, \mathbf{x}) = \text{Cat}(\mathbf{z}_s; \mathbf{x}). \quad (19)$$

情况 2. 考虑情况  $\mathbf{z}_t = \mathbf{m}$ 。将  $\mathbf{z}_t = \mathbf{m}$  和  $\pi = \mathbf{m}$  代入 (16) 中， $q(\mathbf{z}_s|\mathbf{z}_t, \mathbf{x})$  简化为：

$$\begin{aligned} q(\mathbf{z}_s|\mathbf{z}_t = \mathbf{m}, \mathbf{x}) &= \text{Cat}\left(\mathbf{z}_s; \frac{(\alpha_{t|s}\mathbf{m} + (1-\alpha_{t|s})\mathbf{1}) \odot (\alpha_s\mathbf{x} + (1-\alpha_s)\mathbf{m})}{(1-\alpha_t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{(\alpha_{t|s}(1-\alpha_s)\mathbf{m} + (1-\alpha_{t|s})(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x})}{(1-\alpha_t)}\right) \end{aligned}$$

$$= \text{Cat}\left(\mathbf{z}_s; \frac{(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}}{1 - \alpha_t}\right) \quad (20)$$

Note that the above categorical distribution is non-zero for  $\mathbf{z}_s \in \{\mathbf{x}, \mathbf{m}\}$  and zero for every other value. The non-zero values are specified as follows:

$$q(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) = \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \quad (21)$$

$$q(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) = \frac{1 - \alpha_s}{1 - \alpha_t} \quad (22)$$

Combining Cases 1 and 2, we get:

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}}{1 - \alpha_t}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases} \quad (23)$$

### A.2.2 $p_\theta(\mathbf{z}_s | \mathbf{z}_t)$

For the absorbing state diffusion process with  $\pi = \mathbf{m}$ , we want to simplify the (17). For this reason, we consider 2 cases: first, when  $\mathbf{z}_t \neq \mathbf{m}$  (**case 1**), second, when  $\mathbf{z}_t = \mathbf{m}$  (**case 2**).

**Case 1.** Consider the case when  $\mathbf{z}_t \neq \mathbf{m}$ . (17) simplifies to the following:

$$\begin{aligned} p_\theta(\mathbf{z}_s | \mathbf{z}_t \neq \mathbf{m}) &= \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{z}_t \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{\mathbf{z}_t^\top Q_t^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{z}_t \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{[Q_t \mathbf{z}_t]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t] \odot [\alpha_s \mathbf{I} + (1 - \alpha_s)\mathbf{m}\mathbf{1}^\top] \mathbf{x}_\theta(\mathbf{z}_t, t)}{[\alpha_t \mathbf{z}_t]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s)\mathbf{m}\langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle]}{\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle}\right) \\ &\text{since } \langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle = 1, \text{ we have the following:} \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s)\mathbf{m}]}{\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle}\right) \\ &\text{since } \mathbf{z}_t \odot \mathbf{m} = \mathbf{0}, \text{ we have the following:} \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{\alpha_t \mathbf{z}_t \odot \mathbf{x}_\theta(\mathbf{z}_t, t)}{\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle}\right) \\ &= \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) \end{aligned} \quad (24)$$

**Case 2.** Consider the case when  $\mathbf{z}_t = \mathbf{m}$ . (17) simplifies to the following:

$$\begin{aligned} p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}) &= \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{m} \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{\mathbf{m}^\top Q_t^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{m} \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{[Q_t \mathbf{m}]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{m} + (1 - \alpha_{t|s})\mathbf{1}] \odot [\alpha_s \mathbf{I} + (1 - \alpha_s)\mathbf{m}\mathbf{1}^\top] \mathbf{x}_\theta(\mathbf{z}_t, t)}{[\alpha_t \mathbf{m} + (1 - \alpha_t)\mathbf{1}]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{m} + (1 - \alpha_{t|s})\mathbf{1}] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s)\mathbf{m}\langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle]}{\alpha_t \langle \mathbf{m}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle + (1 - \alpha_t) \langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{m} + (1 - \alpha_{t|s})\mathbf{1}] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s)\mathbf{m}]}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}\right) \end{aligned}$$

$$= \text{Cat}\left(\mathbf{z}_s; \frac{(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}}{1 - \alpha_t}\right) \quad (20)$$

请注意，上述分类分布对于  $\mathbf{z}_s \in \{\mathbf{x}, \mathbf{m}\}$  是非零的，对于其他每个值都是零。非零值指定如下：

$$q(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) = \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \quad (21)$$

$$q(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) = \frac{1 - \alpha_s}{1 - \alpha_t} \quad (22)$$

结合案例 1 和案例 2，我们得到：

$$q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{(1-\alpha_s)\mathbf{m} + (\alpha_s - \alpha_t)\mathbf{x}}{1 - \alpha_t}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases} \quad (23)$$

### A.2.2 $p_\theta(\mathbf{z}_s | \mathbf{z}_t)$

对于  $\pi = \mathbf{m}$  的吸收态扩散过程，我们要简化(17)。为此，我们考虑两种情况：第一，当  $\mathbf{z}_t \neq \mathbf{m}$  (情况1)时，第二，当  $\mathbf{z}_t = \mathbf{m}$  (情况2)时。

情况 1. 考虑  $\mathbf{z}_t \neq \mathbf{m}$  时的情况。(17) 简化为：

$$\begin{aligned} p_\theta(\mathbf{z}_s | \mathbf{z}_t \neq \mathbf{m}) &= \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{z}_t \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{\mathbf{z}_t^\top Q_t^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{z}_t \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{[Q_t \mathbf{z}_t]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t] \odot [\alpha_s \mathbf{I} + (1 - \alpha_s)\mathbf{m}\mathbf{1}^\top] \mathbf{x}_\theta(\mathbf{z}_t, t)}{[\alpha_t \mathbf{z}_t]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s)\mathbf{m}\langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle]}{\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle}\right) \\ &\text{since } \langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle = 1, \text{ we have the following:} \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{z}_t] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s)\mathbf{m}]}{\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle}\right) \\ &\text{since } \mathbf{z}_t \odot \mathbf{m} = \mathbf{0}, \text{ we have the following:} \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{\alpha_t \mathbf{z}_t \odot \mathbf{x}_\theta(\mathbf{z}_t, t)}{\alpha_t \langle \mathbf{z}_t, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle}\right) \\ &= \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) \end{aligned} \quad (24)$$

情况 2. 考虑  $\mathbf{z}_t = \mathbf{m}$  时的情况。(17) 简化为：

$$\begin{aligned} p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}) &= \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{m} \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{\mathbf{m}^\top Q_t^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{Q_{t|s}\mathbf{m} \odot Q_s^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}{[Q_t \mathbf{m}]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{m} + (1 - \alpha_{t|s})\mathbf{1}] \odot [\alpha_s \mathbf{I} + (1 - \alpha_s)\mathbf{m}\mathbf{1}^\top] \mathbf{x}_\theta(\mathbf{z}_t, t)}{[\alpha_t \mathbf{m} + (1 - \alpha_t)\mathbf{1}]^\top \mathbf{x}_\theta(\mathbf{z}_t, t)}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{m} + (1 - \alpha_{t|s})\mathbf{1}] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s)\mathbf{m}\langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle]}{\alpha_t \langle \mathbf{m}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle + (1 - \alpha_t) \langle \mathbf{1}, \mathbf{x}_\theta(\mathbf{z}_t, t) \rangle}\right) \\ &= \text{Cat}\left(\mathbf{z}_s; \frac{[\alpha_{t|s}\mathbf{m} + (1 - \alpha_{t|s})\mathbf{1}] \odot [\alpha_s \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s)\mathbf{m}]}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}\right) \end{aligned}$$

$$= \text{Cat}\left(\mathbf{z}_s; \frac{\alpha_t \mathbf{m} \odot \mathbf{x}_\theta(\mathbf{z}_t, t) + (\alpha_s - \alpha_t) \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m}}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}\right) \quad (25)$$

Note that the above categorical distribution, we can obtain the values for  $p_\theta(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m})$  and  $p_\theta(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m})$  which are as follows:

$$p_\theta(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}) = \frac{(\alpha_s - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)} \quad (26)$$

$$p_\theta(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}) = \frac{\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s)}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)} \quad (27)$$

As a sanity check, we can verify that (26) reduces to (21), and (27) reduces to (22) if our denoising network can reconstruct  $\mathbf{x}$  perfectly, i.e.,  $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{x}$ .

Combining (24) and (25), we get the following expression for the reverse process parameterization:

$$p_\theta(\mathbf{z}_s | \mathbf{z}_t) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{\alpha_t \mathbf{m} \odot \mathbf{x}_\theta(\mathbf{z}_t, t) + (\alpha_s - \alpha_t) \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m}}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases} \quad (28)$$

### A.2.3 Diffusion Loss

For a given  $T$ , Let  $\mathcal{L}_T = \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$  denote the diffusion loss. We break down the computation of  $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$  into 2 cases:  $\mathbf{z}_t = \mathbf{x}$  (**case 1**) and  $\mathbf{z}_t = \mathbf{m}$  (**case 2**).

**Case 1:** consider the case  $\mathbf{z}_t = \mathbf{x}$ . Let's simplify  $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}))$ .

$$\begin{aligned} & \text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x})) \\ &= \text{D}_{\text{KL}}(\mathbf{z}_t \| \mathbf{z}_t) \quad \text{From (23) and (24)} \\ &= 0 \end{aligned} \quad (29)$$

**Case 2:** Consider the case  $\mathbf{z}_t = \mathbf{m}$ . Let's simplify  $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}))$ .

$$\begin{aligned} & \text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m})) \\ &= \sum_{\mathbf{z}_s} q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m})} \\ &= \sum_{\mathbf{z}_s \in \{\mathbf{x}, \mathbf{m}\}} q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m})} \\ &= \underbrace{q(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m})}}_{\text{Simplify using (21) and (26)}} \\ &\quad + \underbrace{q(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m})}}_{\text{Simplify using (22) and (27)}} \\ &= \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \\ &\quad + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s) (\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t) (\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \end{aligned} \quad (30)$$

Thus,  $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$  can be written in the following manner where  $\langle \mathbf{z}_t, \mathbf{x} \rangle$  evaluates to 1 if  $\mathbf{z}_t = \mathbf{x}$  and  $\langle \mathbf{z}_t, \mathbf{m} \rangle$  evaluates to 1 if  $\mathbf{z}_t = \mathbf{m}$ :

$$\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$$

$$= \text{Cat}\left(\mathbf{z}_s; \frac{\alpha_t \mathbf{m} \odot \mathbf{x}_\theta(\mathbf{z}_t, t) + (\alpha_s - \alpha_t) \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m}}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}\right) \quad (25)$$

请注意，根据上述分类分布，我们可以获得  $p_\theta(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m})$  和  $p_\theta(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m})$  的值，如下所示：

$$p_\theta(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}) = \frac{(\alpha_s - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)} \quad (26)$$

$$p_\theta(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}) = \frac{\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s)}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)} \quad (27)$$

作为健全性检查，如果我们的去噪网络可以完美地重建  $\mathbf{x}$ ，即  $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{x}$ ，我们可以验证 (26) 减少到 (21)，并且 (27) 减少到 (22)。

结合(24)和(25)，我们得到逆向过程参数化的以下表达式：

$$p_\theta(\mathbf{z}_s | \mathbf{z}_t) = \begin{cases} \text{Cat}(\mathbf{z}_s; \mathbf{z}_t) & \mathbf{z}_t \neq \mathbf{m}, \\ \text{Cat}\left(\mathbf{z}_s; \frac{\alpha_t \mathbf{m} \odot \mathbf{x}_\theta(\mathbf{z}_t, t) + (\alpha_s - \alpha_t) \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \alpha_s) \mathbf{m}}{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}\right) & \mathbf{z}_t = \mathbf{m}. \end{cases} \quad (28)$$

### A.2.3 扩散损耗

对于给定的  $T$ ，令  $\mathcal{L}_T = \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$  表示扩散损失。我们将  $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$  的计算分解为 2 种情况： $\mathbf{z}_t = \mathbf{x}$  (情况 1) 和  $\mathbf{z}_t = \mathbf{m}$  (情况 2)。

情况 1：考虑情况  $\mathbf{z}_t = \mathbf{x}$ 。让我们简化  $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}))$ 。

$$\begin{aligned} & \text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x})) \\ &= \text{D}_{\text{KL}}(\mathbf{z}_t \| \mathbf{z}_t) \quad \text{From (23) and (24)} \\ &= 0 \end{aligned} \quad (29)$$

情况 2：考虑情况  $\mathbf{z}_t = \mathbf{m}$ 。让我们简化  $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}))$ 。

$$\begin{aligned} & \text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m})) \\ &= \sum_{\mathbf{z}_s} q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m})} \\ &= \sum_{\mathbf{z}_s \in \{\mathbf{x}, \mathbf{m}\}} q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m})} \\ &= \underbrace{q(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s = \mathbf{x} | \mathbf{z}_t = \mathbf{m})}}_{\text{Simplify using (21) and (26)}} \\ &\quad + \underbrace{q(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \log \frac{q(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m}, \mathbf{x})}{p_\theta(\mathbf{z}_s = \mathbf{m} | \mathbf{z}_t = \mathbf{m})}}_{\text{Simplify using (22) and (27)}} \\ &= \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \\ &\quad + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s) (\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t) (\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \end{aligned} \quad (30)$$

因此， $\text{D}_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$  可以写成以下方式，其中如果  $\mathbf{z}_t = \mathbf{x}$  则  $\langle \mathbf{z}_t, \mathbf{x} \rangle$  计算为 1，如果  $\mathbf{z}_t = \mathbf{m}$  则  $\langle \mathbf{z}_t, \mathbf{m} \rangle$  计算为 1：

$$\text{DKL}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t))$$

$$= \underbrace{D_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}))}_{=0, \text{ from (29)}} \langle \mathbf{z}_t, \mathbf{x} \rangle + \underbrace{D_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}))}_{\text{Given by (30)}} \langle \mathbf{z}_t, \mathbf{m} \rangle \quad (31)$$

Thus, we derive the diffusion loss,  $\mathcal{L}_T$ , in the following manner:

$$\begin{aligned} \mathcal{L}_T &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T D_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t)) \\ &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \left[ \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right. \\ &\quad \left. + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \quad (32) \end{aligned}$$

Note that  $\mathcal{L}_T$  is 0 if  $\mathbf{z}_t$  is an unmasked token i.e.  $\mathbf{z}_t = \mathbf{x}$ .

#### A.2.4 NELBO

Austin et al. [1], Sohl-Dickstein et al. [54] model  $\alpha_i$  as  $(\alpha_i)_{i \in \{1, \dots, T\}} = 1 - \frac{i}{T}$  given latents  $\mathbf{z}_1, \dots, \mathbf{z}_T$ . However, in this paper, we denote the latents as  $\mathbf{z}_{t(0)}, \dots, \mathbf{z}_{t(T)}$ ; and hence, the  $\alpha_{t(i)}$  are given as follows:

$$\begin{aligned} (\alpha_i)_{i \in \{1, \dots, T\}} &= 1 - \frac{i}{T} && \text{From Austin et al. [1], Sohl-Dickstein et al. [54].} \\ \implies (\alpha_i)_{k \in \{1, \dots, T+1\}} &= 1 - \frac{i}{T+1} && \text{For } T+1 \text{ latents} \\ \implies (\alpha_i)_{i \in \{0, \dots, T\}} &= 1 - \frac{i+1}{T+1} && \text{Offsetting the indices by 1.} \\ \implies (\alpha_{t(i)})_{i \in \{0, \dots, T\}} &= 1 - \frac{i+1}{T+1} && \text{Switching the notations from } \alpha_i \text{ to } \alpha_{t(i)}. \quad (33) \end{aligned}$$

Consequently, from Equation 33, we derive that

$$\alpha_{t(0)} = \frac{T}{T+1}, \quad (34)$$

$$\alpha_{t(T)} = 0. \quad (35)$$

Thus we have the following:

$$\mathbf{z}_{t(0)} \sim \text{Cat}(\cdot; \alpha_{t=0} \mathbf{x} + (1 - \alpha_{t=0}) \mathbf{m}) = \text{Cat}\left(\cdot; \frac{T}{T+1} \mathbf{x} + \frac{1}{T+1} \mathbf{m}\right), \quad (36)$$

$$q(\mathbf{z}_{t(T)} | \mathbf{x}) = \text{Cat}(\cdot; \alpha_{t=1} \mathbf{x} + (1 - \alpha_{t=1}) \mathbf{m}) = \text{Cat}(\cdot; \mathbf{m}), \quad (37)$$

$$p_\theta(\mathbf{z}_{t(T)}) = \text{Cat}(\cdot; \mathbf{m}) \quad (38)$$

The NELBO (2) simplifies to the following:

$$\begin{aligned} &\mathbb{E}_q \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + \underbrace{\mathcal{L}_T}_{\text{Compute using (32)}} \right] + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)} | \mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)})]}_{=0 \text{ using (37) and (38)}} \\ &= \mathbb{E}_{q,t} \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + T \left[ \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right. \right. \\ &\quad \left. \left. + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \right] \quad (39) \end{aligned}$$

## Appendix B MDLM

In this section, we show how SUBS parameterization can simplify the functional form of the NELBO as defined in (39).

$$= \underbrace{D_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{x}))}_{=0, \text{ from (29)}} \langle \mathbf{z}_t, \mathbf{x} \rangle + \underbrace{D_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t = \mathbf{m}))}_{\text{Given by (30)}} \langle \mathbf{z}_t, \mathbf{m} \rangle \quad (31)$$

因此，我们通过以下方式导出扩散损失  $\mathcal{L}_T$ ：

$$\begin{aligned} \mathcal{L}_T &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T D_{\text{KL}}(q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x}) \| p_\theta(\mathbf{z}_s | \mathbf{z}_t)) \\ &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \left[ \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right. \\ &\quad \left. + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \quad (32) \end{aligned}$$

请注意，如果  $\mathbf{z}_t$  是未屏蔽的标记，即  $\mathbf{z}_t = \mathbf{x}$ ，则  $\mathcal{L}_T$  为 0。

#### A.2.4 尼尔博

奥斯汀等人。[1], Sohl-Dickstein 等人。[54] 给定潜在变量  $\mathbf{z}_{1, \dots, T}$ ，将  $\alpha_i$  模型为  $(\alpha_i)_{i \in \{1, \dots, T\}} = 1 - \frac{i}{T}$ 。然而，在本文中，我们将潜在变量表示为  $\mathbf{z}_{t(0), \dots, t(T)}$ ；因此， $\alpha_{t(i)}$  给出如下：

$$\begin{aligned} (\alpha_i)_{i \in \{1, \dots, T\}} &= 1 - \frac{i}{T} && \text{From Austin et al. [1], Sohl-Dickstein et al. [54].} \\ \implies (\alpha_i)_{k \in \{1, \dots, T+1\}} &= 1 - \frac{i}{T+1} && \text{For } T+1 \text{ latents} \\ \implies (\alpha_i)_{i \in \{0, \dots, T\}} &= 1 - \frac{i+1}{T+1} && \text{Offsetting the indices by 1.} \\ \implies (\alpha_{t(i)})_{i \in \{0, \dots, T\}} &= 1 - \frac{i+1}{T+1} && \text{Switching the notations from } \alpha_i \text{ to } \alpha_{t(i)}. \quad (33) \end{aligned}$$

因此，从公式 33 中，我们得出

$$\alpha_{t(0)} = \frac{T}{T+1}, \quad (34)$$

$$\alpha_{t(T)} = 0. \quad (35)$$

因此我们有以下内容：

$$\mathbf{z}_{t(0)} \sim \text{Cat}(\cdot; \alpha_{t=0} \mathbf{x} + (1 - \alpha_{t=0}) \mathbf{m}) = \text{Cat}\left(\cdot; \frac{T}{T+1} \mathbf{x} + \frac{1}{T+1} \mathbf{m}\right), \quad (36)$$

$$q(\mathbf{z}_{t(T)} | \mathbf{x}) = \text{Cat}(\cdot; \alpha_{t=1} \mathbf{x} + (1 - \alpha_{t=1}) \mathbf{m}) = \text{Cat}(\cdot; \mathbf{m}), \quad (37)$$

$$p_\theta(\mathbf{z}_{t(T)}) = \text{Cat}(\cdot; \mathbf{m}) \quad (38)$$

NELBO (2) 简化为：

$$\begin{aligned} &\mathbb{E}_q \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + \underbrace{\mathcal{L}_T}_{\text{Compute using (32)}} \right] + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)} | \mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)})]}_{=0 \text{ using (37) and (38)}} \\ &= \mathbb{E}_{q,t} \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + T \left[ \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t)}{(1 - \alpha_t) \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right. \right. \\ &\quad \left. \left. + \frac{1 - \alpha_s}{1 - \alpha_t} \log \frac{(1 - \alpha_s)(\alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_t))}{(1 - \alpha_t)(\alpha_s \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle + (1 - \alpha_s))} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \right] \quad (39) \end{aligned}$$

## 附录 B MDLM

在本节中，我们将展示 SUBS 参数化如何简化 (39) 中定义 NELBO 函数形式。

## B.1 Rao-Blackwellization

We employ the RB techniques as described in Sec. 3.2.3 to simplify the NELBO (39) to (41) using RB2, and further to (43) using RB1.

### B.1.1 Zero Masking Probabilities

Using “Zero Masking Probabilities” (RB2) from Sec. 3.2.3, we set  $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle = 0$  in (32) to obtain the following simplified diffusion loss:

$$\begin{aligned}\mathcal{L}_T^{\text{RB2}} &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \left[ \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{1}{\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \\ &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \left[ \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \langle \mathbf{z}_t, \mathbf{m} \rangle.\end{aligned}\quad (40)$$

The corresponding Rao-Blackwellized NELBO is given as:

$$\begin{aligned}\mathbb{E}_q \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + \underbrace{\mathcal{L}_T^{\text{RB2}}}_{\text{Compute using (40)}} + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)} | \mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)})]}_{=0 \text{ using (37) and (38)}} \right] \\ = \mathbb{E}_{q,t} \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + T \left[ \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \right]\end{aligned}\quad (41)$$

### B.1.2 Carry Over Unmasking

Notice that the term  $\langle \mathbf{z}_t, \mathbf{m} \rangle$  in (40) is intended to reduce the diffusion loss to zero when  $\mathbf{z}_t = \mathbf{x}$ . Now, we will demonstrate that, by applying “Carry Over Unmasking” (RB1) from Sec. 3.2.3,  $\langle \mathbf{z}_t, \mathbf{m} \rangle$  can be removed from (40).

Recall that RB1 guarantees  $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{x}$  when  $\mathbf{z}_t = \mathbf{x}$ . Thus, with the RB1 parameterization, the diffusion loss in (40) becomes zero for  $\mathbf{z}_t = \mathbf{x}$ , as  $\log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle = 0$ . Consequently,  $\langle \mathbf{z}_t, \mathbf{m} \rangle$  can be safely omitted from (41), yielding the following diffusion loss:

$$\mathcal{L}_T^{\text{RB2} + \text{RB1}} = \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \left[ \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right]\quad (42)$$

### B.1.3 NELBO

Thus, we have the following NELBO:

$$\begin{aligned}\mathbb{E}_q \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + \underbrace{\mathcal{L}_T^{\text{RB2} + \text{RB1}}}_{\text{Compute using (42)}} + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)} | \mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)})]}_{=0 \text{ using (37) and (38)}} \right] \\ = \mathbb{E}_{q,t} \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + T \left[ \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \right]\end{aligned}\quad (43)$$

**Comparing (43) and (41).** Note that due to RB1,  $\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)})$  in (43) reduces to 0 every time  $\mathbf{z}_{t(0)} = \mathbf{x}$  as explained in (45). However, this is not the case in (41), even though it has a functionally similar expression to (43). Because of this reason (43) should lead to a better likelihood estimate and we empirically verify this in Table 8.

## B.2 Continuous Time

### B.2.1 Diffusion Loss

To derive the continuous-time diffusion loss,  $\mathcal{L}_{\text{diffusion}}^\infty$ , we consider the limiting case  $\lim_{T \rightarrow \infty} \mathcal{L}_T^{\text{RB2} + \text{RB1}}$  (42):

$$\mathcal{L}_{\text{diffusion}}^\infty = \lim_{T \rightarrow \infty} \mathcal{L}_T^{\text{RB2} + \text{RB1}}$$

## B.1 Rao-Blackwell 化

我们采用第 2 节中描述的 RB 技术。3.2.3 使用 RB2 将 NELBO(39) 简化为 (41)，并使用 RB1 进一步简化为 (43)。

### B.1.1 零掩蔽概率

使用第 2 节中的“零掩蔽概率” (RB2)。3.2.3，我们将 (32) 中的  $\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{m} \rangle$  设置为 0，得到如下简化的扩散损失：

$$\begin{aligned}\mathcal{L}_T^{\text{RB2}} &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \left[ \frac{\alpha_s - \alpha_t}{1 - \alpha_t} \log \frac{1}{\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle} \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \\ &= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \left[ \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \langle \mathbf{z}_t, \mathbf{m} \rangle.\end{aligned}\quad (40)$$

相应的 Rao-Blackwellized NELBO 给出为：

$$\begin{aligned}\mathbb{E}_q \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + \underbrace{\mathcal{L}_T^{\text{RB2}}}_{\text{Compute using (40)}} \right] &+ \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)} | \mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)})]}_{=0 \text{ using (37) and (38)}} \\ &= \mathbb{E}_{q,t} \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + T \left[ \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \langle \mathbf{z}_t, \mathbf{m} \rangle \right]\end{aligned}\quad (41)$$

### B.1.2 遗留揭露

请注意，(40) 中的项  $\langle \mathbf{z}_t, \mathbf{m} \rangle$  旨在将  $\mathbf{z}_t = \mathbf{x}$  时的扩散损耗降至零。现在，我们将通过应用第 2 节中的“结转揭露” (RB1) 来演示这一点。3.2.3、 $\langle \mathbf{z}_t, \mathbf{m} \rangle$  可以从 (40) 中去除。

回想一下，RB1 在  $\mathbf{z}_t = \mathbf{x}$  时保证  $\mathbf{x}_\theta(\mathbf{z}_t, t) = \mathbf{x}$ 。因此，通过 RB1 参数化，(40) 中的扩散损失对于  $\mathbf{z}_t = \mathbf{x}$  变为零，即  $\log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle = 0$ 。因此，可以安全地从 (41) 中省略  $\langle \mathbf{z}_t, \mathbf{m} \rangle$ ，产生以下扩散损失：

$$\mathcal{L}_T^{\text{RB2} + \text{RB1}} = \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}} \mathbb{E}_{q(\mathbf{z}_t | \mathbf{x})} T \left[ \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \quad (42)$$

### B.1.3 尼尔博

因此，我们有以下 NELBO：

$$\begin{aligned}\mathbb{E}_q \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + \underbrace{\mathcal{L}_T^{\text{RB2} + \text{RB1}}}_{\text{Compute using (42)}} \right] &+ \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)} | \mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)})]}_{=0 \text{ using (37) and (38)}} \\ &= \mathbb{E}_{q,t} \left[ -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) + T \left[ \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \right]\end{aligned}\quad (43)$$

比较 (43) 和 (41)。请注意，由于 RB1，每次  $\mathbf{z}_{t(0)} = \mathbf{x}$  时，(43) 中的  $\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)})$  都会减少到 0，如 (45) 中所述。然而，(41) 中的情况并非如此，尽管它具有与 (43) 功能相似的表达式。由于这个原因 (43) 应该会导致更好的似然估计，我们在表 8 中凭经验验证了这一点。

## B.2 连续时间

### B.2.1 扩散损耗

为了导出连续时间扩散损失  $\mathcal{L}_{\text{diffusion}}^\infty$ ，我们考虑极限情况  $\lim_{T \rightarrow \infty} \mathcal{L}_T^{\text{RB2} + \text{RB1}}$  (42)：

$$\mathcal{L}_{\text{diffusion}}^\infty = \lim_{T \rightarrow \infty} \mathcal{L}_T^{\text{RB2} + \text{RB1}}$$

$$\begin{aligned}
&= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}, q(\mathbf{z}_t | \mathbf{x})} \left[ \lim_{T \rightarrow \infty} T \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \\
&= \mathbb{E}_{t \sim \mathcal{U}[0,1], q(\mathbf{z}_t | \mathbf{x})} \left[ \frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \quad \text{Using } \lim_{T \rightarrow \infty} T(\alpha_t - \alpha_s) = \alpha'_t
\end{aligned} \tag{44}$$

### B.2.2 Reconstruction Loss

For the continuous time case, from (36), we have

$$\begin{aligned}
\mathbf{z}_{t(0)} &\sim \lim_{T \rightarrow \infty} \text{Cat} \left( \cdot; \frac{T}{T+1} \mathbf{x} + \frac{1}{T+1} \mathbf{m} \right) \\
&\implies \mathbf{z}_{t(0)} \sim \text{Cat}(\cdot; \mathbf{x}) \\
&\implies \mathbf{z}_{t(0)} = \mathbf{x}
\end{aligned} \tag{45}$$

Thus, the reconstruction loss reduces to 0 in the following manner:

$$\begin{aligned}
\mathcal{L}_{\text{recons}} &= -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) \\
&= -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)} = \mathbf{x}) \quad \text{From (45)} \\
&= -\log \langle \mathbf{x}_\theta(\mathbf{x}, t(0)), \mathbf{x} \rangle \\
&= -\log \langle \mathbf{x}, \mathbf{x} \rangle \quad \text{Due to "carry-over unmasking" } \mathbf{x}_\theta(\mathbf{x}, t(0)) = \mathbf{x} \\
&= 0.
\end{aligned} \tag{46}$$

### B.2.3 NELBO

Thus, we have the following NELBO:

$$\begin{aligned}
&\mathbb{E}_q \left[ \underbrace{-\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)})}_{=0 \text{ from (46)}} + \underbrace{\mathcal{L}_{\text{diffusion}}^\infty}_{\text{Compute using (42)}} + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)} | \mathbf{x}) \| p_\theta(\mathbf{z}_{t(T)}, t)]}_{=0 \text{ using (37) and (38)}} \right] \\
&= \mathbb{E}_{q,t} \left[ \frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right]
\end{aligned} \tag{47}$$

### B.3 Final Algorithm

In Algorithm 1, we present the training algorithm for MDLM.

---

#### Algorithm 1 Training MDLM

---

- 1: **repeat**
- 2:    $\mathbf{x}^{1:L} \sim q(\mathbf{x})$  ▷ Sample a sentence.
- 3:    $t \sim \mathcal{U}[0,1]$  ▷ Sample a time step.
- 4:    $\mathbf{z}_t^\ell \sim \text{Cat}(\mathbf{z}_t^\ell; \alpha_t \mathbf{x}^\ell + (1 - \alpha_t) \mathbf{m}) \forall 1 \leq \ell \leq L$  ▷ Mask Each token  $\mathbf{x}^\ell$  independently to obtain the latent  $\mathbf{z}_t^{1:L}$ .
- 5:   Take gradient descent step on

$$\nabla_\theta \frac{\alpha'_t}{1 - \alpha_t} \sum_\ell \log \langle \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \mathbf{x}^\ell \rangle$$

- 6: **until** converged
- 

## Appendix C Concrete Score Matching

In the previous section, we defined the discrete diffusion process as a Discrete-Time Markov Chain (DTMC) with a finite set of  $T$  states,  $\mathbf{z}_{\{0, \frac{1}{T}, \dots, 1\}}$ , and a state transition matrix  $Q_t$ . To derive the continuous-time ELBO, we simply take the limit as  $T \rightarrow \infty$ .

$$\begin{aligned}
&= \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}, q(\mathbf{z}_t | \mathbf{x})} \left[ \lim_{T \rightarrow \infty} T \frac{\alpha_t - \alpha_s}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \\
&= \mathbb{E}_{t \sim \mathcal{U}[0,1], q(\mathbf{z}_t | \mathbf{x})} \left[ \frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \quad \text{使用 } \lim_{T \rightarrow \infty} T(\alpha_t - \alpha_s) = \alpha'_t
\end{aligned} \tag{44}$$

### B.2.2 重建损失

对于连续时间的情况，根据 (36)，我们有

$$\begin{aligned}
\mathbf{z}_{t(0)} &\sim \lim_{T \rightarrow \infty} \text{Cat} \left( \cdot; \frac{T}{T+1} \mathbf{x} + \frac{1}{T+1} \mathbf{m} \right) \\
&\implies \mathbf{z}_{t(0)} \sim \text{Cat}(\cdot; \mathbf{x}) \\
&\implies \mathbf{z}_{t(0)} = \mathbf{x}
\end{aligned} \tag{45}$$

因此，重建损失通过以下方式减少到 0：

$$\begin{aligned}
\mathcal{L}_{\text{recons}} &= -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)}) \\
&= -\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)} = \mathbf{x}) \quad \text{From (45)} \\
&= -\log \langle \mathbf{x}_\theta(\mathbf{x}, t(0)), \mathbf{x} \rangle \\
&= -\log \langle \mathbf{x}, \mathbf{x} \rangle \quad \text{Due to "carry-over unmasking" } \mathbf{x}_\theta(\mathbf{x}, t(0)) = \mathbf{x} \\
&= 0.
\end{aligned} \tag{46}$$

### B.2.3 尼尔博

因此，我们有以下 NELBO：

$$\begin{aligned}
&\mathbb{E}_q \left[ \underbrace{-\log p_\theta(\mathbf{x} | \mathbf{z}_{t(0)})}_{=0 \text{ from (46)}} + \underbrace{\mathcal{L}_{\text{diffusion}}^\infty}_{\text{Compute using (42)}} + \underbrace{D_{\text{KL}}[q(\mathbf{z}_{t(T)} | \mathbf{x}) || p_\theta(\mathbf{z}_{t(T)}, t)]}_{=0 \text{ using (37) and (38)}} \right] \\
&= \mathbb{E}_{q,t} \left[ \frac{\alpha'_t}{1 - \alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right]
\end{aligned} \tag{47}$$

### B.3 最终算法

在算法 1 中，我们提出了 MDLM 的训练算法。

---

#### 算法1 训练MDLM

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1: 重复 2:  $\mathbf{x}^{1:L} \sim q(\mathbf{x})$  举一个句子。 3:  $t \sim \mathcal{U}[0,1]$  对时间步进行采样。 4:  $\mathbf{z}_t^\ell \sim \text{Cat}(\mathbf{z}_t^\ell; \alpha_t \mathbf{x}^\ell + (1 - \alpha_t) \mathbf{m}) \forall 1 \leq \ell \leq L$  独立屏蔽每个标记  $\mathbf{x}^\ell$  以获得潜在的  $\mathbf{z}_t^{1:L}$ 。 5: 采取梯度下降步骤

$$\nabla_\theta \frac{\alpha'_t}{1 - \alpha_t} \sum_\ell \log \langle \mathbf{x}_\theta^\ell(\mathbf{z}_t^{1:L}, t), \mathbf{x}^\ell \rangle$$

6: 直到收敛

---

## 附录C 具体分数匹配

在上一节中，我们将离散扩散过程定义为具有有限集合  $T$  状态、 $\mathbf{z}_{\{0, \frac{1}{T}, \dots, 1\}}$  和状态转移矩阵  $Q_t$  的离散时间马尔可夫链 (DTMC)。为了导出连续时间 ELBO，我们只需将极限取为  $T \rightarrow \infty$ 。

In contrast, Campbell et al. [5] and Lou et al. [33] defined the discrete diffusion process as a Continuous-Time Markov Chain (CTMC), where the forward corruption process is specified by the rate change matrix  $R_t \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ , which can be thought of as the instantaneous rate at which one state transitions to another. With this formulation, the forward posterior  $q(\mathbf{z}_t | \mathbf{z}_s)$  and the true reverse posterior  $q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x})$  can be expressed in terms of the rate change matrix as follows:

$$q(\mathbf{z}_t = \mathbf{y}' | \mathbf{z}_s = \mathbf{y}) = \delta_{\mathbf{y}', \mathbf{y}} + R_t(\mathbf{y}', \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \quad (48)$$

$$q(\mathbf{z}_s = \mathbf{y}' | \mathbf{z}_t = \mathbf{y}, \mathbf{x}) = \delta_{\mathbf{y}', \mathbf{y}} + \tilde{R}_t(\mathbf{y}', \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \quad (49)$$

where  $\delta$  is the Kronecker delta and  $\mathcal{O}(\frac{1}{T^2})$  represents higher order terms of  $\frac{1}{T^2}$ . In Sec. C.1, we show how to express the rate matrix  $R_t$  in terms of the state transition matrix  $Q_t$ . Lou et al. [33] propose a continuous time ELBO for this process. They mention that this expression can be derived from Benton et al. [4], though they do not provide an explicit derivation. For this reason, we present a rigorous derivation in Sec. C.2 and further demonstrate that, under the SUBS parameterization in Sec. 3.2.3, this formula reduces to our proposed continuous-time ELBO, given by (10).

For the remainder of this section, we switch to the notation  $q_{t|s}(\mathbf{y}' | \mathbf{y})$  to denote  $q(\mathbf{z}_t = \mathbf{y}' | \mathbf{z}_s = \mathbf{y})$ ,  $q_{s|t}(\mathbf{y}' | \mathbf{y})$  for  $q(\mathbf{z}_s = \mathbf{y}' | \mathbf{z}_t = \mathbf{y})$ , and  $q(\mathbf{z}_t = \mathbf{y} | \mathbf{x})$  for  $q_t(\mathbf{y} | \mathbf{x})$ , aligning with the notation typically used in the CTMC literature.

### C.1 Extracting the Rate Matrix

Here, we aim to express the rate change matrix  $R_t$  in terms of the state transition matrix  $Q_t$ . To do this, we first represent the forward transition  $q_{t|s}$  in terms of  $Q_t$  and  $R_t$  separately, allowing us to illustrate their relationship.

Using (13), we can write  $q_{t|s}$  as follows:

$$\begin{aligned} q_{t|s}(\mathbf{y}' | \mathbf{y}) &= [\mathbf{y}']^\top [\alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1} \mathbf{m}^\top]^\top \mathbf{y} \\ &= [\mathbf{y}']^\top [\alpha_{t|s} \mathbf{y} + (1 - \alpha_{t|s}) \mathbf{m} \mathbf{1}^\top \mathbf{y}] \\ &= [\mathbf{y}']^\top [\alpha_{t|s} \mathbf{y} + (1 - \alpha_{t|s}) \mathbf{m}] \\ &= \alpha_{t|s} [\mathbf{y}']^\top \mathbf{y} + (1 - \alpha_{t|s}) [\mathbf{y}']^\top \mathbf{m} \end{aligned} \quad (50)$$

Now let's analyze all possible combinations for the tuple  $(\mathbf{y}', \mathbf{y})$ :

**1. Case  $(\mathbf{y}' = \mathbf{x}, \mathbf{y} = \mathbf{x})$ :** Using (50), we find that  $q_{t|s}(\mathbf{x} | \mathbf{x}) = \alpha_{t|s}$  for the DTMC. By (48), we have  $q_{t|s}(\mathbf{x} | \mathbf{x}) = 1 + R_t(\mathbf{x}, \mathbf{x}) \frac{1}{T}$  as  $T \rightarrow \infty$ , since the higher-order terms  $\mathcal{O}(\frac{1}{T^2})$  vanish in the limit. Thus, we get:

$$\begin{aligned} \lim_{T \rightarrow \infty} \left[ 1 + R_t(\mathbf{x}, \mathbf{x}) \frac{1}{T} \right] &= \lim_{T \rightarrow \infty} \alpha_{t|s} \\ \implies R_t(\mathbf{x}, \mathbf{x}) &= \lim_{T \rightarrow \infty} T(\alpha_{t|s} - 1) = \frac{\alpha'_t}{\alpha_t} \end{aligned} \quad (51)$$

**2. Case  $(\mathbf{y}' = \mathbf{m}, \mathbf{y} \in \mathcal{V} - \{\mathbf{m}\})$ :** Similarly, using (50) and (48), we have  $q_{t|s}(\mathbf{m} | \mathbf{y} \neq \mathbf{m}) = 1 - \alpha_{t|s}$  and  $q_{t|s}(\mathbf{x} | \mathbf{y} \neq \mathbf{m}) = R_t(\mathbf{m}, \mathbf{y} \neq \mathbf{m}) \frac{1}{T}$ . Thus,

$$\begin{aligned} \lim_{T \rightarrow \infty} \left[ R_t(\mathbf{m}, \mathbf{y} \neq \mathbf{m}) \frac{1}{T} \right] &= \lim_{T \rightarrow \infty} [1 - \alpha_{t|s}] \\ \implies R_t(\mathbf{m}, \mathbf{y} \neq \mathbf{m}) &= \lim_{T \rightarrow \infty} T(1 - \alpha_{t|s}) = -\frac{\alpha'_t}{\alpha_t} \end{aligned} \quad (52)$$

**3. Case  $(\mathbf{y}' = \mathbf{m}, \mathbf{y} = \mathbf{m})$ :** Using (50) and (48), we find  $q_{t|s}(\mathbf{m} | \mathbf{m}) = 1$  and  $q_{t|s}(\mathbf{m} | \mathbf{m}) = 1 + R_t(\mathbf{m}, \mathbf{m}) \frac{1}{T} + \mathcal{O}(\frac{1}{T^2})$ . Since these two expressions must be equal for any  $T$ , it follows that

$$R_t(\mathbf{m}, \mathbf{m}) = 0. \quad (53)$$

Note that when  $R_t(\mathbf{m}, \mathbf{m})$  is constant, the term  $\mathcal{O}(\frac{1}{T^2})$  reduces to zero, as it includes higher-order time derivatives of  $R_t$ .

相比之下，坎贝尔等人。[5] 和卢等人。[33]将离散扩散过程定义为连续时间马尔可夫链（CTMC），其中前向腐败过程由速率变化矩阵  $R_t \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$  指定，可以将其视为一种状态转换到另一种状态的瞬时速率。通过这个公式，前向后验  $q(\mathbf{z}_t | \mathbf{z}_s)$  和真实后验  $q(\mathbf{z}_s | \mathbf{z}_t, \mathbf{x})$  可以用速率变化矩阵表示如下：

$$q(\mathbf{z}_t = \mathbf{y}' | \mathbf{z}_s = \mathbf{y}) = \delta_{\mathbf{y}', \mathbf{y}} + R_t(\mathbf{y}', \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \quad (48)$$

$$q(\mathbf{z}_s = \mathbf{y}' | \mathbf{z}_t = \mathbf{y}, \mathbf{x}) = \delta_{\mathbf{y}', \mathbf{y}} + \tilde{R}_t(\mathbf{y}', \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \quad (49)$$

其中  $\delta$  是克罗内克增量， $\mathcal{O}(\frac{1}{T^2})$  表示  $\frac{1}{T^2}$  的高阶项。在秒。C.1，我们展示了如何用状态转移矩阵  $Q_t$  来表达速率矩阵  $R_t$ 。卢等人。[33] 为该过程提出了连续时间 ELBO。他们提到这个表达可以来自 Benton 等人。[4]，尽管他们没有提供明确的推导。为此，我们在第 2 节中提出了严格的推导。C.2 并进一步证明，在第 2 节中的 SUBS 参数化下。3.2.3，这个公式简化为我们提出的连续时间 ELBO，由 (10) 给出。

在本节的其余部分中，我们切换到符号  $q_{t|s}(\mathbf{y}' | \mathbf{y})$  来表示  $q(\mathbf{z}_t = \mathbf{y}' | \mathbf{z}_s = \mathbf{y})$ ，用  $q_{s|t}(\mathbf{y}' | \mathbf{y})$  来表示  $q(\mathbf{z}_s = \mathbf{y}' | \mathbf{z}_t = \mathbf{y})$ ，用  $q(\mathbf{z}_t = \mathbf{y} | \mathbf{x})$  来表示  $q_t(\mathbf{y} | \mathbf{x})$ ，与 CTMC 文献中通常使用的符号保持一致。

### C.1 提取速率矩阵

在这里，我们的目标是用状态转移矩阵  $Q_t$  来表达速率变化矩阵  $R_t$ 。为此，我们首先分别用  $Q_t$  和  $R_t$  表示前向转换  $q_{t|s}$ ，以便我们说明它们的关系。

利用(13)，我们可以将  $q_{t|s}$  写成如下：

$$\begin{aligned} q_{t|s}(\mathbf{y}' | \mathbf{y}) &= [\mathbf{y}']^\top [\alpha_{t|s} \mathbf{I} + (1 - \alpha_{t|s}) \mathbf{1} \mathbf{m}^\top]^\top \mathbf{y} \\ &= [\mathbf{y}']^\top [\alpha_{t|s} \mathbf{y} + (1 - \alpha_{t|s}) \mathbf{m} \mathbf{1}^\top \mathbf{y}] \\ &= [\mathbf{y}']^\top [\alpha_{t|s} \mathbf{y} + (1 - \alpha_{t|s}) \mathbf{m}] \\ &= \alpha_{t|s} [\mathbf{y}']^\top \mathbf{y} + (1 - \alpha_{t|s}) [\mathbf{y}']^\top \mathbf{m} \end{aligned} \quad (50)$$

现在让我们分析元组  $(\mathbf{y}', \mathbf{y})$  的所有可能组合：

1. 情况  $(\mathbf{y}' = \mathbf{x}, \mathbf{y} = \mathbf{x})$ ：使用(50)，我们发现DTMC为  $q_{t|s}(\mathbf{x} | \mathbf{x}) = \alpha_{t|s}$ 。根据 (48)，我们将  $q_{t|s}(\mathbf{x} | \mathbf{x}) = 1 + R_t(\mathbf{x}, \mathbf{x}) \frac{1}{T}$  视为  $T \rightarrow \infty$ ，因为高阶项  $\mathcal{O}(\frac{1}{T^2})$  在极限中消失。因此，我们得到：

$$\begin{aligned} \lim_{T \rightarrow \infty} \left[ 1 + R_t(\mathbf{x}, \mathbf{x}) \frac{1}{T} \right] &= \lim_{T \rightarrow \infty} \alpha_{t|s} \\ \implies R_t(\mathbf{x}, \mathbf{x}) &= \lim_{T \rightarrow \infty} T(\alpha_{t|s} - 1) = \frac{\alpha'_t}{\alpha_t} \end{aligned} \quad (51)$$

2. 情况  $(\mathbf{y}' = \mathbf{m}, \mathbf{y} \in \mathcal{V} - \{\mathbf{m}\})$ ：类似地，使用 (50) 和 (48)，我们有  $q_{t|s}(\mathbf{m} | \mathbf{y} \neq \mathbf{m}) = 1 - \alpha_{t|s}$  和  $q_{t|s}(\mathbf{x} | \mathbf{y} \neq \mathbf{m}) = R_t(\mathbf{m}, \mathbf{y} \neq \mathbf{m}) \frac{1}{T}$ 。因此，

$$\begin{aligned} \lim_{T \rightarrow \infty} \left[ R_t(\mathbf{m}, \mathbf{y} \neq \mathbf{m}) \frac{1}{T} \right] &= \lim_{T \rightarrow \infty} [1 - \alpha_{t|s}] \\ \implies R_t(\mathbf{m}, \mathbf{y} \neq \mathbf{m}) &= \lim_{T \rightarrow \infty} T(1 - \alpha_{t|s}) = -\frac{\alpha'_t}{\alpha_t} \end{aligned} \quad (52)$$

3. 情况  $(\mathbf{y}' = \mathbf{m}, \mathbf{y} = \mathbf{m})$ ：使用(50)和(48)，我们找到  $q_{t|s}(\mathbf{m} | \mathbf{m}) = 1$  和  $q_{t|s}(\mathbf{m} | \mathbf{m}) = 1 + R_t(\mathbf{m}, \mathbf{m}) \frac{1}{T} + \mathcal{O}(\frac{1}{T^2})$ 。由于这两个表达式对于任何  $T$  都必须相等，因此

$$R_t(\mathbf{m}, \mathbf{m}) = 0. \quad (53)$$

请注意，当  $R_t(\mathbf{m}, \mathbf{m})$  恒定时，项  $\mathcal{O}(\frac{1}{T^2})$  减少为零，因为它包含  $R_t$  的高阶时间导数。

**4. Case ( $\mathbf{y}' = \mathbf{x}, \mathbf{y} \in \mathcal{V} - \{\mathbf{x}\}$ ):** In the context of absorbing state diffusion, these states are never observed. Thus,

$$R_t(\mathbf{y}' = \mathbf{x}, \mathbf{y} \in \mathcal{V} - \{\mathbf{m}, \mathbf{x}\}) = 0 \quad (54)$$

**5. Case ( $\mathbf{y}' \in \mathcal{V} - \{\mathbf{m}, \mathbf{x}\}, \mathbf{y} \in \{\mathbf{m}, \mathbf{x}\}$ ):** In the context of absorbing state diffusion, these states are never observed. Thus,

$$R_t(\mathbf{y}' \in \mathcal{V} - \{\mathbf{m}, \mathbf{x}\}, \mathbf{y} \in \{\mathbf{m}, \mathbf{x}\}) = 0 \quad (55)$$

Finally, we can express the forward rate matrix as:

$$R_t = \frac{\alpha'_t}{\alpha_t} (\mathbf{I} - \mathbf{m} \mathbf{1}^\top) \quad (56)$$

It can be seen that the columns of this matrix sum to zero, i.e.,

$$\sum_{\mathbf{y}' \in \mathcal{V}} R_t(\mathbf{y}', \mathbf{y}) = 0 \implies R_t(\mathbf{y}, \mathbf{y}) = \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}', \mathbf{y}), \quad (57)$$

which ensures that the probability mass is preserved in the forward diffusion process. Similarly, the reverse rate matrix  $\tilde{R}_t$  can be written in terms of the forward rate matrix  $R_t$  as follows [33]:

$$\tilde{R}_t(\mathbf{y}', \mathbf{y}) = \begin{cases} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') & \mathbf{y}' \neq \mathbf{y} \\ -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \frac{q_t(\tilde{\mathbf{y}}|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \tilde{\mathbf{y}}) & \mathbf{y}' = \mathbf{y}. \end{cases} \quad (58)$$

## C.2 NELBO

Meng et al. [37] introduced the term “concrete score” for the term  $q_t(\mathbf{y}'|\mathbf{x})/q_t(\mathbf{y}|\mathbf{x})$  that appears in  $\tilde{R}_t$ . Since this quantity is not directly accessible in the reverse diffusion process, we approximate it using a neural network,  $\mathbf{s}_\theta: \mathcal{V} \rightarrow \mathcal{V}$ , with parameters  $\theta$ . The approximate reverse posterior  $p_{s|t}$  can then be expressed in terms of the approximate reverse rate matrix  $\tilde{R}_t$  in the following manner:

$$p_{s|t}(\mathbf{y}'|\mathbf{y}) = \delta_{\mathbf{y}', \mathbf{y}} + \tilde{R}_t^\theta(\mathbf{y}', \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \quad (59)$$

$$\tilde{R}_t^\theta(\mathbf{y}', \mathbf{y}) = \begin{cases} \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') & \mathbf{y}' \neq \mathbf{y} \\ -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \mathbf{s}_\theta(\mathbf{y})_{\tilde{\mathbf{y}}} R_t(\mathbf{y}, \tilde{\mathbf{y}}) & \mathbf{y}' = \mathbf{y} \end{cases} \quad (60)$$

where  $\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'}$  denotes the approximate concrete score  $q_t(\mathbf{y}'|\mathbf{x})/q_t(\mathbf{y}|\mathbf{x})$ . Lou et al. [33] propose the following NELBO to train such a model:

$$\mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \right) \right) \right] \quad (61)$$

where  $K(a) = a \log a - a$ . They mention that this expression can be derived from Benton et al. [4], though they do not provide an explicit derivation. For this reason, we present a rigorous derivation in the following section.

**Proof.** Let’s focus on the diffusion loss for this process. As mentioned in the previous section, the reconstruction and prior loss terms reduce to zero. The continuous-time diffusion loss is given by:

$$\begin{aligned} & \lim_{T \rightarrow \infty} T \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}, \mathbf{y} \sim q_t(\cdot|\mathbf{x})} [\text{D}_{\text{KL}}(q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \| p_{s|t}(\mathbf{y}'|\mathbf{y}))] \\ &= \lim_{T \rightarrow \infty} T \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}, \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \sum_{\mathbf{y}'} q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})} \right] \\ &= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \lim_{T \rightarrow \infty} T \sum_{\mathbf{y}'} q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})} \right] \end{aligned}$$

4. 情况 ( $\mathbf{y}' = \mathbf{x}, \mathbf{y} \in \mathcal{V} - \{\mathbf{x}\}$ ): 在吸收态扩散的情况下, 这些态永远不会被观察到。因此,

$$R_t(\mathbf{y}' = \mathbf{x}, \mathbf{y} \in \mathcal{V} - \{\mathbf{x}\}) = 0 \quad (54)$$

5. 情况 ( $\mathbf{y}' \in \mathcal{V} - \{\mathbf{m}, \mathbf{x}\}, \mathbf{y} \in \{\mathbf{m}, \mathbf{x}\}$ ): 在吸收态扩散的情况下, 这些态永远不会被观察到。因此,

$$R_t(\mathbf{y}' \in \mathcal{V} - \{\mathbf{m}, \mathbf{x}\}, \mathbf{y} \in \{\mathbf{m}, \mathbf{x}\}) = 0 \quad (55)$$

最后, 我们可以将远期速率矩阵表示为:

$$R_t = \frac{\alpha'_t}{\alpha_t} (\mathbf{I} - \mathbf{m} \mathbf{1}^\top) \quad (56)$$

可以看出, 该矩阵的列之和为零, 即

$$\sum_{\mathbf{y}' \in \mathcal{V}} R_t(\mathbf{y}', \mathbf{y}) = 0 \implies R_t(\mathbf{y}, \mathbf{y}) = \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}', \mathbf{y}), \quad (57)$$

这确保了在前向扩散过程中保留概率质量。类似地, 反向速率矩阵  $\tilde{R}_t$  可以用前向速率矩阵  $R_t$  写成如下[33]:

$$\tilde{R}_t(\mathbf{y}', \mathbf{y}) = \begin{cases} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') & \mathbf{y}' \neq \mathbf{y} \\ -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \frac{q_t(\tilde{\mathbf{y}}|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \tilde{\mathbf{y}}) & \mathbf{y}' = \mathbf{y}. \end{cases} \quad (58)$$

## C.2 内尔博

孟等人。[37]为出现在  $\tilde{R}_t$  中的术语  $q_t(\mathbf{y}'|\mathbf{x})/q_t(\mathbf{y}|\mathbf{x})$  引入了术语“具体分数”。由于该量在反向扩散过程中无法直接获取, 因此我们使用带有参数  $\theta$  的神经网络  $\mathbf{s}_\theta: \mathcal{V} \rightarrow \mathcal{V}$  对其进行近似。然后, 近似逆后验概率  $p_{s|t}$  可以用近似逆概率矩阵  $\tilde{R}_t$  来表示, 如下所示:

$$p_{s|t}(\mathbf{y}'|\mathbf{y}) = \delta_{\mathbf{y}', \mathbf{y}} + \tilde{R}_t^\theta(\mathbf{y}', \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \quad (59)$$

$$\tilde{R}_t^\theta(\mathbf{y}', \mathbf{y}) = \begin{cases} \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') & \mathbf{y}' \neq \mathbf{y} \\ -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \mathbf{s}_\theta(\mathbf{y})_{\tilde{\mathbf{y}}} R_t(\mathbf{y}, \tilde{\mathbf{y}}) & \mathbf{y}' = \mathbf{y} \end{cases} \quad (60)$$

其中  $\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'}$  表示近似具体分数  $q_t(\mathbf{y}'|\mathbf{x})/q_t(\mathbf{y}|\mathbf{x})$ 。卢等人。[33]提出以下 NELBO 来训练这样的模型:

$$\mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} + K\left(\frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})}\right) \right) \right] \quad (61)$$

其中  $K(a) = a \log a - a$ 。他们提到这个表达可以来自 Benton 等人。[4], 尽管他们没有提供明确的推导。为此, 我们在下一节中给出了严格的推导。

证明。让我们关注这个过程的扩散损失。正如上一节中提到的, 重建和先验损失项减少到零。连续时间扩散损失由下式给出:

$$\begin{aligned} & \lim_{T \rightarrow \infty} T \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}, \mathbf{y} \sim q_t(\cdot|\mathbf{x})} [\text{D}_{\text{KL}}(q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \| p_{s|t}(\mathbf{y}'|\mathbf{y}))] \\ &= \lim_{T \rightarrow \infty} T \mathbb{E}_{t \in \{\frac{1}{T}, \frac{2}{T}, \dots, 1\}, \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \sum_{\mathbf{y}'} q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})} \right] \\ &= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \lim_{T \rightarrow \infty} T \sum_{\mathbf{y}'} q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})} \right] \end{aligned}$$

$$= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \underbrace{\lim_{T \rightarrow \infty} T q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}}_{\text{Term 1}} + \underbrace{\lim_{T \rightarrow \infty} T \sum_{\mathbf{y}' \neq \mathbf{y}} q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}}_{\text{Term 2}} \right] \quad (62)$$

Let's simplify these two terms separately. For the derivation, we'll rely on two key observations: In the limiting case as  $T \rightarrow \infty$ , it follows from (49) that  $\lim_{T \rightarrow \infty} q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) = 1$  and from (59) that  $\lim_{T \rightarrow \infty} p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) = 1$ .

**Term 1:**

$$\begin{aligned} & \lim_{T \rightarrow \infty} T q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})} \\ & \because \lim_{T \rightarrow \infty} q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) = 1; \text{ hence,} \\ & = \lim_{T \rightarrow \infty} T \log \frac{q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})} \end{aligned}$$

The above term is in  $\infty \times 0$  indeterminate form; therefore,

$$= \lim_{T \rightarrow \infty} T (\log q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) - \log p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}))$$

Substituting  $q_{s|t}$  and  $p_{s|t}$  from (49) and (59), we get:

$$= \lim_{T \rightarrow \infty} T \left[ \log \left( 1 + \tilde{R}_t(\mathbf{y}, \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \right) - \log \left( 1 + \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \right) \right]$$

Applying the Taylor series expansion for  $\log(1+x)$ , we get:

$$\begin{aligned} & = \lim_{T \rightarrow \infty} T \left[ \tilde{R}_t(\mathbf{y}, \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) - \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y}) \frac{1}{T} - \mathcal{O}\left(\frac{1}{T^2}\right) \right] \\ & = \tilde{R}_t(\mathbf{y}, \mathbf{y}) + \lim_{T \rightarrow \infty} T \mathcal{O}\left(\frac{1}{T^2}\right) - \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y}) - \lim_{T \rightarrow \infty} T \mathcal{O}\left(\frac{1}{T^2}\right) \end{aligned}$$

$$\because \lim_{T \rightarrow \infty} T \mathcal{O}\left(\frac{1}{T^2}\right) = 0, \text{ we get:}$$

$$= \tilde{R}_t(\mathbf{y}, \mathbf{y}) - \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y})$$

Using (58) and (60), we get:

$$= - \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} \right) \quad (63)$$

**Term 2:**

$$\lim_{T \rightarrow \infty} T \sum_{\mathbf{y}' \neq \mathbf{y}} q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}$$

Substituting  $q_{s|t}$  and  $p_{s|t}$  from (49) and (59), we get:

$$\begin{aligned} & = \lim_{T \rightarrow \infty} T \sum_{\mathbf{y}' \neq \mathbf{y}} \left[ \left[ \frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \right] \log \frac{\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right)}{\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right)} \right] \\ & = \lim_{T \rightarrow \infty} \sum_{\mathbf{y}' \neq \mathbf{y}} \left[ \left[ \frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') + T \mathcal{O}\left(\frac{1}{T^2}\right) \right] \log \frac{\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') + T \mathcal{O}\left(\frac{1}{T^2}\right)}{\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') + T \mathcal{O}\left(\frac{1}{T^2}\right)} \right] \\ & \because \lim_{T \rightarrow \infty} T \mathcal{O}\left(\frac{1}{T^2}\right) = 0, \text{ we get:} \end{aligned}$$

$$= \sum_{\mathbf{y}' \neq \mathbf{y}} \frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \log \frac{\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}')}{\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}')}$$

$$= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \underbrace{\lim_{T \rightarrow \infty} T q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}}_{\text{Term 1}} + \underbrace{\lim_{T \rightarrow \infty} T \sum_{\mathbf{y}' \neq \mathbf{y}} q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}}_{\text{Term 2}} \right] \quad (62)$$

让我们分别简化这两个术语。对于推导，我们将依赖两个关键的观察结果：在  $T \rightarrow \infty$  的极限情况下，从 (49) 得出  $\lim_{T \rightarrow \infty} q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) = 1$ ，从 (59) 得出  $\lim_{T \rightarrow \infty} p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) = 1$ 。

第 1 期：

$$\begin{aligned} & \lim_{T \rightarrow \infty} T q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})} \\ & \because \lim_{T \rightarrow \infty} q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) = 1; \text{ hence,} \\ & = \lim_{T \rightarrow \infty} T \log \frac{q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x})} \end{aligned}$$

The above term is in  $\infty \times 0$  indeterminate form; therefore,

$$= \lim_{T \rightarrow \infty} T (\log q_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}) - \log p_{s|t}(\mathbf{y}|\mathbf{y}, \mathbf{x}))$$

Substituting  $q_{s|t}$  and  $p_{s|t}$  from (49) and (59), we get:

$$= \lim_{T \rightarrow \infty} T \left[ \log \left( 1 + \tilde{R}_t(\mathbf{y}, \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \right) - \log \left( 1 + \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \right) \right]$$

Applying the Taylor series expansion for  $\log(1+x)$ , we get:

$$\begin{aligned} & = \lim_{T \rightarrow \infty} T \left[ \tilde{R}_t(\mathbf{y}, \mathbf{y}) \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) - \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y}) \frac{1}{T} - \mathcal{O}\left(\frac{1}{T^2}\right) \right] \\ & = \tilde{R}_t(\mathbf{y}, \mathbf{y}) + \lim_{T \rightarrow \infty} T \mathcal{O}\left(\frac{1}{T^2}\right) - \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y}) - \lim_{T \rightarrow \infty} T \mathcal{O}\left(\frac{1}{T^2}\right) \end{aligned}$$

$$\because \lim_{T \rightarrow \infty} T \mathcal{O}\left(\frac{1}{T^2}\right) = 0, \text{ we get:}$$

$$= \tilde{R}_t(\mathbf{y}, \mathbf{y}) - \tilde{R}_t^\theta(\mathbf{y}, \mathbf{y})$$

Using (58) and (60), we get:

$$= - \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} \right) \quad (63)$$

第 2 学期：

$$\lim_{T \rightarrow \infty} T \sum_{\mathbf{y}' \neq \mathbf{y}} q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x}) \log \frac{q_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}{p_{s|t}(\mathbf{y}'|\mathbf{y}, \mathbf{x})}$$

Substituting  $q_{s|t}$  and  $p_{s|t}$  from (49) and (59), we get:

$$\begin{aligned} & = \lim_{T \rightarrow \infty} T \sum_{\mathbf{y}' \neq \mathbf{y}} \left[ \left[ \frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right) \right] \log \frac{\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right)}{\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') \frac{1}{T} + \mathcal{O}\left(\frac{1}{T^2}\right)} \right] \\ & = \lim_{T \rightarrow \infty} \sum_{\mathbf{y}' \neq \mathbf{y}} \left[ \left[ \frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') + T \mathcal{O}\left(\frac{1}{T^2}\right) \right] \log \frac{\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') + T \mathcal{O}\left(\frac{1}{T^2}\right)}{\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') + T \mathcal{O}\left(\frac{1}{T^2}\right)} \right] \\ & \because \lim_{T \rightarrow \infty} T \mathcal{O}\left(\frac{1}{T^2}\right) = 0, \text{ we get:} \end{aligned}$$

$$= \sum_{\mathbf{y}' \neq \mathbf{y}} \frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \log \frac{\frac{q_s(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}')}{\mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}')}$$

$$= \sum_{\mathbf{y}' \neq \mathbf{y}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \left( \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \log s_\theta(\mathbf{y})_{\mathbf{y}'} \right) \quad (64)$$

Finally, plugging (63) and (64) into (62) yields us the NELBO as proposed in Lou et al. [33]:

$$\begin{aligned} & \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ - \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - s_\theta(\mathbf{y})_{\mathbf{y}'} \right) \right. \\ & \quad \left. + \sum_{\mathbf{y}' \neq \mathbf{y}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \left( \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \log s_\theta(\mathbf{y})_{\mathbf{y}'} \right) \right] \\ &= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} + s_\theta(\mathbf{y})_{\mathbf{y}'} \right. \right. \\ & \quad \left. \left. + \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log s_\theta(\mathbf{y})_{\mathbf{y}'} \right) \right] \\ &= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( s_\theta(\mathbf{y})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log s_\theta(\mathbf{y})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \right) \right) \right] \quad (65) \end{aligned}$$

where  $K(a) = a \log a - a$ . This concludes the proof.

### C.3 Concrete Score for MDLM

Given a latent variable  $\mathbf{z}_t$  and the output of the denoising model,  $\mathbf{x}_\theta(\mathbf{z}_t, t)$  parameterized using SUBS, we aim to recover the concrete score  $\mathbf{s}_\theta(\mathbf{z}_t) \in (\mathbb{R}^+ + \{0\})^{|\mathcal{V}|}$ . Note that  $\mathbf{s}_\theta(\mathbf{z}_t)_\mathbf{y}$  is the ratio  $\frac{p_t(\mathbf{y})}{p_t(\mathbf{z}_t)}$  in the reverse process. Since  $\mathbf{x}_\theta$  approximates  $\mathbf{x}$ , we use  $p_t(\mathbf{y}) = q_t(\mathbf{y}|\mathbf{x}_\theta(\mathbf{z}_t, t))$ ; therefore,

$$\mathbf{s}_\theta(\mathbf{z}_t)_\mathbf{y} = \frac{p_t(\mathbf{y})}{p_t(\mathbf{z}_t)} = \frac{q_t(\mathbf{y}|\mathbf{x}_\theta(\mathbf{z}_t, t))}{q_t(\mathbf{z}_t|\mathbf{x}_\theta(\mathbf{z}_t, t))}. \quad (66)$$

To obtain the score, we first compute  $q_t(\mathbf{y}|\mathbf{x}_\theta(\mathbf{z}_t, t))$  for all possible  $\mathbf{y}$  and  $\mathbf{z}_t$ . Using (4), we derive the following expressions under the SUBS parameterization:

$$q_t(\mathbf{y} \neq \mathbf{m} | \mathbf{x}_\theta(\mathbf{z}_t = \mathbf{m}, t)) = \alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y} \rangle \quad (67)$$

$$q_t(\mathbf{y} \notin \{\mathbf{m}, \mathbf{z}_t\} | \mathbf{x}_\theta(\mathbf{z}_t \neq \mathbf{m}, t)) = 0 \quad (68)$$

$$q_t(\mathbf{y} = \mathbf{z}_t | \mathbf{x}_\theta(\mathbf{z}_t \neq \mathbf{m}, t)) = \alpha_t \quad (69)$$

$$q_t(\mathbf{y} = \mathbf{m} | \mathbf{x}_\theta(\mathbf{z}_t \in \mathcal{V}, t)) = 1 - \alpha_t \quad (70)$$

Plugging these into (66), we get:

$$\mathbf{s}_\theta(\mathbf{z}_t = \mathbf{m})_{\mathbf{y} \neq \mathbf{m}} = \frac{\alpha_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y} \rangle \quad \text{Using (70) and (67)} \quad (71)$$

$$\mathbf{s}_\theta(\mathbf{z}_t = \mathbf{m})_{\mathbf{y} = \mathbf{m}} = 1 \quad \text{Using (70)} \quad (72)$$

$$\mathbf{s}_\theta(\mathbf{z}_t \neq \mathbf{m})_{\mathbf{y} = \mathbf{m}} = \frac{1 - \alpha_t}{\alpha_t} \quad \text{Using (69) and (70)} \quad (73)$$

$$\mathbf{s}_\theta(\mathbf{z}_t \neq \mathbf{m})_{\mathbf{y} = \mathbf{z}_t} = 1 \quad \text{Using (69)} \quad (74)$$

$$\mathbf{s}_\theta(\mathbf{z}_t \neq \mathbf{m})_{\mathbf{y} \notin \{\mathbf{m}, \mathbf{z}_t\}} = 0 \quad \text{Using (68) and (69)} \quad (75)$$

These can be consolidated into the following expression:

$$\mathbf{s}_\theta(\mathbf{z}_t)_\mathbf{y} = \mathbf{y}^\top \left[ \delta_{\mathbf{z}_t, \mathbf{m}} \frac{\alpha_t}{1 - \alpha_t} \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \delta_{\mathbf{z}_t, \mathbf{m}}) \frac{1 - \alpha_t}{\alpha_t} \mathbf{m} + \mathbf{z}_t \right] \quad (76)$$

$$= \sum_{\mathbf{y}' \neq \mathbf{y}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \left( \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \log s_\theta(\mathbf{y})_{\mathbf{y}'} \right) \quad (64)$$

最后，将 (63) 和 (64) 代入 (62) 即可得到 Lou 等人提出的 NELBO。[33]:

$$\begin{aligned} & \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ - \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - s_\theta(\mathbf{y})_{\mathbf{y}'} \right) \right. \\ & \quad \left. + \sum_{\mathbf{y}' \neq \mathbf{y}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} R_t(\mathbf{y}, \mathbf{y}') \left( \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \log s_\theta(\mathbf{y})_{\mathbf{y}'} \right) \right] \\ &= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} + s_\theta(\mathbf{y})_{\mathbf{y}'} \right. \right. \\ & \quad \left. \left. + \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log s_\theta(\mathbf{y})_{\mathbf{y}'} \right) \right] \\ &= \mathbb{E}_{t \in [0,1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \left[ \sum_{\mathbf{y}' \neq \mathbf{y}} R_t(\mathbf{y}, \mathbf{y}') \left( s_\theta(\mathbf{y})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \log s_\theta(\mathbf{y})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{y}|\mathbf{x})} \right) \right) \right] \quad (65) \end{aligned}$$

其中  $K(a) = a \log a - a$ 。证明到此结束。

### C.3 MDLM 的具体评分

给定潜在变量  $\mathbf{z}_t$  和去噪模型的输出（使用 SUBS 参数化的  $\mathbf{x}_\theta(\mathbf{z}_t, t)$ ），我们的目标是恢复具体分数  $s_\theta(\mathbf{z}_t) \in (\mathbb{R}^+ + \{0\})^{|\mathcal{V}|}$ 。注意， $s_\theta(\mathbf{z}_t)_{\mathbf{y}}$  是  $\frac{p_t(\mathbf{y})}{p_t(\mathbf{z}_t)}$  的相反过程中的比率。由于  $\mathbf{x}_\theta$  近似于  $\mathbf{x}$ ，因此我们使用  $p_t(\mathbf{y}) = q_t(\mathbf{y}|\mathbf{x}_\theta(\mathbf{z}_t, t))$ ；所以，

$$s_\theta(\mathbf{z}_t)_{\mathbf{y}} = \frac{p_t(\mathbf{y})}{p_t(\mathbf{z}_t)} = \frac{q_t(\mathbf{y}|\mathbf{x}_\theta(\mathbf{z}_t, t))}{q_t(\mathbf{z}_t|\mathbf{x}_\theta(\mathbf{z}_t, t))}. \quad (66)$$

为了获得分数，我们首先计算所有可能的  $\mathbf{y}$  和  $\mathbf{z}_t$  的  $q_t(\mathbf{y}|\mathbf{x}_\theta(\mathbf{z}_t, t))$ 。利用(4)，我们在SUBS参数化下推导出以下表达式：

$$q_t(\mathbf{y} \neq \mathbf{m} | \mathbf{x}_\theta(\mathbf{z}_t = \mathbf{m}, t)) = \alpha_t \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y} \rangle \quad (67)$$

$$q_t(\mathbf{y} \notin \{\mathbf{m}, \mathbf{z}_t\} | \mathbf{x}_\theta(\mathbf{z}_t \neq \mathbf{m}, t)) = 0 \quad (68)$$

$$q_t(\mathbf{y} = \mathbf{z}_t | \mathbf{x}_\theta(\mathbf{z}_t \neq \mathbf{m}, t)) = \alpha_t \quad (69)$$

$$q_t(\mathbf{y} = \mathbf{m} | \mathbf{x}_\theta(\mathbf{z}_t \in \mathcal{V}, t)) = 1 - \alpha_t \quad (70)$$

将这些代入 (66)，我们得到：

$$s_\theta(\mathbf{z}_t = \mathbf{m})_{\mathbf{y} \neq \mathbf{m}} = \frac{\alpha_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y} \rangle \quad \text{Using (70) and (67)} \quad (71)$$

$$s_\theta(\mathbf{z}_t = \mathbf{m})_{\mathbf{y} = \mathbf{m}} = 1 \quad \text{Using (70)} \quad (72)$$

$$s_\theta(\mathbf{z}_t \neq \mathbf{m})_{\mathbf{y} = \mathbf{m}} = \frac{1 - \alpha_t}{\alpha_t} \quad \text{Using (69) and (70)} \quad (73)$$

$$s_\theta(\mathbf{z}_t \neq \mathbf{m})_{\mathbf{y} = \mathbf{z}_t} = 1 \quad \text{Using (69)} \quad (74)$$

$$s_\theta(\mathbf{z}_t \neq \mathbf{m})_{\mathbf{y} \notin \{\mathbf{m}, \mathbf{z}_t\}} = 0 \quad \text{Using (68) and (69)} \quad (75)$$

这些可以合并为以下表达式：

$$s_\theta(\mathbf{z}_t)_{\mathbf{y}} = \mathbf{y}^\top \left[ \delta_{\mathbf{z}_t, \mathbf{m}} \frac{\alpha_t}{1 - \alpha_t} \mathbf{x}_\theta(\mathbf{z}_t, t) + (1 - \delta_{\mathbf{z}_t, \mathbf{m}}) \frac{1 - \alpha_t}{\alpha_t} \mathbf{m} + \mathbf{z}_t \right] \quad (76)$$

#### C.4 Reverse Rate Matrix for MDLM

We can formulate the reverse rate matrix for MDLM using (76) and (56). Recall that the reverse rate matrix  $\tilde{R}_t(\mathbf{y}', \mathbf{y})$  is given by:

$$\tilde{R}_t(\mathbf{y}', \mathbf{y}) = \begin{cases} \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') & \mathbf{y}' \neq \mathbf{y} \\ -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \tilde{\mathbf{y}}) & \mathbf{y}' = \mathbf{y}. \end{cases}$$

Let's examine the cases where  $\mathbf{y} = \mathbf{m}$  and  $\mathbf{y} \neq \mathbf{m}$ .

**Case  $\mathbf{y} = \mathbf{m}$ :** For  $\mathbf{y}' \neq \mathbf{m}$ , the reverse rate  $\tilde{R}_t(\mathbf{y}', \mathbf{y} = \mathbf{m})$  is given by:

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' \neq \mathbf{m}, \mathbf{y} = \mathbf{m}) \\ &= \mathbf{s}_\theta(\mathbf{y} = \mathbf{m})_{\mathbf{y}' \neq \mathbf{m}} R_t(\mathbf{y} = \mathbf{m}, \mathbf{y}') \\ & \text{Using (71) and (56), we get:} \\ &= \frac{\alpha_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y}' \rangle \left[ -\frac{\alpha'_t}{\alpha_t} \right] \\ &= -\frac{\alpha'_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y}' \rangle \end{aligned} \tag{77}$$

For  $\mathbf{y}' = \mathbf{m}$ , the reverse rate  $\tilde{R}_t(\mathbf{y}', \mathbf{y} = \mathbf{m})$  is given by:

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' = \mathbf{m}, \mathbf{y} = \mathbf{m}) \\ &= -\sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \tilde{R}_t(\tilde{\mathbf{y}}, \mathbf{y} = \mathbf{m}) \\ & \text{Using (77), we get:} \\ &= \sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \frac{\alpha'_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \tilde{\mathbf{y}} \rangle \\ &= \frac{\alpha'_t}{1 - \alpha_t} \sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \tilde{\mathbf{y}} \rangle \\ & \text{"zero-masking probability" in Sec. 3.2.3} \implies \sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \tilde{\mathbf{y}} \rangle = 1; \text{ hence,} \\ &= \frac{\alpha'_t}{1 - \alpha_t}. \end{aligned} \tag{78}$$

**Case  $\mathbf{y} \neq \mathbf{m}$ :** For  $\mathbf{y}' \neq \mathbf{y}$  we have:

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' \notin \{\mathbf{y}, \mathbf{m}\}, \mathbf{y} \neq \mathbf{m}) \\ &= \mathbf{s}_\theta(\mathbf{y} \neq \mathbf{m})_{\mathbf{y}' \notin \{\mathbf{y}, \mathbf{m}\}} \underbrace{R_t(\mathbf{y} \neq \mathbf{m}, \mathbf{y}' \notin \{\mathbf{y}, \mathbf{m}\})}_{= 0 \text{ from (56)}} \\ &= 0 \end{aligned} \tag{79}$$

For  $\mathbf{y}' = \mathbf{m}$ , we have:

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' = \mathbf{m}, \mathbf{y} \neq \mathbf{m}) \\ &= \mathbf{s}_\theta(\mathbf{y} \neq \mathbf{m})_{\mathbf{y}' = \mathbf{m}} \underbrace{R_t(\mathbf{y} \neq \mathbf{m}, \mathbf{y}' = \mathbf{m})}_{= 0 \text{ from (56)}} \\ &= 0 \end{aligned} \tag{80}$$

Thus, for  $\mathbf{y}' = \mathbf{y}$ , we have:

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' = \mathbf{y}, \mathbf{y} \neq \mathbf{m}) \\ &= -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \tilde{R}_t(\tilde{\mathbf{y}}, \mathbf{y} \neq \mathbf{m}) \end{aligned}$$

#### C.4 MDLM 的反向速率矩阵

我们可以使用 (76) 和 (56) 制定 MDLM 的反向速率矩阵。回想一下，反向速率矩阵  $\tilde{R}_t(\mathbf{y}', \mathbf{y})$  由下式给出：

$$\tilde{R}_t(\mathbf{y}', \mathbf{y}) = \begin{cases} \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \mathbf{y}') & \mathbf{y}' \neq \mathbf{y} \\ -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \mathbf{s}_\theta(\mathbf{y})_{\mathbf{y}'} R_t(\mathbf{y}, \tilde{\mathbf{y}}) & \mathbf{y}' = \mathbf{y}. \end{cases}$$

让我们检查一下  $\mathbf{y} = \mathbf{m}$  和  $\mathbf{y} \neq \mathbf{m}$  的情况。

情况  $\mathbf{y} = \mathbf{m}$ ：对于  $\mathbf{y}' \neq \mathbf{m}$ ，反向速率  $\tilde{R}_t(\mathbf{y}', \mathbf{y} = \mathbf{m})$  由下式给出：

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' \neq \mathbf{m}, \mathbf{y} = \mathbf{m}) \\ &= \mathbf{s}_\theta(\mathbf{y} = \mathbf{m})_{\mathbf{y}' \neq \mathbf{m}} R_t(\mathbf{y} = \mathbf{m}, \mathbf{y}') \\ & \text{Using (71) and (56), we get:} \\ &= \frac{\alpha_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y}' \rangle \left[ -\frac{\alpha'_t}{\alpha_t} \right] \\ &= -\frac{\alpha'_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{y}' \rangle \end{aligned} \tag{77}$$

对于  $\mathbf{y}' = \mathbf{m}$ ，反向速率  $\tilde{R}_t(\mathbf{y}', \mathbf{y} = \mathbf{m})$  由下式给出：

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' = \mathbf{m}, \mathbf{y} = \mathbf{m}) \\ &= -\sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \tilde{R}_t(\tilde{\mathbf{y}}, \mathbf{y} = \mathbf{m}) \\ & \text{Using (77), we get:} \\ &= \sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \frac{\alpha'_t}{1 - \alpha_t} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \tilde{\mathbf{y}} \rangle \\ &= \frac{\alpha'_t}{1 - \alpha_t} \sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \tilde{\mathbf{y}} \rangle \\ & \text{"zero-masking probability" in Sec. 3.2.3} \implies \sum_{\tilde{\mathbf{y}} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \tilde{\mathbf{y}} \rangle = 1; \text{ hence,} \\ &= \frac{\alpha'_t}{1 - \alpha_t}. \end{aligned} \tag{78}$$

案例  $\mathbf{y} \neq \mathbf{m}$ ：对于  $\mathbf{y}' \neq \mathbf{y}$ ，我们有：

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' \notin \{\mathbf{y}, \mathbf{m}\}, \mathbf{y} \neq \mathbf{m}) \\ &= \mathbf{s}_\theta(\mathbf{y} \neq \mathbf{m})_{\mathbf{y}' \notin \{\mathbf{y}, \mathbf{m}\}} \underbrace{R_t(\mathbf{y} \neq \mathbf{m}, \mathbf{y}' \notin \{\mathbf{y}, \mathbf{m}\})}_{= 0 \text{ from (56)}} \\ &= 0 \end{aligned} \tag{79}$$

对于  $\mathbf{y}' = \mathbf{m}$ ，我们有：

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' = \mathbf{m}, \mathbf{y} \neq \mathbf{m}) \\ &= \mathbf{s}_\theta(\mathbf{y} \neq \mathbf{m})_{\mathbf{y}' = \mathbf{m}} \underbrace{R_t(\mathbf{y} \neq \mathbf{m}, \mathbf{y}' = \mathbf{m})}_{= 0 \text{ from (56)}} \\ &= 0 \end{aligned} \tag{80}$$

因此，对于  $\mathbf{y}' = \mathbf{y}$ ，我们有：

$$\begin{aligned} & \tilde{R}_t(\mathbf{y}' = \mathbf{y}, \mathbf{y} \neq \mathbf{m}) \\ &= -\sum_{\tilde{\mathbf{y}} \neq \mathbf{y}} \tilde{R}_t(\tilde{\mathbf{y}}, \mathbf{y} \neq \mathbf{m}) \end{aligned}$$

$$\begin{aligned}
&= -\underbrace{\tilde{R}_t(\tilde{\mathbf{y}}=\mathbf{m}, \mathbf{y} \neq \mathbf{m})}_{=0 \text{ from (80)}} - \underbrace{\sum_{\tilde{\mathbf{y}} \notin \{\mathbf{y}, \mathbf{m}\}} \tilde{R}_t(\tilde{\mathbf{y}}, \mathbf{y} \neq \mathbf{m})}_{=0 \text{ from (79)}} \\
&= 0
\end{aligned} \tag{81}$$

Summarizing (77), (78), (79), (80), (81), we have:

$$\begin{aligned}
\tilde{R}_t(\mathbf{y}', \mathbf{y}) &= \begin{cases} -\langle \mathbf{x}_\theta(\mathbf{y}, t), \mathbf{y}' \rangle \frac{\alpha'_t}{1-\alpha_t} & \mathbf{y}' \neq \mathbf{m}, \mathbf{y} = \mathbf{m} \\ \frac{\alpha'_t}{1-\alpha_t} & \mathbf{y}' = \mathbf{m}, \mathbf{y} = \mathbf{m} \\ 0 & \text{otherwise.} \end{cases} \\
&= \boxed{-\frac{\alpha'_t}{1-\alpha_t} [\mathbf{y}']^\top [\mathbf{x}_\theta(\mathbf{y}, t) - \mathbf{m}] \langle \mathbf{y}, \mathbf{m} \rangle}
\end{aligned} \tag{82}$$

### C.5 Deriving MDLM's NELBO via CTMC

Now, we aim to show that substituting the expression for the rate matrix  $R_t$  in terms of state transition matrix  $Q_t$  from (56) into (65) and switching from score-parameterization to the SUBS parameterization (Sec. 3.2.3) yields the simplified NELBO for MDLM as given by (10). We present the proof below. Recall that the term  $\langle \mathbf{a}, \mathbf{b} \rangle$  denotes the dot product of two vectors  $\mathbf{a}$  and  $\mathbf{b}$ . When  $\mathbf{a}$  and  $\mathbf{b}$  represent two *one-hot* vectors, this quantity evaluates to 1 if  $\mathbf{a} = \mathbf{b}$  and 0 otherwise.

**Proof.** Recall that for absorbing state diffusion,  $\mathbf{y}$  takes only two possible values, i.e.,  $\mathbf{y} \in \{\mathbf{x}, \mathbf{m}\}$ . Thus, we expand (65) as follows:

$$\begin{aligned}
\mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} &\left[ \langle \mathbf{y}, \mathbf{x} \rangle \left[ \sum_{\mathbf{y}' \neq \mathbf{x}} R_t(\mathbf{x}, \mathbf{y}') \left( \mathbf{s}_\theta(\mathbf{x})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{x} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{x})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{x} | \mathbf{x})} \right) \right) \right] \right. \\
&\quad \left. + \langle \mathbf{y}, \mathbf{m} \rangle \left[ \sum_{\mathbf{y}' \neq \mathbf{m}} R_t(\mathbf{m}, \mathbf{y}') \left( \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \right) \right) \right] \right]
\end{aligned}$$

$\because R_t(\mathbf{x}, \mathbf{y}' \neq \mathbf{x}) = 0$  from (54), we get:

$$= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ \sum_{\mathbf{y}' \neq \mathbf{m}} R_t(\mathbf{m}, \mathbf{y}') \left( \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \right) \right) \right]$$

Substituting  $R_t(\mathbf{m}, \mathbf{y}' \neq \mathbf{m})$  from (52), we get:

$$\begin{aligned}
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ \sum_{\mathbf{y}' \neq \mathbf{m}} -\frac{\alpha'_t}{\alpha_t} \left( \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \right) \right) \right] \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ -\frac{\alpha'_t}{\alpha_t} \left( \underbrace{\sum_{\mathbf{y}' \neq \mathbf{m}} \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'}}_{\text{Term 1}} - \underbrace{\sum_{\mathbf{y}' \neq \mathbf{m}} \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'}}_{\text{Term 2}} + \underbrace{\sum_{\mathbf{y}' \neq \mathbf{m}} K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \right)}_{\text{Term 3}} \right) \right]
\end{aligned} \tag{83}$$

**Term 1:**

$$\begin{aligned}
&\sum_{\mathbf{y}' \neq \mathbf{m}} \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} \\
&\text{Using (67), we get,} \\
&= \sum_{\mathbf{y}' \neq \mathbf{m}} \frac{\alpha_t}{1-\alpha_t} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{y}' \rangle
\end{aligned}$$

$$\begin{aligned}
&= -\underbrace{\tilde{R}_t(\tilde{\mathbf{y}}=\mathbf{m}, \mathbf{y} \neq \mathbf{m})}_{=0 \text{ from (80)}} - \underbrace{\sum_{\tilde{\mathbf{y}} \notin \{\mathbf{y}, \mathbf{m}\}} \tilde{R}_t(\tilde{\mathbf{y}}, \mathbf{y} \neq \mathbf{m})}_{=0 \text{ from (79)}} \\
&= 0
\end{aligned} \tag{81}$$

总结(77)、(78)、(79)、(80)、(81)，我们有：

$$\begin{aligned}
\tilde{R}_t(\mathbf{y}', \mathbf{y}) &= \begin{cases} -\langle \mathbf{x}_\theta(\mathbf{y}, t), \mathbf{y}' \rangle \frac{\alpha'_t}{1-\alpha_t} & \mathbf{y}' \neq \mathbf{m}, \mathbf{y} = \mathbf{m} \\ \frac{\alpha'_t}{1-\alpha_t} & \mathbf{y}' = \mathbf{m}, \mathbf{y} = \mathbf{m} \\ 0 & \text{otherwise.} \end{cases} \\
&= \boxed{-\frac{\alpha'_t}{1-\alpha_t} [\mathbf{y}']^\top [\mathbf{x}_\theta(\mathbf{y}, t) - \mathbf{m}] \langle \mathbf{y}, \mathbf{m} \rangle}
\end{aligned} \tag{82}$$

### C.5 通过 CTMC 导出 MDLM 的 NELBO

现在，我们的目标是证明，将速率矩阵  $R_t$  的表达式从 (56) 替换为状态转换矩阵  $Q_t$  到 (65)，并从分数参数化切换到 SUBS 参数化（第 3.2.3 节），产生 MDLM 的简化 NELBO，如 (10) 所示。我们在下面给出证明。回想一下，术语  $\langle \mathbf{a}, \mathbf{b} \rangle$  表示两个向量  $\mathbf{a}$  和  $\mathbf{b}$  的点积。当  $\mathbf{a}$  和  $\mathbf{b}$  表示两个 *one-hot* 向量时，如果  $\mathbf{a} = \mathbf{b}$  则该量计算为 1，否则计算为 0。

证明。回想一下，对于吸收态扩散， $\mathbf{y}$  仅采用两个可能的值，即  $\mathbf{y} \in \{\mathbf{x}, \mathbf{m}\}$ 。因此，我们将(65)展开如下：

$$\begin{aligned}
&\mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \left[ \langle \mathbf{y}, \mathbf{x} \rangle \left[ \sum_{\mathbf{y}' \neq \mathbf{x}} R_t(\mathbf{x}, \mathbf{y}') \left( \mathbf{s}_\theta(\mathbf{x})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{x} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{x})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{x} | \mathbf{x})} \right) \right) \right] \right. \\
&\quad \left. + \langle \mathbf{y}, \mathbf{m} \rangle \left[ \sum_{\mathbf{y}' \neq \mathbf{m}} R_t(\mathbf{m}, \mathbf{y}') \left( \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \right) \right) \right] \right] \\
&\because R_t(\mathbf{x}, \mathbf{y}' \neq \mathbf{x}) = 0 \text{ 从 (54) 中我们得到:} \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ \sum_{\mathbf{y}' \neq \mathbf{m}} R_t(\mathbf{m}, \mathbf{y}') \left( \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \right) \right) \right] \\
&\text{将 (52) 中的 } R_t(\mathbf{m}, \mathbf{y}' \neq \mathbf{m}) \text{ 代入, 我们得到:} \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ \sum_{\mathbf{y}' \neq \mathbf{m}} -\frac{\alpha'_t}{\alpha_t} \left( \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} - \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'} + K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \right) \right) \right] \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ -\frac{\alpha'_t}{\alpha_t} \left( \underbrace{\sum_{\mathbf{y}' \neq \mathbf{m}} \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'}}_{\text{Term 1}} - \underbrace{\sum_{\mathbf{y}' \neq \mathbf{m}} \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \log \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}'}}_{\text{Term 2}} + \underbrace{\sum_{\mathbf{y}' \neq \mathbf{m}} K \left( \frac{q_t(\mathbf{y}' | \mathbf{x})}{q_t(\mathbf{m} | \mathbf{x})} \right)}_{\text{Term 3}} \right) \right]
\end{aligned} \tag{83}$$

第 1 期：

$$\begin{aligned}
&\sum_{\mathbf{y} \neq \mathbf{m}} \mathbf{s}_\theta(\mathbf{m})_{\mathbf{y}} \\
&\text{Using (67), we get,} \\
&= \sum_{\mathbf{y} \neq \mathbf{m}} \frac{\alpha_t}{1-\alpha_t} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{y} \rangle
\end{aligned}$$

$$\begin{aligned}
&= \frac{\alpha_t}{1-\alpha_t} \sum_{\mathbf{y} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{y} \rangle \\
&\text{“zero-masking probability” in Sec. 3.2.3} \implies \sum_{\mathbf{y} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{y} \rangle = 1; \text{ hence,} \\
&= \frac{\alpha_t}{1-\alpha_t}
\end{aligned} \tag{84}$$

**Term 2:**

$$\begin{aligned}
&\sum_{\mathbf{y}' \neq \mathbf{m}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log s_\theta(\mathbf{m})_{\mathbf{y}'} \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log s_\theta(\mathbf{m})_{\mathbf{x}} + \sum_{\mathbf{y}' \notin \{\mathbf{m}, \mathbf{x}\}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log s_\theta(\mathbf{m})_{\mathbf{y}'} \\
&\because q_t(\mathbf{y}'|\mathbf{x}) = 0 \text{ for } \mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\} \text{ from (4) we get:} \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log s_\theta(\mathbf{m})_{\mathbf{x}} \\
&\text{Using (4), we get:} \\
&= \frac{\alpha_t}{1-\alpha_t} \log s_\theta(\mathbf{m})_{\mathbf{x}} \\
&\text{Using (67), we get:} \\
&= \frac{\alpha_t}{1-\alpha_t} \log \left[ \frac{\alpha_t}{1-\alpha_t} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right] \\
&= \frac{\alpha_t}{1-\alpha_t} \log \frac{\alpha_t}{1-\alpha_t} + \frac{\alpha_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle
\end{aligned} \tag{85}$$

**Term 3:**

$$\begin{aligned}
&\sum_{\mathbf{y}' \neq \mathbf{m}} K \left( \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right) \\
&= \sum_{\mathbf{y}' \neq \mathbf{m}} \left[ \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right] \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} + \sum_{\mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\}} \left[ \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right] \\
&\because q_t(\mathbf{y}'|\mathbf{x}) = 0 \text{ for } \mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\}, \text{ we get,} \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \\
&\text{Substituting the values using (4), we get:} \\
&= \frac{\alpha_t}{1-\alpha_t} \log \frac{\alpha_t}{1-\alpha_t} - \frac{\alpha_t}{1-\alpha_t}
\end{aligned} \tag{86}$$

Substituting (84), (85), and (86) in (83) we get,

$$\begin{aligned}
&\mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ -\frac{\alpha'_t}{\alpha_t} \left( \frac{\alpha_t}{1-\alpha_t} - \frac{\alpha_t}{1-\alpha_t} \log \frac{\alpha_t}{1-\alpha_t} - \frac{\alpha_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right. \right. \\
&\quad \left. \left. + \frac{\alpha_t}{1-\alpha_t} \log \frac{\alpha_t}{1-\alpha_t} - \frac{\alpha_t}{1-\alpha_t} \right) \right] \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ -\frac{\alpha'_t}{\alpha_t} \left( -\frac{\alpha_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right) \right] \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ \frac{\alpha'_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right]
\end{aligned}$$

$$\begin{aligned}
&= \frac{\alpha_t}{1-\alpha_t} \sum_{\mathbf{y} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{y} \rangle \\
&\text{"zero-masking probability" in Sec. 3.2.3} \implies \sum_{\mathbf{y} \neq \mathbf{m}} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{y} \rangle = 1; \text{ hence,} \\
&= \frac{\alpha_t}{1-\alpha_t}
\end{aligned} \tag{84}$$

第 2 学期:

$$\begin{aligned}
&\sum_{\mathbf{y}' \neq \mathbf{m}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log s_\theta(\mathbf{m})_{\mathbf{y}'} \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log s_\theta(\mathbf{m})_{\mathbf{x}} + \sum_{\mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\}} \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log s_\theta(\mathbf{m})_{\mathbf{y}'} \\
&\because q_t(\mathbf{y}'|\mathbf{x}) = 0 \text{ for } \mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\} \text{ from (4)} \text{ we get:} \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log s_\theta(\mathbf{m})_{\mathbf{x}} \\
&\text{Using (4), we get:} \\
&= \frac{\alpha_t}{1-\alpha_t} \log s_\theta(\mathbf{m})_{\mathbf{x}} \\
&\text{Using (67), we get:} \\
&= \frac{\alpha_t}{1-\alpha_t} \log \left[ \frac{\alpha_t}{1-\alpha_t} \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right] \\
&= \frac{\alpha_t}{1-\alpha_t} \log \frac{\alpha_t}{1-\alpha_t} + \frac{\alpha_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle
\end{aligned} \tag{85}$$

第 3 学期:

$$\begin{aligned}
&\sum_{\mathbf{y}' \neq \mathbf{m}} K \left( \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right) \\
&= \sum_{\mathbf{y}' \neq \mathbf{m}} \left[ \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right] \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} + \sum_{\mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\}} \left[ \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{y}'|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \right] \\
&\because q_t(\mathbf{y}'|\mathbf{x}) = 0 \text{ for } \mathbf{y}' \notin \{\mathbf{x}, \mathbf{m}\}, \text{ we get,} \\
&= \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \log \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} - \frac{q_t(\mathbf{x}|\mathbf{x})}{q_t(\mathbf{m}|\mathbf{x})} \\
&\text{Substituting the values using (4), we get:} \\
&= \frac{\alpha_t}{1-\alpha_t} \log \frac{\alpha_t}{1-\alpha_t} - \frac{\alpha_t}{1-\alpha_t}
\end{aligned} \tag{86}$$

将 (84)、(85) 和 (86) 代入 (83) 我们得到,

$$\begin{aligned}
&\mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ -\frac{\alpha'_t}{\alpha_t} \left( \frac{\alpha_t}{1-\alpha_t} - \frac{\alpha_t}{1-\alpha_t} \log \frac{\alpha_t}{1-\alpha_t} - \frac{\alpha_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right. \right. \\
&\quad \left. \left. + \frac{\alpha_t}{1-\alpha_t} \log \frac{\alpha_t}{1-\alpha_t} - \frac{\alpha_t}{1-\alpha_t} \right) \right] \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ -\frac{\alpha'_t}{\alpha_t} \left( -\frac{\alpha_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right) \right] \\
&= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot|\mathbf{x})} \langle \mathbf{y}, \mathbf{m} \rangle \left[ \frac{\alpha'_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{m}, t), \mathbf{x} \rangle \right]
\end{aligned}$$

Under the SUBS parameterization,  $\log\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle = 0$  when  $\mathbf{z}_t = \mathbf{x}$ ; hence  $\langle \mathbf{y}, \mathbf{m} \rangle$  can be dropped:

$$= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \left[ \frac{\alpha'_t}{1 - \alpha_t} \log\langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \right] \quad (87)$$

This concludes the proof.

## Appendix D Experimental details

### D.1 Likelihood Evaluation

We use a single monte-carlo estimate for  $t$  to evaluate the likelihood. The low discrepancy sampler (D.3) plays a key role in reducing the variance of the estimate as seen in Table 8.

### D.2 Avg. Number of Tokens seen

Given `training_steps`, `batch_size`, `context_length`, the number of tokens seen by the AR model is given as:

$$\text{training\_steps} \times \text{batch\_size} \times \text{context\_length}. \quad (88)$$

However, this expression doesn't hold for a diffusion model, since at each training step, a fraction of the input tokens are masked before being fed to the model. Let  $p_m$  be the probability of a token being masked at a timestep  $t$ . For the log-linear schedule in our experiments,  $p_m = t$ . Thus, the expected number of tokens seen by the diffusion model is:

$$\begin{aligned} & \mathbb{E}_{t \sim \mathcal{U}[0, 1]} [\text{training\_steps} \times \text{batch\_size} \times \text{context\_length} \times p_m] \\ &= \text{training\_steps} \times \text{batch\_size} \times \text{context\_length} \times \mathbb{E}_{t \sim \mathcal{U}[0, 1]} [p_m] \\ &= \text{training\_steps} \times \text{batch\_size} \times \text{context\_length} \times \mathbb{E}_{t \sim \mathcal{U}[0, 1]} [t] \quad \because p_m = t \\ &= \text{training\_steps} \times \text{batch\_size} \times \text{context\_length} \times 0.5. \quad \because \mathbb{E}_{t \sim \mathcal{U}[0, 1]} [t] = 0.5 \end{aligned} \quad (89)$$

**LM1B.** Following [1, 33, 26], we train MDLM for 1M training steps with a `batch_size` = 512, and a context length of 128. Like [33] we use a log-linear schedule and hence the number of tokens seen by our model is  $\approx 33\text{B}$  (89). Similarly, MDLM trained for 10M steps, saw 327B tokens in expectation. The corresponding AR baseline was trained for 0.5M and 5M steps to ensure a similar number of tokens was seen.

**OWT.** We train SEDD and MDLM for 1M training steps with a `batch_size` = 512, `context_length` = 1024, and log-linear schedule. Hence, these models saw 262B tokens during training. Similarly, the AR model saw the same number of tokens when trained for 0.5M steps with the same `batch_size` and `context_length`.

### D.3 Low discrepancy sampler

To reduce variance during training we use a low-discrepancy sampler, similar to that proposed in Kingma et al. [29]. Specifically, when processing a minibatch of  $N$  samples, instead of independently sampling  $N$  from a uniform distribution, we partition the unit interval and sample the time step for each sequence  $i \in \{1, \dots, N\}$  from a different portion of the interval  $t_i \sim U[\frac{i-1}{N}, \frac{i}{N}]$ . This ensures that our sampled timesteps are more evenly spaced across the interval  $[0, 1]$ , reducing the variance of the ELBO.

### D.4 Language Modeling

For our forward noise process, we use a log-linear noise schedule similar to Lou et al. [33].

We detokenize the One Billion Words dataset following Lou et al. [33], whose code can be found [here](https://github.com/louaaron/Score-Entropy-Discrete-Diffusion/blob/main/data.py)<sup>4</sup>. We tokenize the One Billion Words dataset with the `bert-base-uncased` tokenizer, following He et al. [26]. We pad and truncate sequences to a length of 128.

<sup>4</sup><https://github.com/louaaron/Score-Entropy-Discrete-Diffusion/blob/main/data.py>

SUBS参数化下, 当 $\mathbf{z}_t = \mathbf{x}$ 时 $\log(\mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x}) = 0$ ; 因此可以删除  $\langle \mathbf{y}, \mathbf{m} \rangle$ :

$$= \mathbb{E}_{t \in [0, 1], \mathbf{y} \sim q_t(\cdot | \mathbf{x})} \left[ \frac{\alpha'_t}{1 - \alpha_t} \log(\mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x}) \right] \quad (87)$$

证明到此结束。

## 附录 D 实验细节

### D.1 可能性评估

我们使用  $t$  的单个蒙特卡罗估计来评估可能性。低差异采样器 (D.3) 在减少估计方差方面发挥着关键作用, 如表 8 所示。

### D.2 平均看到的代币数量

给定 training\_steps、batch\_size、context\_length, AR 模型看到的 token 数量如下:

$$\text{training\_steps} \times \text{batch\_size} \times \text{context\_length}. \quad (88)$$

然而, 这个表达式不适用于扩散模型, 因为在每个训练步骤中, 一部分输入标记在输入模型之前会被屏蔽。令  $p_m$  为令牌在时间步长  $t$  处被屏蔽的概率。对于我们实验中的对数线性时间表,  $p_m = t$ 。因此, 扩散模型看到的预期代币数量为:

$$\begin{aligned} \mathbb{E}_{t \sim \mathcal{U}[0, 1]}[\text{training\_steps} \times \text{batch\_size} \times \text{context\_length} \times p_m] &= \text{training\_steps} \times \text{batch\_size} \times \text{context\_length} \times \mathbb{E}_{t \sim \mathcal{U}[0, 1]}[p_m] \\ &= \text{training\_steps} \times \text{batch\_size} \times \text{context\_length} \times \mathbb{E}_{t \sim \mathcal{U}[0, 1]}[t] \\ \therefore p_m = t &= \text{训练步骤} \times \text{批量大小} \times \text{上下文长度} \times 0.5. \quad \therefore \mathbb{E}_{t \sim \mathcal{U}[0, 1]}[t] = 0.5 \end{aligned} \quad (89)$$

LM1B。按照[1, 33, 26], 我们使用 batch\_size = 512 和上下文长度 128 来训练 MDLM 进行 1M 训练。像[33]一样, 我们使用对数线性调度, 因此我们的模型看到的令牌数量是  $\approx 33\text{B}$  (89)。同样, MDLM 训练了 10M 步, 预计会出现 327B 个令牌。相应的 AR 基线经过 0.5M 和 5M 步骤的训练, 以确保看到相似数量的令牌。

守望先锋。我们使用 batch\_size = 512、context\_length = 1024 和对数线性调度来训练 SEDD 和 MDLM 以实现 1M 训练步骤。因此, 这些模型在训练期间看到了 262B 个令牌。类似地, 当使用相同的批量大小和上下文长度进行 0.5M 步骤训练时, AR 模型看到了相同数量的标记。

### D.3 低差异采样器

为了减少训练期间的方差, 我们使用低差异采样器, 类似于 Kingma 等人提出的采样器。[29]。具体来说, 当处理小批量的  $N$  样本时, 我们不是从均匀分布中独立采样  $N$ , 而是对单位间隔进行划分, 并从间隔  $t_i \sim U[\frac{i-1}{N}, \frac{i}{N}]$  的不同部分对每个序列  $i \in \{1, \dots, N\}$  的时间步进行采样。这确保了我们的采样的时间步在区间  $[0, 1]$  上分布更均匀, 从而减少了 ELBO 的方差。

### D.4 语言建模

为了我们的福尔摩斯过程, 我们使用类似于 L 的对数线性噪声表 欧等人。[33]。

我们跟随 Lou 等人对 10 亿字数据集进行去标记化。[33], 其代码可以在这里找到<sup>4</sup>。我们遵循 He 等人的建议, 使用 bert-base-uncased 分词器对十亿字数据集进行分词。[26]。我们将序列填充并截断为长度 128。

<sup>4</sup><https://github.com/louaaron/Score-Entropy-Discrete-Diffusion/blob/main/data.py>

We tokenize OpenWebText with the GPT2 tokenizer. We do not pad or truncate sequences – we concatenate and wrap them to a length of 1,024. When wrapping, we add the eos token in-between concatenated. We additionally set the first and last token of every batch to be eos. Since OpenWebText does not have a validation split, we leave the last 100k docs as validation.

We parameterize our autoregressive baselines, SEDD, and MDLM with the transformer architecture from Lou et al. [33]. We use 12 layers, a hidden dimension of 768, 12 attention heads, and a timestep embedding of 128 when applicable. Word embeddings are not tied between the input and output.

We use the AdamW optimizer with a batch size of 512, constant learning rate warmup from 0 to a learning rate of  $3e-4$  for 2,500 steps. We use a constant learning rate for 1M, 5M, or 10M steps on One Billion Words, and 1M steps for OpenWebText. We use a dropout rate of 0.1.

## D.5 Zeroshot Likelihood

We evaluate zeroshot likelihoods by taking the models trained on OpenWebText and evaluating likelihoods on the validation splits of 7 datasets: Penn Tree Bank (PTB; Marcus et al. [36]), Wikitext [38], One Billion Word Language Model Benchmark (LM1B; Chelba et al. [8]), Lambada [41], AG News [68], and Scientific Papers (Pubmed and Arxiv subsets; Cohan et al. [10]). We detokenize the datasets following Lou et al. [33]. For the AG News and Scientific Papers (Pubmed and Arxiv), we apply both the Wikitext and One Billion Words detokenizers. Since the zeroshot datasets have different conventions for sequence segmentation, we wrap sequences to 1024 and do not add eos tokens in between sequences.

## D.6 Representation Learning

Following Devlin et al. [15], we evaluate on all GLUE tasks [65], but exclude WNLI.

We pre-train a MosaicBERT model on C4 [46] for 70k steps, corresponding to 36B tokens. We pad and truncate the data to 128 tokens using the bert-base-uncased tokenizer.

MosaicBERT [43] has a similar architecture to bert-base-uncased and has 137M parameters, 12 layers, 12 attention heads, a hidden dimension of 768, an intermediate size of 3072, and ALiBi attention bias [44].

For pre-training, we use the following hyperparameters: A global batch size of 4096 with gradient accumulation, a learning rate of  $5e-4$ , linear decay to  $0.02x$  of the learning rate with a warmup of  $0.06x$  of the full training duration, and the decoupled AdamW optimizer with  $1e-5$  weight decay and betas 0.9 and 0.98.

For diffusion fine-tuning we use AdamW with a warmup of 2,500 steps from a learning rate of 0 to  $5e-5$ , betas 0.95 and 0.999, and batch size 512. We train for 5k steps total, corresponding to 32M tokens.

For GLUE evaluation, we use the HuggingFace script found [here](#)<sup>5</sup>. We use the default parameters for all datasets, except for a batch size of 16, which we found helped with smaller datasets. This includes the default of 3 epochs for all datasets and learning rate of  $2e-5$ .

## D.7 Diffusion DNA Models

**Dataset** We pre-train the Caduceus MLM [50] on the HG38 human reference genome [11]. Following Schiff et al. [50], we use character- / base pair-level tokenization. The dataset is based on the splits used in Avsec et al. [3]: the training split comprises of 35 billion tokens covering the human genome. This consists of 34,021 segments extended to a maximum length of 1,048,576 (220 segments). We maintain a constant  $2^{20}$  tokens per batch. For the Genomics Benchmark tasks, we use 5-fold cross-validation where we split the training set into 90/10 train/validation splits.

**Architecture** The Caduceus MLM uses as a backbone a bi-directional variant of the data-dependent SSM Mamba block proposed in Gu et al. [22]. This architecture is ideal as it contains inductive biases that preserve reverse complement (RC) equivariance, respecting the inherent symmetry of double-stranded DNA molecules [35, 50, 70].

<sup>5</sup><https://github.com/huggingface/transformers/tree/main/examples/pytorch/text-classification>

我们使用 GPT2 分词器对 OpenWebText 进行分词。我们不会填充或截断序列 - 我们将它们连接并包装到长度为 1,024。包装时，我们在连接之间添加 eos 令牌。我们另外将每个批次的第一个和最后一个代币设置为 eos。由于 OpenWebText 没有验证拆分，因此我们保留最后 100k 文档作为验证。

我们使用 Lou 等人的变压器架构参数化自回归基线、SEDD 和 MDLM。[33]。我们使用 12 层、768 个隐藏维度、12 个注意力头以及 128 个时间步嵌入（如果适用）。词嵌入在输入和输出之间没有联系。

我们使用 AdamW 优化器，批量大小为 512，持续学习率预热从 0 到  $3e-4$  的学习率，持续 2,500 个步骤。我们对十亿字使用 1M、5M 或 10M 步的恒定学习率，对 OpenWebText 使用 1M 步的恒定学习率。我们使用 0.1 的辍学率。

## D.5 零概率

我们通过采用在 OpenWebText 上训练的模型并评估 7 个数据集的验证分割的可能性来评估零样本可能性：Penn Tree Bank (PTB; Marcus 等人[36])、Wikitext [38]、十亿字语言模型基准 (LM1B; Chelba 等人[8])、Lambada [41]、AG News [68] 和科学论文 (Pubmed 和 Arxiv 子集; Cohan 等人)。[10])。我们按照 Lou 等人的方法对数据集进行去标记。[33]。对于 AG 新闻和科学论文 (Pubmed 和 Arxiv)，我们应用了 Wikitext 和 10 亿字去标记器。由于零样本数据集具有不同的序列分割约定，因此我们将序列包装为 1024，并且不在序列之间添加 eos 标记。

## D.6 表征学习

继 Devlin 等人之后。[15]，我们评估所有 GLUE 任务 [65]，但排除 WNLI。

我们在 C4 [46] 上预训练 MosaicBERT 模型 70k 步骤，对应 36B 个令牌。我们使用 bert-base-uncased 分词器将数据填充并截断为 128 个分词。

MosaicBERT [43] 具有与 bert-base-uncased 类似的架构，具有 137M 个参数、12 层、12 个注意力头、隐藏维度 768、中间大小 3072 和 ALiBi 注意力偏差 [44]。

对于预训练，我们使用以下超参数：全局批量大小为 4096，梯度累积，学习率为  $5e-4$ ，线性衰减到学习率的 0.02 倍，预热为整个训练持续时间的 0.06 倍，解耦 AdamW 优化器具有  $1e-5$  权重衰减和 beta 0.9 和 0.98。

对于扩散微调，我们使用 AdamW，从 0 到  $5e-5$ 、beta 0.95 和 0.999、批量大小 512 进行 2,500 步的预热。我们总共训练 5k 步，对应于 32M 令牌。

对于 GLUE 评估，我们使用此处找到的 HuggingFace 脚本<sup>5</sup>。我们对所有数据集使用默认参数，但批量大小 16 除外，我们发现这对于较小的数据集很有帮助。这包括所有数据集的默认 3 epoch 和  $2e-5$  的学习率。

## D.7 扩散 DNA 模型

数据集 我们在 HG38 人类参考基因组 [11] 上预训练 Caduceus MLM [50]。跟随 Schiff 等人。[50]，我们使用字符/碱基对级标记化。该数据集基于 Avsec 等人中使用的分割。[3]：训练分割由覆盖人类基因组的 350 亿个令牌组成。这由 34,021 个段组成，最大长度扩展为 1,048,576 (220 个段)。我们每批次保持恒定的  $2^{20}$  令牌。对于基因组基准测试任务，我们使用 5 折交叉验证，将训练集分成 90/10 的训练/验证部分。

架构 Caduceus MLM 使用 Gu 等人提出的数据相关 SSM Mamba 块的双向变体作为主干。[22]。这种架构非常理想，因为它包含保留反向补体 (RC) 等价性的归纳偏差，尊重双链 DNA 分子固有的对称性 [35,50,70]。

<sup>5</sup><https://github.com/huggingface/transformers/tree/main/examples/pytorch/text-classification>

**Training details** All models are pre-trained on 10B tokens (10K steps) and fine-tuned on a generative objective for an additional 50B tokens (50K steps). We use a global batch size of 1024 for a context length of 1024 tokens. Downstream task fine-tuning is performed for 16K steps (1B tokens).

For performing Caduceus MLM pre-training, we follow Schiff et al. [50] for the model size configuration, and hyperparameter selection. For pre-training, we use a fixed 15% mask rate as done in Devlin et al. [15]. Of the 'masked' tokens, 80% are replaced with [MASK], 10% are replaced with a random token from the vocabulary, and 10% are left unchanged.

For fine-tuning all Mamba-based models (including Caduceus) on diffusion objectives, we lower the learning rate from  $8e-3$  to  $1e-3$ . For fine-tuning HyenaDNA [39], we lower the learning rate from  $6e-4$  to  $5e-5$ . Similar to Gu et al. [22], Schiff et al. [50], we found that Mamba-based models were robust to higher learning rates. We exclude timestep embeddings for all Diffusion DNA experiments, as we show it has minimal impact on generative performance (see Table 12, Suppl. E.5).

We perform downstream task fine-tuning on the final hidden state embedding from pre-training. We perform mean pooling across the sequence length, which may vary from 200 to approximately 2,000 bps. We report the mean and  $\pm$  on max/min classification accuracy over 5-fold cross-validation (CV) using different random seeds, with early stopping on validation accuracy. For each task, we do a hyperparameter sweep over batch size and learning rate and report the values of the 5-fold CV for the best configuration.

**Genomic Benchmark Task Distributions** We use a subset of the Genomic Benchmark tasks with an emphasis on tasks from Human data. The positive samples for each dataset were generated by selecting samples that were annotated, either computationally or experimentally, in previous work (e.g enhancers, promoters, open chromatin regions (OCR)) [20]. These annotations each correspond to subsets of the genome of varying sizes that may exhibit different distributions of DNA than those observed globally over the reference genome. Due to this, the observed dataset may have a different distribution than the data used for pre-training and calculating perplexity. This might in turn lead to a case where perplexity and downstream performance may not necessarily correlate.

## Appendix E Additional Experiments

### E.1 Noise schedule parameterization

As described in Sec. 3.4, the ELBO is invariant to the functional form of  $\alpha_t$ . To demonstrate this, we evaluate MDLM, initially trained using a log-linear schedule on OWT, by replacing the noise schedule with various other noise schedules as mentioned below. Following prior works [1, 33, 54], we parameterize  $\alpha_t = e^{-\sigma(t)}$ , where  $\sigma(t) : [0, 1] \rightarrow \mathbb{R}^+$ . Various functional forms of  $\sigma(t)$  are listed below:

**Log Linear** [1, 33, 54]. The log linear schedule is given as:

$$\sigma(t) = -\log(1-t) \quad (90)$$

**Cosine Squared schedule** [24]. The Cosine Squared schedule is given as:

$$\sigma(t) = -\log \cos^2\left(\frac{\pi}{2}(1-t)\right) \quad (91)$$

**Cosine schedule.** The Cosine schedule is given as:

$$\sigma(t) = -\log \cos\left(\frac{\pi}{2}(1-t)\right) \quad (92)$$

**Linear.** The Linear schedule is given as:

$$\sigma(t) = \sigma_{\max} t \quad (93)$$

where  $\sigma_{\max}$  is a very large number. In our experiments we set it to  $10^8$ .

#### E.1.1 ELBO Invariance

The function  $\alpha_t$  is invertible due to the monotonicity assumption in Sec. 3.1, and so we can perform the following change of variables in (10):  $\gamma \equiv \log(1 - \alpha_t)$ . Let  $f : [0, 1] \rightarrow \mathbb{R}^+$  be a function such

训练细节 所有模型都在 10B 令牌（10K 步骤）上进行预训练，并针对额外 50B 令牌（50K 步骤）的生成目标进行微调。对于 1024 个令牌的上下文长度，我们使用全局批量大小 1024。下游任务微调执行 16K 步骤（1B 令牌）。

为了执行 Caduceus MLM 预训练，我们遵循 Schiff 等人的方法。[50]用于模型大小配置和超参数选择。对于预训练，我们使用固定的 15% 掩码率，如 Devlin 等人所做的那样。[15]。在“屏蔽”标记中，80% 被替换为 [MASK]，10% 被替换为词汇表中的随机标记，10% 保持不变。

为了在扩散目标上微调所有基于 Mamba 的模型（包括 Caduceus），我们将学习率从  $8e-3$  降低到  $1e-3$ 。为了微调 HyenaDNA [39]，我们将学习率从  $6e-4$  降低到  $5e-5$ 。与顾等人类似。[22]，希夫等人。[50]，我们发现基于 Mamba 的模型对于更高的学习率具有鲁棒性。我们排除了所有扩散 DNA 实验的时间步嵌入，因为我们表明它对生成性能的影响最小（参见表 12，补充材料 E.5）。

我们对预训练的最终隐藏状态嵌入进行下游任务微调。我们在序列长度上进行平均池化，序列长度可能从 200 到大约 2,000 bps 不等。我们使用不同的随机种子报告 5 倍交叉验证 (CV) 的最大/最小分类精度的平均值和  $\pm$ ，并提前停止验证精度。对于每项任务，我们都会对批量大小和学习率进行超参数扫描，并报告最佳配置的 5 倍 CV 值。

基因组基准任务分布 我们使用基因组基准任务的子集，重点关注来自人类数据的任务。每个数据集的阳性样本是通过选择在之前的工作中通过计算或实验进行注释的样本（例如增强子、启动子、开放染色质区域（OCR））来生成的[20]。这些注释各自对应于不同大小的基因组子集，这些子集可能表现出与在参考基因组上全局观察到的DNA分布不同的DNA分布。因此，观察到的数据集可能与用于预训练和计算困惑度的数据具有不同的分布。这可能反过来导致困惑度和下游性能不一定相关的情况。

## 附录 E 附加实验

### E.1 噪声表参数化

如第 2 节所述。3.4，ELBO 对于  $\alpha_t$  的函数形式是不变的。为了证明这一点，我们评估 MDLM，最初使用 OWT 上的对数线性计划进行训练，通过用下面提到的各种其他噪声计划替换噪声计划。遵循先前的工作 [1, 33, 54]，我们参数化  $\alpha_t = e^{-\sigma(t)}$ ，其中  $\sigma(t) : [0,1] \rightarrow \mathbb{R}^+$ 。下面列出了  $\sigma(t)$  的各种函数形式：

对数线性 [1, 33, 54]。对数线性时间表给出如下：

$$\sigma(t) = -\log(1-t) \quad (90)$$

余弦平方时间表[24]。余弦平方时间表给出如下：

$$\sigma(t) = -\log \cos^2\left(\frac{\pi}{2}(1-t)\right) \quad (91)$$

余弦时间表。余弦时间表给出如下：

$$\sigma(t) = -\log \cos\left(\frac{\pi}{2}(1-t)\right) \quad (92)$$

线性。线性时间表给出如下：

$$\sigma(t) = \sigma_{\max} t \quad (93)$$

其中  $\sigma_{\max}$  是一个非常大的数字。在我们的实验中，我们将其设置为  $10^8$ 。

#### E.1.1 ELBO 不变性

由于第 2 节中的单调性假设，函数  $\alpha_t$  是可逆的。3.1，因此我们可以对 (10) 中的变量进行以下更改： $\gamma \equiv \log(1 - \alpha_t)$ 。令  $f : [0,1] \rightarrow \mathbb{R}^+$  为这样的函数

that  $\gamma = f(t)$ . Note that  $\alpha_t$  goes through a monotonic transformation to obtain  $\gamma$ ; hence,  $\gamma$  is also monotonic in  $t$  since  $\alpha_t$  is monotonic in  $t$ . This implies that the function  $f$  is invertible. Let  $t = f^{-1}(\gamma)$ . Then, we can have the following diffusion loss:

$$\begin{aligned}
\mathcal{L}_{\text{NELBO}}^\infty &= \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle dt \\
&= -\mathbb{E}_q \int_{t=0}^{t=1} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \frac{d}{dt} [\log(1-\alpha_t)] dt \\
&= -\mathbb{E}_q \int_{t=0}^{t=1} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \frac{d}{dt} [f(t)] dt && \text{Substituting } f(t) = \log(1-\alpha_t) \\
&= -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \mathbf{x}_\theta(\mathbf{z}_{f^{-1}(\gamma)}, f^{-1}(\gamma)), \mathbf{x} \rangle d\gamma && \text{Change of variables } \gamma \equiv f(t) \\
&= -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \mathbf{x}_\theta(\tilde{\mathbf{z}}_\gamma, f^{-1}(\gamma)), \mathbf{x} \rangle d\gamma && \tilde{\mathbf{z}}_\gamma \equiv \mathbf{z}_{f^{-1}(\gamma)} \\
&= -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \tilde{\mathbf{x}}_\theta(\tilde{\mathbf{z}}_\gamma, \gamma), \mathbf{x} \rangle d\gamma && \tilde{\mathbf{x}}_\theta(\tilde{\mathbf{z}}_\gamma, \gamma) \equiv \mathbf{x}_\theta(\tilde{\mathbf{z}}_\gamma, f^{-1}(\gamma)) \quad (94)
\end{aligned}$$

This new formulation demonstrates that the diffusion loss is invariant to the functional form of  $\alpha_t$ . In Table 9, we demonstrate empirically that noise schedules with different functional forms evaluate to the same Likelihood which is consistent with our theory. However, different schedules lead to different per data point variance. Notably, the log-linear schedule exhibits the lowest variance among all the noise schedules considered.

Table 9: Likelihood in bits per dimension (BPD) for different noise schedules on OWT dataset, is reported along with the mean and variance associated with each noise schedule per data point. We empirically observe that noise schedules with different functional forms yield the same likelihood, consistent with our theory in Sec. 3.4; however, different schedules result in different variances.

| $\sigma(t)$         | Mean | Variance per datapoint |
|---------------------|------|------------------------|
| Log Linear (90)     | 3.30 | 1.81                   |
| Cosine (92)         | 3.30 | 3.30                   |
| Cosine Squared (91) | 3.30 | 3.30                   |
| Linear (93)         | 3.30 | 7.57                   |

## E.2 Faster sampling with caching

In Figure 10, we compare the wall clock times of various methods: AR, SEDD, MDLM with caching, and MDLM without caching for generating 64 samples on a single GPU. When sampling in batches, a change of 1 token would necessitate a call to the denoising model. Therefore, smaller batch sizes have a lower likelihood of a token being unmasked. This might lead one to prefer generating samples in smaller batches, as opposed to using a larger batch size that fully saturates the GPU. Table 10 shows that generating samples with a batch size of 1 and using caching is twice as fast as generating samples without caching while fully utilizing the GPU. In Fig. 2, we observe that MDLM without caching yields samples that consistently get better generative perplexity than SEDD. For  $T = \{5k, 10k\}$ , both SEDD and MDLM get better generative perplexity than the AR model.

Table 10: Wall clock time reported in minutes to generate 64 samples on a single A5000 GPU.

|           | $T = 5k(\downarrow)$ | $T = 10k(\downarrow)$ |
|-----------|----------------------|-----------------------|
| SEDD      | 85.3                 | 155.2                 |
| MDLM      | 70.3                 | 127.9                 |
| + caching | <b>40.1</b>          | <b>60.4</b>           |

那个 $\gamma = f(t)$ 。注意， $\alpha_t$ 经过单调变换得到 $\gamma$ ；因此， $\gamma$ 在 $t$ 中也是单调的，因为 $\alpha_t$ 在 $t$ 中是单调的。这意味着函数 $f$ 是可逆的。让 $t = f^{-1}(\gamma)$ 。那么，我们可以得到以下扩散损失：

$$\begin{aligned}
\mathcal{L}_{\text{NELBO}}^\infty &= \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1-\alpha_t} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle dt \\
&= -\mathbb{E}_q \int_{t=0}^{t=1} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \frac{d}{dt} [\log(1-\alpha_t)] dt \\
&= -\mathbb{E}_q \int_{t=0}^{t=1} \log \langle \mathbf{x}_\theta(\mathbf{z}_t, t), \mathbf{x} \rangle \frac{d}{dt} [f(t)] dt && \text{Substituting } f(t) = \log(1-\alpha_t) \\
&= -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \mathbf{x}_\theta(\mathbf{z}_{f^{-1}(\gamma)}, f^{-1}(\gamma)), \mathbf{x} \rangle d\gamma && \text{Change of variables } \gamma \equiv f(t) \\
&= -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \mathbf{x}_\theta(\tilde{\mathbf{z}}_\gamma, f^{-1}(\gamma)), \mathbf{x} \rangle d\gamma && \tilde{\mathbf{z}}_\gamma \equiv \mathbf{z}_{f^{-1}(\gamma)} \\
&= -\mathbb{E}_q \int_{\gamma=-\infty}^{\gamma=0} \log \langle \tilde{\mathbf{x}}_\theta(\tilde{\mathbf{z}}_\gamma, \gamma), \mathbf{x} \rangle d\gamma && \tilde{\mathbf{x}}_\theta(\tilde{\mathbf{z}}_\gamma, \gamma) \equiv \mathbf{x}_\theta(\tilde{\mathbf{z}}_\gamma, f^{-1}(\gamma)) \quad (94)
\end{aligned}$$

这个新公式表明扩散损失对于 $\alpha_t$ 的函数形式是不变的。在表9中，我们凭经验证明，具有不同函数形式的噪声表评估为相同的可能性，这与我们的理论一致。然而，不同的计划会导致每个数据点的差异不同。值得注意的是，对数线性计划在所有考虑的噪声计划中表现出最低的方差。

表 9: OWT 数据集上不同噪声表的每维位数 (BPD) 的可能性，以及与每个数据点的每个噪声表相关的平均值和方差一起报告。我们凭经验观察到，具有不同函数形式的噪声表产生相同的可能性，这与我们在第二节中的理论一致。3.4；然而，不同的时间表会导致不同的差异。

| $\sigma(t)$         | Mean | Variance per datapoint |
|---------------------|------|------------------------|
| Log Linear (90)     | 3.30 | 1.81                   |
| Cosine (92)         | 3.30 | 3.30                   |
| Cosine Squared (91) | 3.30 | 3.30                   |
| Linear (93)         | 3.30 | 7.57                   |

## E.2 使用缓存加快采样速度

在图 10 中，我们比较了各种方法的挂钟时间：AR、SEDD、带缓存的 MDLM 和不带缓存的 MDLM 在单个 GPU 上生成 64 个样本。批量采样时，1 个标记的变化就需要调用去噪模型。因此，较小的批量大小具有较低的标记被揭露的可能性。这可能会导致人们更喜欢以较小的批次生成样本，而不是使用使 GPU 完全饱和的较大批次大小。表 10 显示，在充分利用 GPU 的情况下，生成批量大小为 1 并使用缓存的样本的速度是不使用缓存生成样本的两倍。在图 2 中，我们观察到不带缓存的 MDLM 生成的样本始终比 SEDD 获得更好的生成困惑度。对于  $T = \{5k, 10k\}$ ，SEDD 和 MDLM 都比 AR 模型获得更好的生成困惑度。

表 10: 在单个 A5000 GPU 上生成 64 个样本的挂钟时间（以分钟为单位）。

|           | $T=5k(\downarrow)$ | $T=10k(\downarrow)$ |
|-----------|--------------------|---------------------|
| SEDD      | 85.3               | 155.2               |
| MDLM      | 70.3               | 127.9               |
| + caching | <b>40.1</b>        | <b>60.4</b>         |

Generative perplexities across sample times on OpenWebText

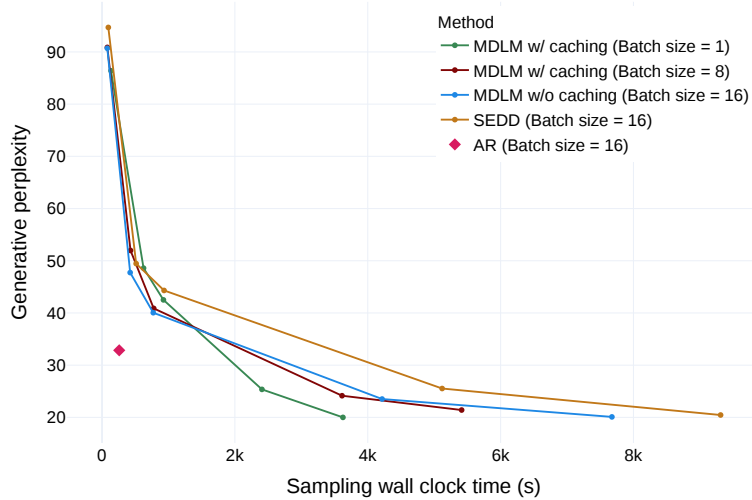


Figure 2: Generative perplexities across wall clock time for generating 64 samples on OWT using a single 32GB A5000 GPU are compared by varying  $T \in \{100, 500, 1000, 5000, 10000\}$  in the reverse diffusion process. The samples are generated in mini-batches with a batch size of 16 for AR, SEDD, and MDLM without caching, as it is the largest batch size that fits on this GPU. For MDLM with caching, we vary the batch size.

### E.3 LM1B ablations

We assess the importance of our continuous-time framework by performing ablation on diffusion steps  $T$ . In Table 11, we compare NLL and PPL under continuous and discrete  $T$  in MDLM. We find that NLL consistently decreases as  $T \rightarrow \infty$ .

Table 11: Discrete vs continuous time evaluation for MDLM w/o time-conditioning on OWT. MDLM was trained with  $T = \infty$ . We report test perplexity for a discrete  $T$ .

| T        | PPL( $\leq$ ) |
|----------|---------------|
| $\infty$ | <b>23.05</b>  |
| 10       | 42.18         |
| 20       | 30.70         |
| 50       | 25.77         |
| 100      | 24.35         |
| 200      | 23.66         |
| 500      | 23.26         |
| 1000     | 23.15         |

### E.4 Train NLL curves on OWT

In Figure 3, we show that MDLM achieves lower variance loss during training compared to a previous diffusion language model, SEDD. Training is performed over 1M steps on OWT (which corresponds to 524B tokens).

Generative perplexities across sample times on OpenWebText

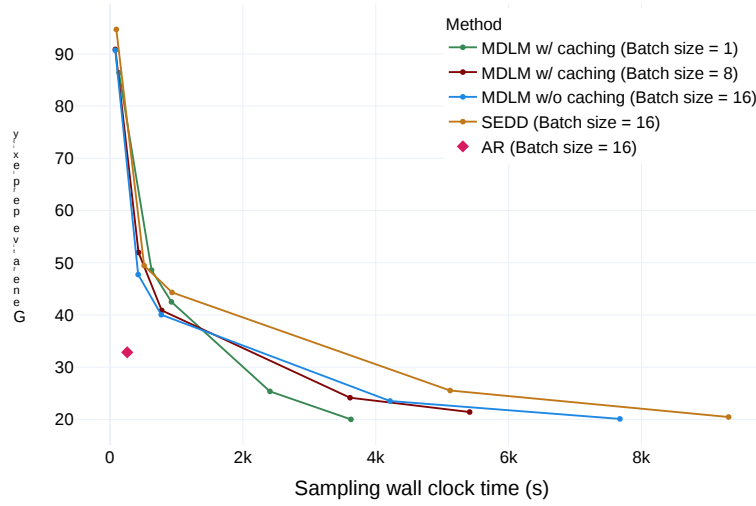


图 2: 通过在反向扩散过程中改变  $T \in \{100, 500, 1000, 5000, 10000\}$  来比较使用单个 32GB A5000 GPU 在 OWT 上生成 64 个样本的跨挂钟时间的生成困惑度。AR、SEDD 和 MDLM 的样本以小批量形式生成, 批量大小为 16, 无需缓存, 因为这是适合该 GPU 的最大批量大小。对于具有缓存的 MDLM, 我们改变批量大小。

### E.3 LM1B 消融

我们通过对扩散步骤  $T$  进行消融来评估连续时间框架的重要性。在表11中, 我们比较了MDLM中连续和离散 $T$ 下的NLL和PPL。我们发现 NLL 随着  $T \rightarrow \infty$  持续减小。

表 11: OWT 上不帶时间调节的 MDLM 的离散时间与连续时间评估。MDLM 使用  $T = \infty$  进行训练。我们报告离散  $T$  的测试困惑度。

| T        | PPL( $\leq$ ) |
|----------|---------------|
| $\infty$ | <b>23.05</b>  |
| 10       | 42.18         |
| 20       | 30.70         |
| 50       | 25.77         |
| 100      | 24.35         |
| 200      | 23.66         |
| 500      | 23.26         |
| 1000     | 23.15         |

### E.4 在 OWT 上训练 NLL 曲线

在图 3 中, 我们表明与之前的扩散语言模型 SEDD 相比, MDLM 在训练期间实现了更低的方差损失。在 OWT 上执行超过 1M 个步骤的训练 (相当于 524B 个令牌)。

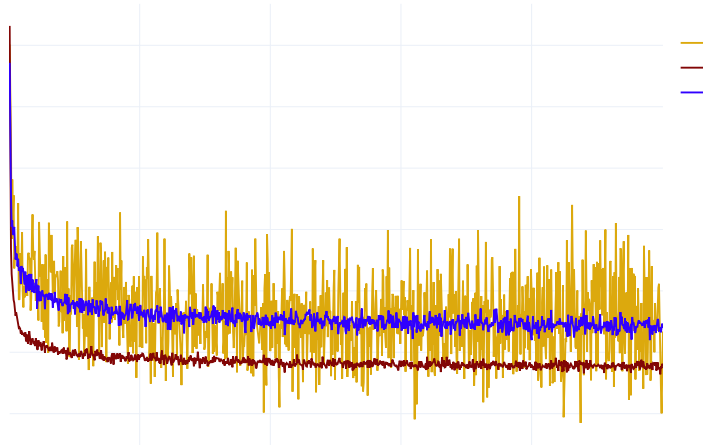


Figure 3: Train negative log-likelihood (NLL) curves across 1M gradient steps (524B tokens) on OpenWebText [18]. NLL is logged every 1K steps without value smoothing.

### E.5 Time-conditioning ablation on OWT

In Table 12, we assess the importance of time conditioning in MDLM on OWT. We observe that time-conditioning has minimal impact on perplexity. Training is performed over 1M steps on OWT (which corresponds to 524B tokens).

Table 12: Ablation on time-conditioning in MDLM on OWT.

| Method                     | PPL          |
|----------------------------|--------------|
| MDLM w/ time-conditioning  | 23.21        |
| MDLM w/o time-conditioning | <b>23.05</b> |

### E.6 Unconditional Samples

Here, we present some unconditional samples generated by MDLM trained on OWT with a context length of  $L=1024$  for  $T=\{1000,10000\}$ .

#### E.6.1 $T=1000$

**Example 1** `<|endoftext|>` a 17-10 victory and a trip to the playoffs.

The last wildcard seed: New York Jets, Houston and the last potable playoff spot. The last-second home wins: New Orleans and Carolina, 21-21.

The Saints finish sixth with the highest regular season (42) NFC wins. They lost 14 of their 13 games in the conference playoffs.

The Cardinals were in Group A in Round 1 with Game 2, Round 3, Round 4 and Quarter Game 5, but they made their last trip to the playoffs off North Carolina on the road as even North Carolina.

True to their reputation, the Cards swept the Saints in the first round, but knocked it out at home. No Panthers went to the playoffs more than the Saints.

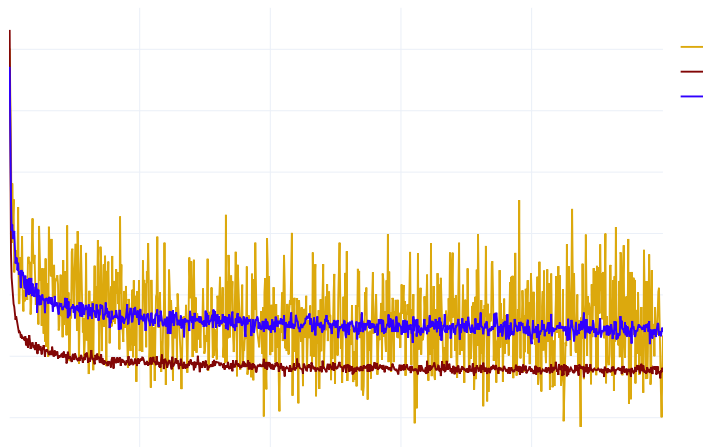


图 3: 在 OpenWebText [18] 上跨 1M 梯度步长 (524B 标记) 训练负对数似然 (NLL) 曲线。NLL 每 1K 步记录一次, 不进行值平滑。

#### E.5 OWT 上的时间调节消融

在表 12 中, 我们评估了 OWT 上 MDLM 中时间调节的重要性。我们观察到时间调节对困惑度的影响很小。在 OWT 上执行超过 1M 个步骤的训练 (相当于 524B 个令牌)。

表 12: OWT 上 MDLM 中时间调节的消融。

| Method                     | PPL          |
|----------------------------|--------------|
| MDLM w/ time-conditioning  | 23.21        |
| MDLM w/o time-conditioning | <b>23.05</b> |

#### E.6 无条件样本

在这里, 我们展示了一些由在 OWT 上训练的 MDLM 生成的无条件样本, 上下文长度为  $L = 1024$  (对于  $T = \{1000, 10000\}$ )。

##### E.6.1 $T = 1000$

示例 1  $\langle \text{endoftext} \rangle$  以 17-10 获胜并进入季后赛。

最后一支外卡种子队: 纽约喷气机队、休斯顿队和最后一个季后赛席位。最后一秒主场获胜: 新奥尔良队和卡罗莱纳队, 21-21。

圣徒队以 NFC 常规赛最高胜场数 (42 场) 排名第六。他们在分区季后赛的 13 场比赛中输掉了 14 场。

红雀队在第一轮、第二轮、第三轮、第四轮和四分之一场第五场比赛中分在 A 组, 但他们最后一次进入季后赛是在北卡罗来纳州的客场。

正如他们的声誉一样, 卡牌队在第一轮横扫了圣徒队, 但在主场将其淘汰。没有黑豹队比圣徒队更多地进入季后赛。

Don Jean no longer is the South Carolina Panther.

The Cardinals thought that provided that he had a chance to be an NFL player.

"I did," said defensive end Lorenzo Williams with a laugh as he exited his car at the airport. "Also, I won that game."

KC win brings Carson back home.

Griffin made promise on Sunday to never exactly give up the dunk. Although he failed to score 40 points in the playoffs, he has had better luck in them this year.

With turnovers and fumbles returning, he has to play out because the team doesn't trust him. He's long years of injuries, turnovers and calls because he knows he can play that way for everybody.

Griffin is no stranger to Saints fans.

"Players want him to know them," someone from South Carolina said after coming out against the Panthers - in their best home home Week 7 win Sunday - in an 11-9 rout. South Carolina did win its final three and passed the Saints, 24-1.

Although the Cardinals are in the South, they am a step behind.

They still have little time left to take down the West Coast wild card. There is no chance they get another victory.

The West was out by Beshear in their first round games last season, losing by 63 to the 49ers.

The outcome will be tough.

"Now we're so close, let's figure out the time to win," Brees said. "We still have a few games left; I'm glad about that."

South Carolina takes the revenge.

When asked about his second time since Super 4, Brees shot back that he understood.

"You can doubt the answer but I think that was a no-brainer. In time, you try to prove an answer wrong," Brees said. "I think his ability will be as cool as Julio Jones' ability, but having that time [out] to my season was difficult overall. I did what I was expecting to do. Hopefully they'll tell me to try again."

After their late win, the NFL calls the Saints reschedule'must try.'

"Because Saints," those who am there still say it, "focus on defense and, offense is defense."

ESPN said Saints's star receiver Dashon Jeffishard, turning heads on long passes and connecting with open defenders, already had their 20-yard overall score from the field. When Jeffishard finished with three passing he set up. He obviously had no difference; it was his first snap-off.

With his changes in his starting lineup, Brees was just hoping they had little to prove against Carolina over the weekend.

"I felt like we didn't have quite enough focus on and there was so puny coverage that I wanted more of our guys at the same position so we'd up our game," he said.

Brees said South Carolina was well at linebacker.

唐·让不再是南卡罗来纳黑豹队的球员。

红雀队认为只要他有机会成为一名 NFL 球员就可以了。

“我做到了，”防守端锋洛伦佐·威廉姆斯在机场下车时笑着说道。“而且，那场比赛我赢了。”

KC 的胜利让卡森回到了家。

格里芬周日承诺永远不会完全放弃扣篮。虽然他在季后赛中没能得到40分，但今年他的运气更好了。

随着失误和掉球的回归，他必须上场比赛，因为球队不信任他。他多年来一直遭受伤病、失误和判罚，因为他知道他可以为每个人打球。

格里芬对于圣徒球迷来说并不陌生。

“球员们希望他了解他们，”一位来自南卡罗来纳州的人在对阵黑豹队后说道，黑豹队在周日以 11-9 击败了黑豹队，这是黑豹队第七周最好的主场胜利。南卡罗来纳州确实赢得了最后三场比赛，并以 24-1 击败了圣徒队。

红雀虽然在南方，但还是落后了一步。

他们距离拿下西海岸外卡的时间已经所剩无几了。他们没有机会再次取得胜利。

上赛季西部队在首轮比赛中被 Beshear 淘汰，以 63 分输给了 49 人队。

结果将会很艰难。

“现在我们已经很接近了，让我们找出获胜的时间，”布里斯说。“我们还剩下几场比赛；我对此很高兴。”

南卡罗来纳州报仇。

当被问及自《超级4》以来他的第二次经历时，布里斯回击说他理解。

“你可以怀疑这个答案，但我认为这是理所当然的。随着时间的推移，你会试图证明答案是错误的，”布里斯说。“我认为他的能力将和胡里奥·琼斯的能力一样酷，但总的来说，在我的赛季中度过这段时期是很困难的。我做了我期望做的事情。希望他们会告诉我再试一次。”

在他们最后获胜后，美国国家橄榄球联盟称圣徒队的重新安排“必须尝试”。

“因为圣徒，”在场的人至今仍这么说，“专注于防守，进攻就是防守。”

ESPN 表示，圣徒队的明星接球手达尚·杰菲沙德 (Dashon Jeffishard) 在长传中转身并与空位防守球员联系，已经在场上取得了 20 码的总得分。当杰菲沙德完成三次传球后，他起球。他显然没有什么不同；这是他第一次突然爆发。

随着首发阵容的变化，布里斯只是希望他们在周末对阵卡罗莱纳的比赛中没有什么可以证明的。

“我觉得我们没有足够的关注，而且报道也太少了，所以我希望我们更多的人处于同一位置，这样我们就可以提高比赛水平，”他说。

布里斯说南卡罗来纳州的线卫表现出色。

"If that's part of it, if you're going to try and stay with what you're going [out with]. What'd you want? Smart play," Brees said. "What would you say? That you're always ready to play. So you're going out there strong and ready to go to play football."

That said, New Orleans was damned shy when it came to Carolina.

"My guys admit to feeling it a little bit [Sunday but] I say to them that they always knew, 'I don't think that was necessarily how I would beat you, that will give them their confidence," Brees said.

"It was really hard because I've obviously learned a lot of detail about how to deal with everyone and as hard as I have to be, I also feel part of the stuff that they've been through on the team, like they're still going to go through things they know are somewhat right, but they feel a lot of pressure so it's got to be important to get it right now to get it in the future."

Could all ask for more roses?

Let's just take a slip, South Carolina, and face the NFC<|endoftext|>

## **Example 2** <|endoftext|> Memorial Hospital.

Valia and Hill had been working with the Coast Guard in response to public questions, and when they were reached couldn't comment on the new information, Chapman said.

People referred to Valia during the years from Hill's family in Ants, and she cut in contact with their family and friends in 2016.

"Each day they stepped on the bus, when they left they saw me on TV," she said.

After separating from their family recently, Valia, 32, also moved into a Richmond house last October.

Read or Share this story: [<|endoftext|>CIVIL C. "Marky" Hogan](http://usat.ly/1NNC4zY) has been charged with homicides with a few days remaining after the April 2 purdade high school shooting where an undercover medical examiner and two other state and Illinois police officers was using heroin to go see a therapist.

DICEZ TV's Zach Putler reported Tuesday that Hogan was charged with felony drug possession by the Chicago Police Department at the preliminary hearing on Monday. Putler interviewed on Monday. Authorities could offer a limit until Cook County takes Tuesday afternoon or they have to assign plea agreements.

Dogan said in a news conference he made during a conference call Wednesday in Chicago that he believes the people who used him as a legal tool in the killing and fired employees hired for suffering also participated.

He said the couple's request to an attorney Monday will let the charges finally play out. Their lawyer did not respond Wednesday.

Dogan would not give away to possibility that he speculated in a statement that he would escape and return unless shot.

Chicago police initially said the other drug charges failed to raise enough evidence to establish why the killers were charged last year, raising the possibility that the drug mix contributed to a reason for their arrest. But a new statement was made Tuesday by a man

“如果这就是其中的一部分，如果你想尝试并坚持你要做的事情。你想要什么？聪明的比赛，”布里斯说。“你会说什么？你总是准备好比赛。所以你会坚强地出去并准备好踢球。”

也就是说，新奥尔良在面对卡罗莱纳州时非常害羞。

“我的队员们承认（周日）有一点感觉，但我对他们说，他们一直都知道，‘我不认为这一定是我击败你的方式，这会给他们信心，’布里斯说。

“这真的很难，因为我显然已经学到了很多关于如何与每个人打交道的细节，尽管我必须如此努力，但我也感觉到他们在团队中经历过的事情的一部分，就像他们仍然会经历他们知道有些正确的事情，但他们感到很大的压力，所以现在得到它对于将来得到它必须很重要。”

所有人都可以要求更多的玫瑰吗？

南卡罗来纳州，让我们来面对 NFC<|endoftext|>

示例 2 <|endoftext|> 纪念医院。

查普曼说，瓦利亚和希尔一直在与海岸警卫队合作回答公众问题，当我们联系到他们时，他们无法对新信息发表评论。

那些年来，人们从希尔在蚂蚁的家人那里提到了瓦莉娅，她在 2016 年切断了与他们的家人和朋友的联系。

“他们每天踏上公共汽车，离开时都会在电视上看到我，”她说。

32 岁的瓦利亚最近与家人分居，去年 10 月也搬进了里士满的一所房子。

阅读或分享这个故事：<http://usat.ly/1NNC4zY><|endoftext|>CIVIL C. “Marky” Hogan 被指控犯有杀人罪，4 月 2 日普德德高中枪击事件发生后几天，一名卧底法医和另外两名州和伊利诺伊州警察使用海洛因去看治疗师。

DICEZ TV 的扎克·普特勒 (Zach Putler) 周二报道称，霍根在周一的预审听证会上被芝加哥警察局指控犯有持有毒品重罪。普特勒周一接受了采访。当局可以在库克县周二下午之前提供限制，否则他们必须签署认罪协议。

多根在周三于芝加哥举行的电话会议期间举行的新闻发布会上表示，他相信那些利用他作为杀戮的法律工具并解雇受雇受苦的员工的人也参与了其中。

他说，这对夫妇周一向律师提出的请求将使指控最终得到解决。他们的律师周三没有回应。

多根不愿透露他在一份声明中推测，除非被枪杀，否则他将逃跑并返回的可能性。

芝加哥警方最初表示，其他毒品指控未能提供足够的证据来确定凶手去年被指控的原因，这增加了毒品混合导致他们被捕的可能性。但周二一名男子发表了新声明

who worked as the shooter in his unit on campus, and suggested the charges might be related to his work at university supervision.

His attorney and the university's president last week signaled that the incident of the bat gun was not a police investigation at Wednesday's conference.

Michael Durin, Illinois State University spokesman said he did not meet with university officials at the conference, and that university officials don't have any updates yet either.

"The fact that the Defendants were charged is a major factor in why it would get this much attention," the university spokesperson said. "Given that all the matters are not being resolved for months and months, any new specific information and other concerns they may be tasked with investigating now are understandable."

After city police began looking for evidence in connection to Hogan's April shooting, Durin said he had not noticed it until the Chicago Police Department found a person who was producing a bind gun on Illinois State campus. That same department found that 14 officers shot and were injured during a standoff, but it led to the launch of a combination of unrelated and related investigations leading to homicides charges in May 2015.

A memo from special school investigators suggests it had identified the drug fentanyl, and says the department had described the individual-oriented and inconsistent use of the gun, as well as the substance administered by CODC.

Dogan's allegations claim that a 2009 police gravesite package showed water running over campus and shows that the dental show photos of supposed victims were reinterpreted.

David Mann, a member of the Police Department, said he spoke exclusively to News 1 on condition of anonymity because university officials can't review documents immediately, and university officials had to change information that had been a consideration.

"We didn't change our information until he personally told the drug overdose problem," he said. "He said that drug dealing wasn't really a focal point."

He however, in a May 15 statement, also was "proleased" with university and state officials in the U.S. Attorney's Office. "We do not have any way to estimate the crime syndicate, given that the finding in the case does not preclude a separate review of standard CPD policy directed at any school employee, and whether that employee worked or went out of work," Mann said.

The lawsuit alleges that donations from a full visit with Hogan's workers and thousands of dollars, spent on gift cards, cigars, stock tickets to trips across the country, as well as other financial accounts and income from social members of former employees, were missing from buyers bought in stock.

Both employees and co-participants remain employed at the university.

On the Illinois State University Facebook page<|endoftext|>Image Steve Healey, the Cook Electronics Co. Ltd., general manager of Aug. 8-Feb. 16 at a recent internal session meeting (see link) held in The Apple Building in London. This July 1939 photo shows Peggy Deaver in 1986. (Photo courtesy the Cook Elec-

他在校园内的单位担任枪手，并表示这些指控可能与他在大学监督工作有关。

他的律师和大学校长上周在周三的会议上表示，蝙蝠枪事件并不是警方的调查。

伊利诺伊州立大学发言人迈克尔·杜林 (Michael Durin) 表示，他没有在会议上与大学官员会面，大学官员也没有任何最新消息。

该大学发言人表示：“被告受到指控的事实是该事件引起如此多关注的一个主要因素。”  
“鉴于所有问题在几个月内都没有得到解决，他们现在可能负责调查的任何新的具体信息和其他担忧都是可以理解的。”

在市警察开始寻找与霍根四月份枪击事件有关的证据后，杜林说，直到芝加哥警察局在伊利诺伊州立大学校园内发现一个人在制造绑定枪时，他才注意到这一点。该部门发现 14 名警察在对峙中开枪受伤，但这导致启动了一系列不相关和相关的调查，并于 2015 年 5 月提出了杀人指控。

特殊学校调查人员的一份备忘录表明，他们已经识别出了芬太尼药物，并表示该部门已经描述了以个人为导向且不一致的枪支使用情况，以及 CODC 管理的物质。

多根的指控称，2009 年警方墓地包裹显示校园内有水流过，并显示所谓受害者的牙科展示照片被重新解读。

警察局成员戴维·曼 (David Mann) 表示，他不愿透露姓名的情况下接受了 News 1 独家采访，因为大学官员无法立即审查文件，而且大学官员必须更改已考虑的信息。

“直到他亲自讲述药物过量问题后，我们才更改信息，”他说。“他说毒品交易并不是真正的焦点。”

然而，他在 5 月 15 日的一份声明中也对美国检察官办公室的大学和州官员表示“不满”。曼恩说：“鉴于本案的调查结果并不排除针对任何学校员工的标准 CPD 政策以及该员工是否工作或失业进行单独审查，我们没有任何方法来估计犯罪集团。”

该诉讼称，购买股票的买家缺少对霍根员工进行全面拜访的捐款，以及用于购买礼品卡、雪茄、全国旅行股票票以及其他财务账户和前员工社会成员的数千美元的收入。

员工和共同参与者均仍受雇于大学。

伊利诺伊州立大学 Facebook 页面<|endoftext|>图片 Steve Healey，库克电子有限公司 8 月 8 日至 2 月 2 日总经理 16 日在伦敦苹果大厦举行的最近一次内部会议（见链接）上。这张 1939 年 7 月的照片拍摄的是 1986 年的佩吉·迪弗 (Peggy Deaver)。（照片由库克电子公司提供）

tronics Co. Ltd., displayed here under G. Healey in prison dress.)<|endoftext|>Steve GIRO's H-P<|endoftext|>

## E.6.2 T=10000

**Example 1** <|endoftext|> has been presenting a number of temporary measures to help help resolve a crisis.

The last crisis was like this, but we have to resolve it at the specified level; we have to stay low. The people of Greece will be interested in the effectiveness of the measures. They will not only manage these measures, but also they will help in order to cope with the problems of the fiscal stability.

However, we do not want to dispose assets for the treasury. This also, so we will work on developing the national economy, and also paying on the national debt.

This affects the national incomes

So, as of 2007-2011, we use the government's temporary measures as a measure, helping resolve the crisis. In addition, we are able to pay for the borrowing costs. Additionally, we pay \$440 billion to settle debt issues, which can never be settled by default in a country.

These temporary measures will be aimed on several fronts, because the government will have three different partners in the system, in my right.

Firstly, we will be allowed to borrow a lot more upon the addition of the emergency measures and these temporary measures will help provide for the repayment of our debts before we are forced into a crisis as a result of our borrowing bills. Secondly, in the case of this, we will have resort to temporary measures in the revenue budget for Greece. The budget costs the government another \$5.2 billion a year.

So what you propose - if it happens again, does this mean that, since 2010, will you resolve the deficits which will occur on the basis of what we already have?

The fiscal situation

We would be able to settle our debts by the end of June which is the end. That said, we are taking our part as one of the most important countries in Europe, not only to make a proper transfer of the money but also to rely on it in the economy. However, first of all, we cannot achieve this on a day-to-day basis.

It is still true that we have decided to be able to deal with the economic situation of the country, but there might be another change in the fiscal situation, and therefore, we will try to negotiate on the situation at the end of June and over the summer.

The changes in the fiscal situation would be up to the parliament of management, bureaucrats, judges and a legitimate parliament of Greece.

Is the government planning to talk about the 'temporary measures' of Greece?

We will continue the process to operate through the temporary measures. This is not a temporary measure at this point, because after a crisis, not there yet at crisis level, you can still have enough investment purchases until the end of the month.

tronics Co. Ltd., 此处展示的是身着囚服的 G. Healey。) <|endoftext|>Steve GI  
RO 的 H-P<|endoftext|>

E.6.2 T =10000

示例 1 <|endoftext|> 提出了一些临时措施来帮助解决危机。

上次危机就是这样，但是我们要在规定的层面上解决，我们必须保持低调。希腊人民将对这些措施的有效性感兴趣。他们不仅将管理这些措施，还将帮助解决财政稳定问题。

然而，我们不想为财政部处置资产。所以我们还要致力于发展国民经济，还好国债。

这影响国民收入

所以，从2007年到2011年，我们用政府的临时措施作为措施，帮助解决危机。此外，我们有能力支付借款费用。此外，我们还支付了 4400 亿美元来解决债务问题，而在一个国家违约的情况下，这些问题永远无法得到解决。

这些临时措施将针对多个方面，因为政府将在系统中拥有三个不同的合作伙伴，就我而言。

首先，在增加紧急措施后，我们将被允许借更多的钱，这些临时措施将有助于在我们因借贷账单而被迫陷入危机之前偿还债务。其次，在这种情况下，我们将在希腊的收入预算中采取临时措施。该预算每年让政府另外花费 52 亿美元。

那么您的建议是——如果再次发生这种情况，这是否意味着自2010年以来，您会在我们已有的基础上解决将出现的赤字吗？

财政状况

我们将能够在六月底之前还清我们的债务。也就是说，我们作为欧洲最重要的国家之一，不仅要进行适当的资金转移，而且要在经济中依赖资金。然而，首先，我们无法在日常工作中实现这一目标。

确实，我们已经决定能够应对国家的经济形势，但财政形势可能会出现另一个变化，因此，我们将尝试在六月底和整个夏季就形势进行谈判。

财政状况的变化将取决于希腊管理层、官僚、法官和合法议会的议会。

政府是否计划谈论希腊的“临时措施”？

我们将继续通过临时措施进行运营。目前这不是临时措施，因为危机过后，尚未达到危机水平，您仍然可以在月底之前有足够的投资购买。

Again the government decided to create a temporary measure and now it depends upon a particular event, such as that there's another liquidity crunch. It is better that the government and the authorities decided to create a temporary measure effective in June at fair sum monthly bond rates.

The temporary measures will also enhance the government's economic status, especially when following the measures at the end of the month.

Temporary measures is a real tool for growth, not just for the economy.

Knowing that there are several measures in place to increase our supply, for example, the level of profit on public sector enterprises is certain, under all of these temporary measures the increases in output, after that, will increase the external demand and the internal demand.

We will be able to create the demand, and also strengthen the government's credibility through fiscal organization. What is important here, here is that we will apply these measures to our reserves, and at the same time, we apply these measures to the debt level, which will also be the aim of debt-free Greece.

So, first of all, everything is certain of what continues to be collected by the government. Given the situation and after the release of the last data on October 16, you also recognize that this will not be any kind of non-payment.

In the case of the payment against the equipment, we will be able to manage with the measures.

What does government expect in its plans to create a fiscal consolidation for the public sector and the new budget.

Regarding this is the temporary measure, we will be able to cope with the troubled finances. However, I do not think it is any measure which threatens the fiscal stability of the economy. However, that is not a temporary measure, a permanent measure.

On the other hand, there will be our ongoing work on construction in the ministry. If this falls, we will continue work on job creation, the expansion of the economy.

Also, also the government mentioned the new government reforms, which increased labor hours for the employees, which will further the economic growth, and the second aspect of budget as well and this is government welfare, which will improve the quality of life. We will<|endoftext|>

**Example 2** <|endoftext|> him. He said: "What are you doing?"

I hesitated before answering. "Boy, this is so exciting. You need a better girl. Is she?"

And I said, "You don't have a brain. You have no brain anymore."

After a minute, he had walked back and said on his own, working through that, he thought he had got himself going in a new direction.

He could've been a better boy in the first three years.

"You'll only have once before it starts."

MVP

政府再次决定制定一项临时措施，现在这取决于特定事件，例如另一次流动性紧缩。政府和当局最好决定制定一项临时措施，并于 6 月生效，按月公允债券利率计算。

临时措施还将提高政府的经济地位，特别是在月底采取措施后。

临时措施是促进增长的真正工具，而不仅仅是经济。

知道有一些措施来增加我们的供给，例如，公共部门企业的利润水平是确定的，在所有这些临时措施下，产出的增加，之后将增加外部需求和内部需求。

我们将能够创造需求，并通过财政组织增强政府的信誉。这里重要的是，我们将这些措施应用于我们的储备，同时，我们将这些措施应用于债务水平，这也将是希腊无债的目标。

因此，首先，政府继续收集的数据是确定的。鉴于这种情况，并且在 10 月 16 日发布最后数据之后，您也认识到这不会是任何形式的不付款。

如果是针对设备付款，我们将能够采取措施进行管理。

政府对公共部门财政整顿计划和新预算有何期望？

这是临时措施，我们将能够应对陷入困境的财务。然而，我认为这不会威胁到经济的财政稳定。但这不是临时措施，而是永久性措施。

另一方面，我们部门将继续进行建设工作。如果这一数字下降，我们将继续致力于创造就业机会和扩张经济。

此外，政府还提到了新的政府改革，增加了雇员的劳动时间，这将进一步促进经济增长，预算的第二个方面是政府福利，这将提高生活质量。我们将<|结束文本|>

示例 2 <|endoftext|> 他。他说：“你在做什么？”

我犹豫了一下才回答。“男孩，这太令人兴奋了。你需要一个更好的女孩。是吗？”

我说：“你没有大脑。你已经没有大脑了。”

一分钟后，他走回来并独自说道，通过这些努力，他认为自己已经走向了一个新的方向。

在最初的三年里，他本可以成为一个更好的男孩。

“在开始之前你只能有一次。”

MVP

The story is always, "That's what the other guy has to do." He was the guy who had to do anything. He had to reason with school officials. My cousin mentioned to me that some of my friends almost doubled over at one meeting.

I'd picked up a lot of the money I owed him from high people in me; he liked my grades. Drop-outs didn't consider me high enough to let me go hang out. He hung up when I challenged him after practice to show a new talk. He started making, and, quite, never

I first saw him C. morning, in the sixth grade class. He wouldn't hang up with him on point at team meetings. He started talking about things about me: "I'm an M, I'll get an A. Tonight." Having had that conversation over lunch, my heart touched mine with pride. He came, my boy. Now he looks like he's going back to school. I don't know if he's going to sue. Let's just have a two-bedroom apartment, a \$500-dollar condo for renting, and a pool. And then he was back.

That was a part of my life as I think about it. It was the school year.

I never saw a guy come up at the locker room and show a new talk. That day one day, I told the high school, "We'll show up one day right here, we can have a little fun," and after this, I remember a small handful of the boys made friends, and they never, ever showed up for a new talk.

I call them "M's kids. I always remember him, I remember his ass up his ass, getting ready for a freshman orientation out there. He'll show everything.

He'll show if I'm freshman, I'll act I was going to play junior. In a few years I'll try it, then he'll make sure he's going to judge me. He will come over to me one day.

Then one day, the senior class was sitting on the bench, pressing his ball on the floor of the locker room, the referee was just standing the knelt it down.

And when he heard about that, another boy, three of his friends, and one of his cousins were on the other side of the room. The boys' class was filling in with his new brother and his new cousin and his new M.M's player.

The senior class watched me walk me through the chair to the bench. Everyone passed by the boy. Just on his toes on a foot, too.

He [and a girl] passed over his head and, as I looked at them, he carried me into the locker room. And the biggest part of the story, was a mistake.

With his elbows out, he pulled me down on my shoe, my other, sort of a- don't know what they were; palebelly somethings, like bleeding very much, or on little toes. He was up on the stairs and everybody watches, with men and high school kids, who saw him in the locker room. And he caught a breath. Then his old man approached me and, disappeared into the middle of the room. He took off his vest, fast enough as to herd him into the locker room. I walked into the room and read him little cards with my own eyes to make notes, to pull him under my shoes.

I told them: "Listen, because I say this today, when you talk to 'em today, "Just make sure you talk is more than you'll show. He's

故事总是这样的：“这就是另一个人必须做的事情。”他是那个必须做任何事情的人。他必须与学校官员讲道理。我的表弟向我提到，我的一些朋友在一次会议上几乎弯下腰来。

我欠他的很多钱都是从我内心的高人那里讨来的。他喜欢我的成绩。退学的人认为我的地位不够高，不能让我出去玩。当我在练习后挑战他展示新演讲时，他挂断了电话。他开始制作，而且，实际上，从来没有

我第一次见到他是在早上，在六年级的班级里。他不会在团队会议上挂断电话。他开始谈论关于我的事情：“我是 M，今晚我会得到 A。”午餐时的谈话让我感到自豪。他来了，我的孩子。现在他看起来要回学校了。我不知道他是否会起诉。我们只拥有一套两居室的公寓、一套价值 500 美元的出租公寓和一个游泳池。然后他回来了。

我认为那是我生活的一部分。那是学年。

我从来没有见过一个人出现在更衣室并发表新的演讲。那天，我告诉高中，“有一天我们会出现在这里，我们可以玩得开心。”此后，我记得一小部分男孩交了朋友，但他们从来没有出现过新的演讲。

我称他们为“M的孩子”。我一直记得他，我记得他的屁股上他的屁股，为新生准备在那里。他会展示一切。

他会告诉我，如果我是一年级新生，我会表现出我要打三年级。几年后我会尝试一下，然后他会确保他会评判我。总有一天他会来找我的。

然后有一天，高年级学生坐在替补席上，把球按在更衣室的地板上，裁判只是站着把球跪了下来。

当他听说这个消息时，另一个男孩、他的三个朋友和他的一个表兄弟在房间的另一边。男孩班上有他的新兄弟、他的新表弟和他的新 M.M's 球员。

高年级的同学看着我穿过椅子走到长凳上。所有人都从男孩身边经过。也只是踮起脚尖。

他（和一个女孩）从他的头上走过，当我看着他们时，他把我抱进了更衣室。这个故事最重要的部分是一个错误。

他伸出手肘，把我拉倒在我的鞋子上，我的另一只，有点不知道它们是什么；腹部苍白的东西，比如流血很多，或者小脚趾上流血。他在楼梯上，所有人都在观看，包括男人和高中生，他们在更衣室里看到了他。他屏住了呼吸。然后他的老人向我走来，消失在房间中央。他脱下背心，速度之快，把他赶进了更衣室。我走进房间，亲眼给他念小卡片，做笔记，把他拉到我的鞋子里。

我告诉他们：“听着，因为我今天这么说，当你今天和他们谈话时，‘只要确保你说的比你表现的多。他是

just listening to me, and he's telling me I'm going to be there for him."

I'm always the one who wants to do something important about you than I show up. I walk around and ask, I want a message from you, "Keep it going. It<|endoftext|>

只是听我说话，他告诉我我会在他身边。”

我总是那个想为你做一些比我出现的更重要的事情的人。我四处走动并询问，我想要你的消息，“继续下去。它<|endoftext|>

## NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [NA].
- [NA] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for NA).

**The checklist answers are an integral part of your paper submission.** They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

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### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: Claims are addressed

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Our method under-performs compared to autoregressive models. We also discuss other limitations in the paper.

### 3. Theory Assumptions and Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: They are in the proofs.

Guidelines:

- The answer NA means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.

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问题：摘要和引言中提出的主要主张是否准确反映了论文的贡献和范围？

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理由：索赔已得到解决

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问题：本文是否讨论了作者所做工作的局限性？

答案：[是]

理由：与自回归模型相比，我们的方法表现不佳。我们还在本文中讨论了其他限制。

### 3 理论假设与证明

问题：对于每个理论结果，论文是否提供了全套假设和完整（且正确）的证明？

答案：[是]

理由：它们在样张中。

指南：

- 答案 NA 意味着该论文不包含理论结果。
- 论文中的所有定理、公式和证明都应编号并交叉引用。
- 所有假设都应在任何定理的陈述中清楚地陈述或引用。
- 证明可以出现在主论文或补充材料中，但如果它们出现在补充材料中，则鼓励作者提供简短的证明草图以提供直觉。

- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

#### 4. Experimental Result Reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: We provide all hyperparameters necessary to reproduce the experiments and will provide code.

#### 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: We will release all code after the paper is accepted. The datasets are already public.

Guidelines:

- The answer NA means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so “No” is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://nips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

#### 6. Experimental Setting/Details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

Answer: [Yes]

Justification: We provide detailed hyperparameters for all experiments.

Guidelines:

- The answer NA means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

#### 7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

- 相反，论文核心中提供的任何非正式证明都应由附录或补充材料中提供的正式证明进行补充。
- 应正确引用证明所依赖的定理和引理。

#### 4. 实验结果重现性

问：论文是否充分披露了重现论文主要实验结果所需的所有信息，并达到影响论文主要主张和/或结论的程度（无论是否提供代码和数据）？

答案：[是]

理由：我们提供重现实验所需的所有超参数，并将提供代码。

#### 5. 开放数据和代码的访问

问题：论文是否提供对数据和代码的开放访问，并提供足够的说明来忠实地再现主要实验结果，如补充材料中所述？

答案：[否]

理由：论文被接受后，我们将发布所有代码。数据集已经公开。

指南：

- 答案 NA 意味着论文不包含需要代码的实验。
- 请参阅 NeurIPS 代码和数据提交指南 (<https://nips.cc/public/guides/CodeSubmissionPolicy>) 了解更多详细信息。
- 虽然我们鼓励发布代码和数据，但我们知道这可能是不可能的，因此“否”是一个可以接受的答案。论文不能仅仅因为不包含代码而被拒绝，除非这是贡献的核心（例如，对于新的开源基准）。
- 说明应包含运行以重现结果所需的确切命令和环境。有关更多详细信息，请参阅 NeurIPS 代码和数据提交指南 (<https://nips.cc/public/guides/CodeSubmissionPolicy>)。
- 作者应提供数据访问和准备的说明，包括如何访问原始数据、预处理数据、中间数据和生成数据等。
- 作者应提供脚本来重现新提出的方法和基线的所有实验结果。如果只有一部分实验是可重复的，他们应该说明脚本中省略了哪些实验以及原因。
- 在提交时，为了保持匿名，作者应发布匿名版本（如果适用）。
- 建议在补充材料（附加到论文中）中提供尽可能多的信息，但允许包含数据和代码的 URL。

#### 6. 实验设置/细节

问题：论文是否指定了理解结果所需的所有训练和测试细节（例如，数据分割、超参数、如何选择它们、优化器的类型等）？

答案：[是]

理由：我们为所有实验提供详细的超参数。

指南：

- 答案 NA 意味着该论文不包含实验。
- 实验设置应在论文的核心部分详细介绍，以便欣赏结果并理解结果。
- 完整的详细信息可以在代码中、附录中或作为补充材料提供。

#### 7. 实验统计意义

问题：论文是否报告了适当且正确定义的误差线或有关实验统计显著性的其他适当信息？

Answer: [Yes]

Justification: Many of our labels include error bars and standard deviations

Guidelines:

- The answer NA means that the paper does not include experiments.
- The authors should answer "Yes" if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g. negative error rates).

#### 8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes] .

Justification: We conduct all experiments on 8x 3090s, 8xA6000s, 8xA100s, or 8xH100s. The largest models on OpenWebText take 2 weeks to train on 8xA100, the LM1B models only take 2 days to train on the same hardware

Guidelines:

- The answer NA means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

#### 9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: We follow standard practices

Guidelines:

- The answer NA means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer No, they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

#### 10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

答案: [是]

理由: 我们的许多表格都包含误差线和标准差指南:

- 答案 NA 意味着该论文不包含实验。
- 如果结果附有误差线、置信区间或统计显著性检验, 至少对于支持论文主要主张的实验, 作者应该回答“是”。
- 应明确说明误差线捕获的变异因素(例如, 训练/测试分割、初始化、某些参数的随机绘制或给定实验条件下的总体运行)。
- 应解释计算误差线的方法(封闭式公式、调用库函数、引导程序等)
- 应给出所做的假设(例如, 正态分布误差)。
- 应明确误差条是标准差还是平均值的标准误差。
- 报告 1-sigma 误差线是可以的, 但应该说明这一点。如果错误正态性假设未得到验证, 作者最好报告 2-sigma 误差线, 而不是声明其 CI 为 96%。
- 对于不对称分布, 作者应注意不要在表格或图中显示对称误差条, 否则会产生超出范围的结果(例如负误差率)。

#### 8. 实验计算资源

问题: 对于每个实验, 论文是否提供了重现实验所需的计算机资源(计算工作者类型、内存、执行时间)的足够信息?

答案: 【是】。

理由: 我们在 8x 3090s、8xA6000s、8xA100s 或 8xH100s 上进行所有实验。OpenWebText 上最大的模型在 8xA100 上训练需要 2 周, LM1B 模型在相同硬件上训练只需要 2 天

指南:

- 答案 NA 意味着该论文不包含实验。
- 论文应指出计算工作人员 CPU 或 GPU、内部集群或云提供商的类型, 包括相关内存和存储。
- 论文应提供每次实验运行所需的计算量并估计总计算量。
- 论文应披露整个研究项目是否需要比论文中报告的实验更多的计算(例如, 未纳入论文的初步或失败的实验)。

#### 9. 道德准则

问题: 论文中进行的研究在各个方面都符合 NeurIPS 道德准则 <https://neurips.cc/public/EthicsGuidelines> 吗?

答案: [是]

理由: 我们遵循标准做法

指南:

- 答案 NA 意味着作者尚未审查 NeurIPS 道德准则。
- 如果作者回答“否”, 他们应该解释需要偏离《道德准则》的特殊情况。
- 作者应确保保持匿名(例如, 如果由于其管辖范围内的法律或法规而有特殊考虑)。

#### 10. 更广泛的影响

问题: 本文是否讨论了所开展工作的潜在积极社会影响和消极社会影响?

Answer: [Yes]

Justification: Our model will allow for more controllable text generation models, and do not increase the capability of current autoregressive models

Guidelines:

- The answer NA means that there is no societal impact of the work performed.
- If the authors answer NA or No, they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

#### 11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [Yes]

Justification: These models are trained on trivial datasets and unlikely to cause any harm compared to state of the art language models.

Guidelines:

- The answer NA means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

#### 12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: All assets are publically available and we respect the licenses for all the data.

Guidelines:

- The answer NA means that the paper does not use existing assets.

答案: [是]

理由: 我们的模型将允许更可控的文本生成模型, 并且不会增加当前自回归模型的能力

指南:

- 答案 NA 意味着所做的工作没有社会影响。
- 如果作者回答“不适用”或“否”, 他们应该解释为什么他们的工作没有社会影响, 或者为什么论文没有解决社会影响。
- 负面社会影响的例子包括潜在的恶意或非预期用途(例如, 虚假信息、生成虚假个人资料、监视)、公平性考虑(例如, 部署可能做出不公平地影响特定群体的决策的技术)、隐私考虑和安全考虑。
- 会议预计许多论文将是基础研究, 不与特定应用程序相关, 更不用说部署了。然而, 如果存在任何负面应用的直接路径, 作者应该指出。例如, 可以合理地指出, 生成模型质量的改进可用于生成虚假信息的深度伪造品。另一方面, 无需指出优化神经网络的通用算法可以使人们能够更快地训练生成 Deepfakes 的模型。
- 作者应考虑当技术按预期使用并正常运行时可能出现的危害、当技术按预期使用但给出不正确的结果时可能出现的危害、以及(有意或无意)滥用技术而造成的危害。
- 如果存在负面社会影响, 作者还可以讨论可能的缓解策略(例如, 模型的门控发布、除了攻击之外还提供防御、监控滥用的机制、监控系统如何随着时间的推移从反馈中学习的机制、提高机器学习的效率和可访问性)。

#### 11. 保障措施

问题: 本文是否描述是为负责任地发布滥用风险较高的数据或模型(例如预先训练的语言模型、图像生成器或抓取的数据集)而采取的保障措施?

答案: [是]

理由: 这些模型是在简单的数据集上进行训练的, 与最先进的语言模型相比, 不太可能造成任何损害。

指南:

- 答案 NA 意味着该论文不存在此类风险。
- 已发布的具有高误用或双重用途风险的模型应在发布时采取必要的保障措施, 以允许模型的受控使用, 例如要求用户遵守使用指南或访问模型的限制或实施安全过滤器。
- 从互联网上抓取的数据集可能会带来安全风险。作者应该描述他们如何避免发布不安全的图像。
- 我们认识到提供有效的保障措施具有挑战性, 许多论文并不要求这样做, 但我们鼓励作者考虑到这一点并尽最大努力。

#### 12. 现有资产的许可证

问题: 论文中使用的资产(例如代码、数据、模型)的创建者或原始所有者是否得到了适当的认可, 许可和使用条款是否明确提及并得到适当尊重?

答案: [是]

理由: 所有资产都是公开的, 我们尊重所有数据的许可证。

指南:

- 答案 NA 意味着该论文不使用现有资产。

- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, [paperswithcode.com/datasets](https://paperswithcode.com/datasets) has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

### 13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [NA]

Justification: We provide no new assets.

Guidelines:

- The answer NA means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

### 14. Crowdsourcing and Research with Human Subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NA]

Justification: [NA]

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

### 15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [NA]

Justification: [NA]

- 作者应该引用生成代码包或数据集的原始论文。
- 作者应说明使用的是哪个版本的资产，如果可能，还应包含 URL。
- 每项资产均应包含许可证名称（例如 CC-BY 4.0）。
- 对于从特定来源（例如网站）抓取的数据，应提供该来源的版权和服务条款。
- 如果发布资产，则应提供包中的许可证、版权信息和使用条款。对于流行的数据集，paperswithcode.com/datasets 已为某些数据集策划了许可证。他们的许可指南可以帮助确定数据集的许可。
- 对于重新打包的现有数据集，应提供原始许可证和派生资产的许可证（如果已更改）。
- 如果无法在线获取此信息，我们鼓励作者联系该资产的创建者。

### 13. 新资产

问题：本文中引入的新资产是否有详细记录，并且文档是否与资产一起提供？

答案：[不适用]

理由：我们没有提供新资产。

指南：

- 答案 NA 意味着该论文不发布新资产。
- 研究人员应通过结构化模板将数据集/代码/模型的详细信息作为其提交内容的一部分进行交流。这包括有关培训、许可证、限制等的详细信息。
- 文件应讨论是否以及如何获得资产被使用者的同意。
- 提交时，请记住对您的资产进行匿名化（如果适用）。您可以创建匿名 URL 或包含匿名 zip 文件。

### 14. 众包和人类受试者研究

问题：对于以人类为对象的众包实验和研究，论文是否包含向参与者提供的说明全文和屏幕截图（如果适用）以及有关补偿的详细信息（如果有）？

答案：[不适用]

理由：[不适用]

指南：

- 答案 NA 意味着该论文不涉及众包，也不涉及人类受试者的研究。
- 在补充材料中包含这些信息很好，但如果论文的主要贡献涉及人类受试者，那么主论文中应包含尽可能多的细节。
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### 15. 人体受试者研究的机构审查委员会 (IRB) 批准或同等批准

问题：论文是否描述了研究参与者产生的潜在风险，是否向受试者披露了这些风险，以及是否获得了机构审查委员会 (IRB) 的批准（或根据您所在国家或机构的要求进行的同等批准/审查）？

答案：[不适用]

理由：[不适用]

Guidelines:

- The answer NA means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

End of NEURIPS CHECKLIST. Must be at end of document after appendix

指南:

- 答案 NA 意味着该论文不涉及众包, 也不涉及人类受试者的研究。
- 根据进行研究的国家/地区, 任何人体受试者研究可能需要 IRB 批准 (或同等文件)。如果您获得了 IRB 批准, 您应该在文件中明确说明这一点。
- 我们认识到, 不同机构和地点的程序可能存在很大差异, 我们希望作者遵守 NeurIPS 道德准则及其机构指南。
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End of NEURIPS CHECKLIST. Must be at end of document after appendix